Gene Regulatory Networks in Evolutionary Development: From Deep Conservation to Clinical Translation

Generated at: 2026-04-03
Source URL: https://www.evobiosci.com/posts/gene-regulatory-networks-in-evolutionary-development-from-deep-conservation-to-clinical-translation

Gene Regulatory Networks in Evolutionary Development: From Deep Conservation to Clinical Translation

Abstract

This article synthesizes current research on Gene Regulatory Networks (GRNs) to explore their pivotal role in evolutionary developmental biology (evo-devo) and their growing applications in biomedicine. We first establish the foundational principles of GRN architecture, including conserved kernels and plastic peripheries, illustrating how they govern morphological evolution through concepts like developmental system drift. The review then details cutting-edge computational and experimental methodologies—from single-cell multi-omics to AI-powered network inference—that are revolutionizing GRN reconstruction. A critical troubleshooting section addresses persistent challenges in network validation, benchmarking, and modeling stochasticity. Finally, we present advanced frameworks for GRN validation and comparative analysis across species, highlighting how these approaches uncover core disease mechanisms. This integrated perspective is essential for researchers and drug development professionals aiming to leverage evolutionary insights for therapeutic innovation.

Core Principles of GRNs in Evolutionary Developmental Biology

Gene Regulatory Networks (GRNs) represent the complex control circuitry that governs cellular responses to internal and external cues, characterized by non-linear dynamics, feedback loops, and a modular architecture [1]. Understanding GRN structure is fundamental to deciphering developmental processes and evolutionary mechanisms. The architecture of GRNs can be deconstructed into core functional components: **kernels**, **modules**, and specific **regulatory logic**. These elements work in concert to determine the output of the network, influencing cell fate decisions, developmental robustness, and evolutionary innovation [2] [3]. Kernels constitute the essential, conserved heart of a developmental program, while modules are functional subunits that can be rearranged or rewired. The regulatory logic defines the operational rules governing interactions between components, creating a framework for understanding how complex biological functions emerge from molecular interactions [1].

Kernels: The Conserved Core of Developmental Programs

A GRN kernel is an evolutionarily conserved subcircuit that performs essential upstream functions in a developmental process. It is characterized by its **low propensity for evolutionary change** and its critical role in establishing the foundational identity of a body plan or organ system [3].

Identification and Properties of Kernels

Kernels are typically discovered through comparative evolutionary studies across phylogenetically distant taxa. Research on gastrulation in *Acropora digitifera* and *Acropora tenuis* coral species, which diverged approximately 50 million years ago, revealed a conserved regulatory kernel despite significant divergence in peripheral network components [3]. This kernel consisted of **370 differentially expressed genes** that were consistently up-regulated at the gastrula stage in both species, with enriched functions in axis specification, endoderm formation, and neurogenesis [3].

Table 1: Characteristics of GRN Kernels Identified in Evolutionary Studies

Property	Description	Experimental Evidence
Evolutionary Conservation	Retained across phylogenetically distant species	370-gene kernel conserved in *Acropora* species despite 50 million years of divergence [3]
Developmental Role	Executes essential upstream functions in development	Involved in axis specification, endoderm formation, and neurogenesis [3]
Topological Feature	Occupies upstream positions in hierarchical networks	Acts as a foundational circuit upon which other network components are built [3]
Perturbation Resistance	Resists change and maintains function under mutation	Maintains gastrulation process despite regulatory rewiring in peripheral networks [3]

The resilience of kernels stems from their **topological properties** within the broader network architecture. Their position as upstream regulators buffers them from evolutionary tinkering, as changes to kernel elements would have catastrophic developmental consequences. This conservation

manifests as preserved gene expression patterns and functional roles across vast evolutionary timescales, even when morphological outcomes appear similar but underlying mechanisms diverge through developmental system drift [3].

Modules: Functional Units for Evolutionary Innovation

GRN modules are discrete **functional subunits** that execute specific tasks within the broader network. Unlike kernels, modules exhibit **evolutionary plasticity** and can be rewired, duplicated, or modified to generate phenotypic diversity while preserving the core developmental program [3].

Modularity and Evolutionary Diversification

Modular architecture enables specific aspects of development to be modified without compromising essential kernel functions. The comparative study of *Acropora* species revealed significant temporal and modular expression divergence in orthologous genes, indicating GRN diversification rather than complete conservation [3]. This modular rewiring represents a fundamental mechanism for evolutionary change, allowing organisms to adapt to diverse ecological niches while maintaining core developmental processes.

Table 2: Mechanisms of GRN Module Evolution and Diversification

Mechanism	Functional Impact	Example in Acropora Species
Paralog Divergence	Generation of novel functions through gene duplication	*A. digitifera* exhibits greater paralog divergence consistent with neofunctionalization [3]
Expression Shift	Altered temporal or spatial deployment of existing genes	Significant temporal expression divergence of orthologous genes between species [3]
Alternative Splicing	Expansion of protein diversity from limited gene sets	Species-specific differences in alternative splicing patterns indicating independent peripheral rewiring [3]
Regulatory Rewiring	Changes in connectivity between network components	Conserved kernel with divergent peripheral circuitries [3]

The modular structure of GRNs provides a substrate for evolutionary innovation through several mechanisms. **Gene duplication** events create genetic raw material for new modules, with paralogs diverging in function through mutations in regulatory regions [3]. **Alternative splicing** expands proteomic complexity without requiring genome expansion, allowing single genes to generate multiple proteins that can participate in different modules [3]. This modular plasticity enables the same ancestral molecular toolbox to be readjusted in different organisms according to species-specific environmental pressures, mechanical constraints, and geometrical factors [3].

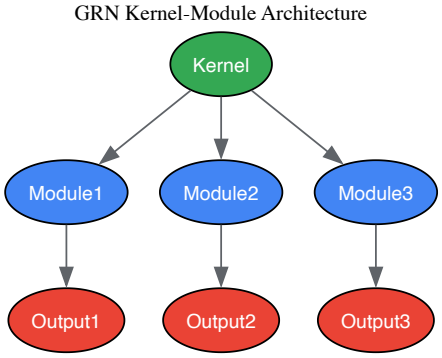


Diagram 1: Hierarchical organization of GRN kernels and modules. The conserved kernel regulates multiple specialized modules, which in turn produce specific developmental outputs.

Regulatory Logic: The Operational Framework

Regulatory logic defines the **combinatorial rules** that govern how transcription factors integrate multiple inputs to determine target gene expression states. This logic is implemented through cis-regulatory elements (CREs) that process information from multiple transcription factors [2].

Logic Functions in GRN Operation

The regulatory logic between transcription factors can follow different Boolean principles: **AND logic**, where both factors must be present for expression; **OR logic**, where either factor is sufficient; or more complex combinations incorporating **NOT logic** (repression) [2]. In the well-studied

Cross-Inhibition with Self-activation (CIS) network motif—found in systems like Gata1-PU.1 in hematopoiesis and Nanog-Gata6 in gastrulation—the logic function between self-activation and cross-inhibition fundamentally influences network dynamics and cell fate outcomes [2].

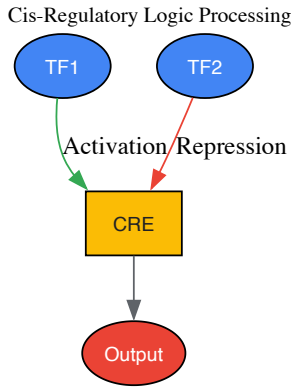


Diagram 2: cis-regulatory element integrating activating and repressive inputs to determine expression output.

Logic Motifs in Cell Fate Decisions

The incorporation of specific logic motifs profoundly influences cell fate decisions by shaping the **dynamics and stability** of developmental states. Research comparing noise-driven versus signal-driven fate decisions has revealed that regulatory logic determines how cells interpret intrinsic noise versus extrinsic signals [2]. In noise-driven mode, fate bias depends on spontaneous heterogeneity of gene expression, with regulatory logic determining how initial variations are amplified. In signal-driven mode, extrinsic signals reshape the epigenetic landscape, with logic gates determining cellular responses to these cues [2].

Table 3: Regulatory Logic Functions and Their Impact on Network Behavior

Logic Function	Mathematical Representation	Biological Impact on Fate Decisions
AND	Output = Input1 ∧ Input2	Creates bistable switches; enables coordinated response to multiple cues [2]
OR	Output = Input1 ∨ Input2	Increases robustness; allows multiple pathways to activate same program [2]
CIS with AND	Self-activation AND NOT cross-inhibition	Enables clear bistability; enhances fate commitment [2]
CIS with OR	Self-activation OR NOT cross-inhibition	Creates primed states; enables plasticity and trans-differentiation [2]

The relationship between regulatory logic and cell fate is particularly evident in the **priming stage** of differentiation, where cells enter a transient state of heightened responsiveness to fate-determining signals. This priming is directly controlled by the logic rules embedded in GRN architecture, which create specific attractor states in the developmental landscape [2]. Logic-incorporated GRN models have successfully deciphered distinctive trajectories of reprogramming influenced by logic motifs and characterized the progression-accuracy trade-off in differentiation processes [2].

Experimental Methodologies for GRN Analysis

Deciphering GRN architecture requires specialized experimental and computational approaches that can identify components, map interactions, and quantify dynamics.

Comparative Transcriptomics for Kernel Identification

Objective: Identify conserved kernels and divergent modules through cross-species comparison of gene expression during development [3].

Protocol:

- **Sample Collection**: Collect embryos from multiple species (e.g., *Acropora digitifera* and *A. tenuis*) at equivalent developmental stages (blastula/prawn chip, gastrula, sphere stages) [3].
- **RNA Sequencing**: Extract total RNA and prepare sequencing libraries. Sequence with sufficient depth (typically >30 million reads per sample) and biological replicates [3].
- **Differential Expression Analysis**: Align reads to respective reference genomes. Identify significantly differentially expressed genes between stages within each species [3].
- **Conservation Assessment**: Compare temporal expression patterns of orthologous genes across species. Identify genes with conserved up-regulation at key developmental transitions (e.g., gastrulation) [3].

- **Network Inference**: Construct co-expression networks and identify highly interconnected subnetices that are conserved across species—these represent candidate kernels [3].

Logic-Incorporating GRN Modeling

Objective: Incorporate regulatory logic into quantitative GRN models to simulate and predict cell fate decisions [2].

Protocol:

- **Network Topology Definition**: Based on literature and experimental data, establish network architecture (e.g., CIS topology with cross-inhibition and self-activation) [2].
- **Logic Function Assignment**: Assign specific logic rules (AND, OR) to each node based on cis-regulatory analysis or perturbation data [2].
- **Parameter Estimation**: Use optimization algorithms to fit model parameters to experimental time-course data of gene expression [2].
- **Dynamics Simulation**: Solve differential equations or run stochastic simulations under different initial conditions and parameter sets [2].
- **Landscape Analysis**: Construct potential landscapes to identify attractor states corresponding to different cell fates [2].
- **Validation**: Compare model predictions with experimental observations of fate bias, reprogramming trajectories, and noise patterns [2].

CRISPR-Based Functional Validation

Objective: Experimentally validate predicted GRN architectures and regulatory relationships using precision gene editing [4].

Protocol:

- **Guide RNA Design**: Use computational tools (e.g., CRISPR-GPT) to design high-efficiency guide RNAs with minimal off-target effects for targeting key GRN components [4].
- **Delivery System Selection**: Choose appropriate CRISPR system (Cas9, Cas12a) and delivery method (lentivirus, electroporation) based on cell type and experiment duration [4].
- **Cell Line Engineering**: Transfert target cells (e.g., human lung adenocarcinoma A549 cells) with CRISPR constructs and select successfully modified populations [4].
- **Phenotypic Characterization**: Assess editing efficiency (DNA sequencing), gene expression changes (RNA-seq), and functional consequences (differentiation, proliferation assays) [4].
- **Network Perturbation Analysis**: Systematically perturb multiple network nodes to validate predicted interactions and logic rules [4].

Computational Framework for GRN Analysis

Modern GRN analysis requires sophisticated computational frameworks that can handle the complexity and stochasticity of gene regulatory systems while quantifying uncertainty.

The PC-GRN Framework

The Probabilistic Categorical GRN (PC-GRN) framework represents a novel theoretical approach that integrates three core methodologies to address limitations of traditional modeling paradigms [1]. It combines **category theory** for formalizing modularity and composition of regulatory pathways, **Bayesian Typed Petri Nets (BTPNs)** as interpretable, mechanistic substrates for modeling stochastic cellular processes, and an **end-to-end generative Bayesian inference engine** that learns full posterior distributions over BTPN models directly from data [1].

This framework addresses critical challenges in GRN modeling, including high dimensionality, non-linearity, and stochasticity, by characterizing posterior distributions over ensembles of plausible models rather than identifying a single "correct" network [1]. The PC-GRN framework employs a GFlowNet to sample network topologies and a HyperNetwork to perform amortized inference for predicting corresponding parameter distributions, enabling rigorous uncertainty quantification in GRN inference [1].

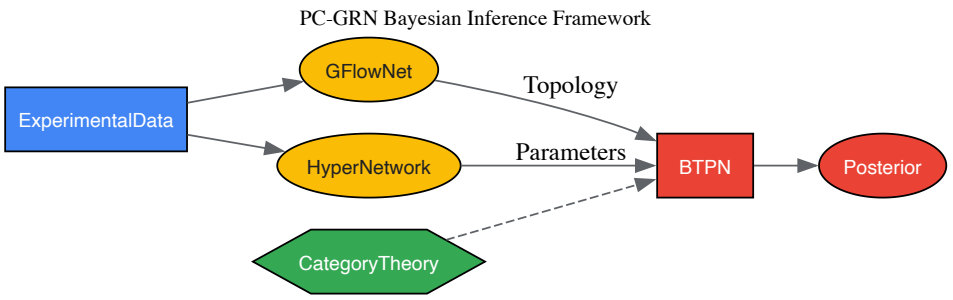


Diagram 3: The PC-GRN framework integrates category theory, GFlowNets, and HyperNetworks to infer Bayesian Typed Petri Net models from experimental data.

Research Reagent Solutions

Advanced GRN research requires specialized reagents and tools for precise manipulation and analysis of regulatory networks.

Table 4: Essential Research Reagents for GRN Architecture Studies

Reagent/Tool Category	Specific Examples	Function in GRN Analysis
Gene Editing Systems	CRISPR-Cas9, CRISPR-Cas12a, CRISPR-dCas9 [4]	Precise perturbation of network components; epigenetic modulation of gene expression
Computational Design Tools	CRISPR-GPT AI agent system [4]	Automated experiment planning, gRNA design, off-target evaluation, and protocol generation
Lineage Tracing Systems	Single-cell RNA sequencing, barcoding technologies	Mapping fate decisions and lineage relationships in developing systems
Logic Incorporation Models	Boolean networks with AND/OR gates; ODE models [2]	Simulating network dynamics and predicting fate outcomes under different regulatory logics
Comparative Genomics Tools	Ortholog identification; phylogenetic footprinting [3]	Identifying conserved kernels and divergent modules across species

The integration of AI-assisted tools like CRISPR-GPT has significantly accelerated GRN research by automating complex design tasks and making sophisticated gene-editing approaches accessible to non-specialists. This system leverages large language models with domain-specific knowledge for tasks including CRISPR system selection, guide RNA design, delivery method recommendation, and experimental protocol optimization [4]. Such tools enable researchers to systematically dissect GRN architecture by providing end-to-end experimental design and analysis capabilities.

Developmental System Drift (DSD) describes the phenomenon where deeply conserved morphological traits are maintained despite significant divergence in the underlying gene regulatory networks (GRNs) that control their development [3] [5]. This evolutionary paradox reveals that different genetic pathways can arrive at functionally similar phenotypic outcomes, highlighting the remarkable plasticity and robustness of biological systems. The concept has emerged from comparative studies across diverse taxa, from corals to nematodes, demonstrating that conservation at the phenotypic level often masks substantial molecular evolution. This whitepaper synthesizes current research on DSD, with particular emphasis on its implications for understanding evolutionary developmental biology and its relevance to biomedical research where conserved physiological processes may involve divergent genetic mechanisms across species.

Empirical Evidence for Developmental System Drift

Gastrulation Divergence in Acropora Corals

Gastrulation represents a fundamental morphogenetic process conserved across animals, yet its underlying molecular mechanisms exhibit striking divergence. Research comparing two coral species, *Acropora digitifera* and *Acropora tenuis*, which diverged approximately 50 million years ago, reveals compelling evidence for DSD [3] [5]. Despite nearly identical gastrulation morphology, comparative transcriptomics demonstrates extensive divergence in their gene regulatory networks.

Table 1: Quantitative Comparison of GRN Components in Acropora Species

GRN Component	A. digitifera	A. tenuis	Functional Implication
Conserved Gastrula-Upregulated Genes	370	370	Forms conserved regulatory "kernel" for axis specification, endoderm formation, and neurogenesis
Paralog Expression Pattern	High divergence, neofunctionalization	Redundant expression	Differential evolutionary trajectories following gene duplication
Alternative Splicing Patterns	Species-specific isoforms	Species-specific isoforms	Peripheral network rewiring around conserved core
Overall GRN Architecture	Divergent	Divergent	Developmental system drift despite morphological conservation

The study identified only 370 differentially expressed genes that were consistently up-regulated during gastrulation in both species, representing a conserved regulatory "kernel" responsible for core developmental processes [5]. This kernel operates within extensively rewired peripheral networks, demonstrating how conserved morphology can persist despite significant GRN evolution.

Mouth-Form Plasticity in Nematodes

Studies of mouth-form developmental plasticity in diplogastrid nematodes provide additional compelling evidence for DSD [6]. Research comparing *Pristionchus pacificus* and *Allodiplogaster sudhausi*, which diverged approximately 180 million years ago, reveals deep conservation of phenotypic plasticity in feeding structures despite significant functional divergence in the underlying genetic regulators.

Table 2: Functional Divergence of Conserved Developmental Plasticity Genes in Nematodes

Gene/Pathway	P. pacificus Function	A. sudhausi Function	Conservation Pattern
sulfatase (eud-1/sul-2)	Eu mouth-form switch	Eu mouth-form switch + novel Te morph regulation	Conserved function with additional novel role
sulfotransferase	St morph regulation	St morph regulation	Functional conservation as switch gene
Other mouth-form regulators	Quantitative effects on mouth-form	Distinct phenotypic effects	Functional divergence despite conservation
Overall GRN Architecture	Integrated plastic response	Integrated plastic response + novel morph	Conservation with lineage-specific elaboration

Despite the evolutionary distance, key regulatory genes maintain their role in mouth-form determination, though their precise functions and genetic interactions have diverged [6]. Some genes retain their switch-like function, while others exhibit quantitative effects, demonstrating different modes of regulatory evolution within the same developmental network.

Experimental Approaches for Investigating DSD

Comparative Transcriptomics Protocol

The primary methodology for identifying DSD involves comparative transcriptomics across species and developmental stages. The following protocol outlines the key steps based on current research [3] [5]:

- **Biological Material Collection**: Collect embryos at matched developmental stages across multiple species. For Acropora studies, samples included blastula (PC), gastrula (G), and sphere (S) stages collected in biological triplicates [5].
- **RNA Extraction and Library Preparation**: Extract total RNA using validated kits (e.g., Zymo RNA Isolation kits). Treat with DNase to remove genomic DNA contamination. Prepare sequencing libraries using standard protocols (e.g., Illumina TruSeq).
- **Sequencing and Read Processing**: Sequence libraries on appropriate platforms (e.g., Illumina NovaSeq). Process raw reads through quality control (FastQC), adapter trimming (Trimmomatic), and filtering.
- **Read Mapping and Transcript Assembly**: Map filtered reads to reference genomes using splice-aware aligners (STAR, HISAT2). For Acropora, mapping rates of 68.1-89.6% to *A. digitifera* and 67.51-73.74% to *A. tenuis* genomes were achieved [5]. Assemble transcripts using reference-based approaches (StringTie, Cufflinks).
- **Differential Expression Analysis**: Identify differentially expressed genes between species and stages using statistical frameworks (DESeq2, edgeR). In Acropora, this analysis revealed orthologous genes with significant temporal and modular expression divergence [3].
- **GRN Reconstruction and Comparison**: Construct co-expression networks (WGCNA) and identify conserved and divergent modules. Test for enriched functional categories among conserved elements.

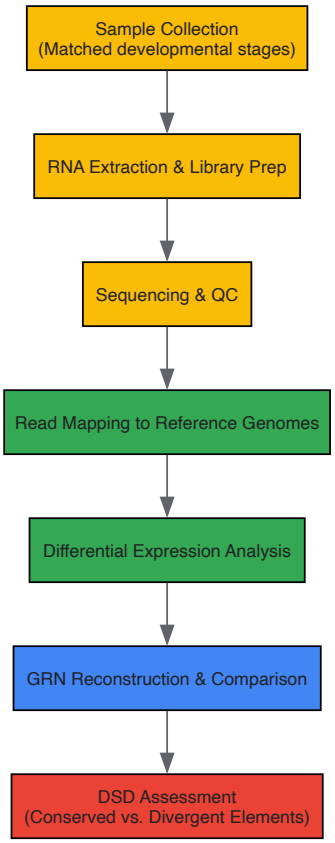


Figure 1: Experimental workflow for identifying developmental system drift through comparative transcriptomics

Functional Validation Using CRISPR-Cas9

Functional validation of DSD observations requires genetic manipulation to test hypotheses about conserved and divergent regulatory elements [6]:

- **Target Selection**: Identify candidate genes showing expression divergence between species but functioning in conserved developmental processes.
- **Guide RNA Design**: Design sgRNAs targeting specific genes of interest. For nematode studies, single guide RNAs targeting each pair of whole-genome duplication-derived genes were used [6].
- **CRISPR Delivery**: Microinject CRISPR-Cas9 components into embryos. For *A. sudhausi*, sgRNAs were injected into wild-type worms to generate knock-out mutants.
- **Mutant Selection**: Identify homozygous frameshift mutations in subsequent progeny through genotyping.
- **Phenotypic Assessment**: Score developmental phenotypes under controlled conditions. In nematode studies, mouth-form phenotypes were assessed across different dietary conditions [6].
- **Comparative Functional Analysis**: Compare mutant phenotypes across species to determine conservation or divergence of gene function.

Mechanistic Insights into DSD

Molecular Mechanisms Underlying DSD

Research across multiple systems has revealed several molecular mechanisms that facilitate developmental system drift:

Gene Duplication and Paralog Divergence Lineage-specific gene duplication events followed by paralog divergence represent a key mechanism for GRN rewiring. In Acropora, *A. digitifera* exhibits greater paralog divergence consistent with neofunctionalization, while *A. tenuis* shows more redundant expression patterns, suggesting different evolutionary trajectories following duplication events [3].

Alternative Splicing and Isoform Usage Species-specific differences in alternative splicing patterns provide another mechanism for GRN evolution without changes in gene content. Different isoforms can significantly impact phenotypes and promote morphological innovation by expanding molecular network connectivity [5].

Cis-Regulatory Element Evolution Changes in cis-regulatory elements, including promoters and enhancers, enable divergence in gene expression patterns without altering protein coding sequences. In dipteran insects, the evolution of overlapping expression between buttonhead and even-skipped transcription factors enabled the formation of a novel morphological structure (cephalic furrow) in cyclorrhaphan flies [7].

Post-translational Modifications Divergence in post-translational modification systems can alter protein function and interaction networks. Studies of actin in sponges reveal that different actin variants exhibit distinct post-translational modifications, cellular localizations, and functions despite similar sequences [8].

Theoretical Framework: The Hourglass Model

The hourglass model of developmental evolution provides a theoretical framework for understanding DSD. This model predicts early and late phases of developmental divergence within a phylum, linked by a morphologically conserved period of mid-embryonic development known as the phylotypic period [3]. DSD represents the molecular underpinnings of this model, revealing how conserved phenotypes can be maintained despite upstream and downstream regulatory divergence.

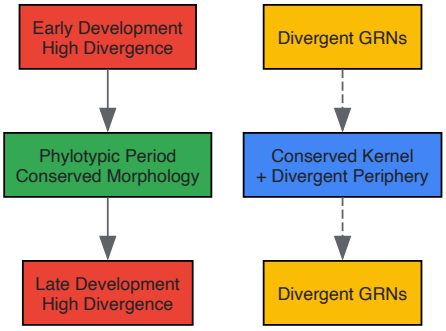


Figure 2: The hourglass model illustrating developmental system drift

The Scientist's Toolkit: Research Reagent Solutions

Table 3: Essential Research Reagents for Investigating Developmental System Drift

Reagent/Category	Specific Examples	Function in DSD Research
RNA Isolation Kits	Zymo RNA Isolation Kit (#R2051)	High-quality RNA extraction from limited embryonic material
DNase Treatment	Turbo DNase (#AM2238)	Genomic DNA removal for clean transcriptomic data
RNase Inhibitors	RiboLock RNase Inhibitor (#EO0381)	RNA integrity preservation during processing
Library Prep Kits	Illumina TruSeq Stranded mRNA	Strand-specific RNA-seq library preparation
CRISPR Components	Cas9 protein, synthetic sgRNAs	Gene knockout generation for functional validation
Reference Genomes	Species-specific genome assemblies	Read mapping and transcript quantification
Cell Dissociation Reagents	Collagenase, Trypsin-EDTA	Tissue separation for single-cell approaches

Implications and Future Directions

Biomedical Relevance

Understanding DSD has significant implications for biomedical research and drug development. The phenomenon explains why animal models may show different molecular pathways underlying conserved physiological processes, complicating translational research. Additionally, DSD informs our understanding of disease resistance and susceptibility across species, as divergent networks may respond differently to similar perturbations.

Conservation Biology Applications

In conservation biology, DSD highlights the importance of genetic diversity for evolutionary potential. Populations with greater genetic variation in their developmental networks may possess enhanced capacity to adapt to changing environments through GRN rewiring, even when morphological conservation is maintained.

Future Research Directions

Key future research directions include:

- Single-cell resolution studies of DSD across tissues and developmental trajectories
- Integration of epigenomic data to understand regulatory element evolution
- Investigation of DSD in response to anthropogenic environmental change
- Development of computational models predicting evolutionary outcomes of GRN architectures

The continued investigation of developmental system drift will undoubtedly yield deeper insights into the fundamental principles governing the relationship between genotypic and phenotypic evolution, with broad applications across biological disciplines.

The conceptual journey of the **epigenetic landscape**, first proposed by Conrad Waddington as a metaphor for cellular development, has evolved into a rigorous quantitative framework central to evolutionary developmental biology (evo-devo). Waddington's powerful visual analogy depicted a ball rolling down a furrowed hillside, representing a cell's progression from pluripotency to a committed differentiated state [9]. This metaphor captured the essence of developmental robustness and canalization—the tendency of developmental processes to produce consistent phenotypes despite environmental or genetic perturbations [9]. Today, this conceptual framework has been formalized mathematically through **gene regulatory networks (GRNs)**, transforming a qualitative analogy into a predictive, quantitative science that bridges the gap between genetic programs and emergent phenotypic patterns [9] [10].

Within modern evo-devo research, the integration of Waddington's landscape with GRN dynamics has created what is now termed the **Epigenetic Attractors Landscape (EAL)** [9]. This formal framework interprets the valleys and ridges of Waddington's landscape as **attractor states** within a high-dimensional state space defined by gene expression profiles [9]. These attractors correspond to stable cell fates—transient progenitor states or terminal differentiated phenotypes—while the topography of the landscape itself emerges from the underlying regulatory interactions [9]. This synthesis has provided evo-devo researchers with powerful computational tools to explore how evolutionary changes in GRN architecture manifest as alterations in developmental trajectories and phenotypic outcomes.

Theoretical Foundations: Mathematical Formalization of the Epigenetic Landscape

The Dynamical Systems Approach to Gene Regulatory Networks

The formalization of Waddington's landscape rests on a **dynamical systems view** of cell biology, where a cell's state is described by a vector of gene expression values: $x(t) = [x_1(t), x_2(t) \ldots, x_n(t)]$ for a GRN comprising n genes [9]. The **state space** encompasses all theoretically possible expression profiles, with each point representing a unique cellular configuration [9]. The temporal evolution of the cellular state follows a dynamical equation:

$$x(t+\delta t) = F(x(t), u, \delta t)$$

where F represents the **transition map** that connects a cell's current state to its future state, and u denotes parameters representing external signals [9]. This map F is physically implemented by the **GRN architecture**, which specifies both the topology of regulatory interactions and the nature of gene regulations [9]. The causal structure imposed by F constrains the possible behaviors of the system, leading to the emergence of stable gene expression configurations.

A fundamental insight from this perspective is that GRNs naturally give rise to a limited number of **stationary or quasi-stationary states**—the attractors that correspond to distinct cell fates [9]. When these steady states (x*) demonstrate resilience to perturbations (returning to x* after disturbance), they are classified as **attractors** in the dynamical systems framework [9]. The collection of these attractors and their basins of attraction constitutes the EAL, providing a quantitative incarnation of Waddington's original metaphor [9].

Quantitative Modeling Approaches for Epigenetic Landscapes

Several mathematical frameworks have been developed to quantitatively model the EAL, each with distinct strengths and applications in evo-devo research. The table below summarizes the primary approaches:

Table 1: Mathematical Frameworks for Modeling Epigenetic Landscapes

Modeling Approach	Key Features	Advantages	Limitations	Representative Applications
Hopfield Networks	Auto-associative neural networks; energy minimization; pattern completion [10]	Computationally efficient; direct from expression data; identifies driver TFs [10]	Discrete states; simplified dynamics	Cell differentiation trajectories; cancer subtypes [10]
Boolean Networks	Binary gene states (on/off); logical rules for transitions [9]	Handles large networks; captures essential dynamics	Oversimplified; lacks quantitative gradients	Pluripotency and differentiation [9]

Modeling Approach	Key Features	Advantages	Limitations	Representative Applications
Ordinary Differential Equations (ODEs)	Continuous gene expression; biochemical parameters [10]	Quantitative predictions; detailed dynamics	Requires kinetic parameters; computationally intensive	Binary cell fate decisions (GATA1/PU.1) [10]
Stochastic Models	Incorporates molecular noise; probabilistic transitions [11]	Explains heterogeneity; realistic dynamics	Computationally expensive; complex analysis	Cell fate priming; bet-hedging strategies [11]

The **Hopfield network (HN)** approach has proven particularly valuable for modeling EALs from empirical data [10]. In this framework, gene co-expression patterns are used to compute connection weights between nodes (genes/TFs), resulting in a weight matrix (W) that stores interactions across the network [10]. Each expression pattern is associated with an **energy score (E)**, with stable phenotypic states occupying lower energy values and transitional states exhibiting higher energy [10]. This energy landscape recapitulates the arc-shaped trajectory of differentiation, with cells progressing from stable primitive states through higher-energy transitional states to reach stable differentiated fates [10].

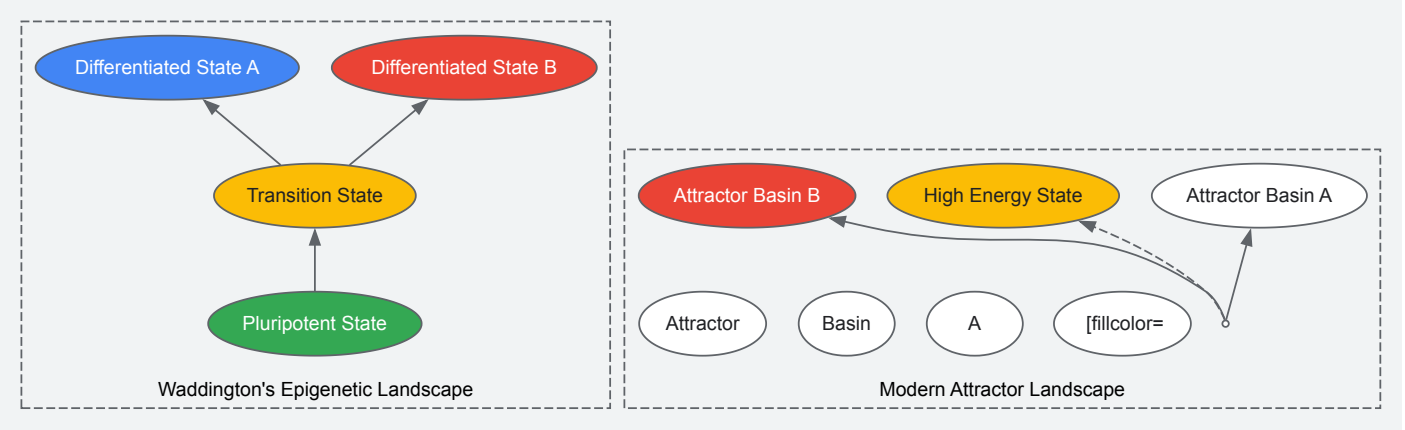


Figure 1: The conceptual evolution from Waddington's metaphorical landscape to the modern attractor landscape model. The ball represents a cell's gene expression state, with attractor basins corresponding to stable cell fates.

Current Research Applications: EAL Models in Evolutionary Developmental Biology

Modeling the Evolution of Phenotypic Plasticity

Recent research has leveraged the EAL framework to understand the evolution of **phenotypic plasticity**—the ability of a single genotype to produce different phenotypes in response to environmental conditions. In the nematode *Pristionchus pacificus*, which exhibits a **mouth-form dimorphism** (predatory versus bacterial-feeding forms), a detailed GRN has been elucidated that controls this plastic trait [12]. Developmental transcriptomics across different environments, genetic backgrounds, and mutants revealed that only two genes in the GRN (**eud-1** and **seud-1/sult-1**) function as environmentally sensitive **switch genes** during the critical window of plasticity [12]. These genes act as sequential checkpoints, with **eud-1** expression being sensitive to environmental conditions earlier in development than **seud-1/sult-1** [12].

This research demonstrates how EAL models can decompose complex plastic responses into **environmentally sensitive decision points** and downstream **phenotype-execution modules**. The differential expression of switch genes across strains and species with different mouth-form biases suggests that evolutionary changes in the regulatory regions of these genes underlie the evolution of plasticity thresholds [12]. Furthermore, comprehensive analysis revealed **metabolism** as a shared pathway regulating mouth-form plasticity, connecting environmental sensing to the GRN governing morphology [12].

Evolution of Displasticity Through GRN Architecture

The evolution of **dispersal plasticity**—a critical ecological trait—has been modeled using GRN architectures to understand how density-dependent and sex-biased dispersal evolve during range expansions [13]. Researchers developed an individual-based metapopulation model where the dispersal trait is represented as a GRN that takes population density and an individual's sex as inputs [13]. This approach contrasts with traditional **reaction norm (RN)** models by explicitly representing the genetic architecture underlying the plastic response.

Under equilibrium metapopulation conditions, the GRN model produced emergent density-dependent and sex-biased dispersal plastic response shapes matching theoretical expectations of RN models [13]. However, during range expansion, the GRN model led to **faster range expansion** when mutation effects were sufficiently large, because GRNs maintained higher **adaptive potential** [13]. This research demonstrates that the

genetic architecture of traits, represented as GRNs, significantly influences eco-evolutionary dynamics—a crucial consideration for understanding contemporary evolutionary processes.

Evolutionary Changes in Gene Regulation and Transposable Elements

The EAL framework has also illuminated how evolutionary changes in gene regulation, particularly through **KRAB zinc finger (KRAB-ZNF)** genes and their interactions with **transposable elements (TEs)**, have shaped human brain evolution and disease susceptibility [14]. KRAB-ZNF proteins represent the largest family of transcription factors in higher vertebrates and play a key role in repressing TE expression [14]. Through the development of **TEKRABber**—a computational tool for cross-species comparative analysis of TE expression—researchers have revealed significantly more interactions between KRAB-ZNF genes and TEs in humans than in other primates, particularly with recently evolved TEs [14].

This research demonstrates an evolutionary **arms race** between TEs and their repressors, with rapidly evolving KRAB-ZNF genes acquiring new regulatory relationships that potentially contribute to human-specific brain features [14]. In Alzheimer's disease patients, a subnetwork of interactions between KRAB-ZNF genes and Alu TEs appears diminished or lost, suggesting that the evolutionary legacy of TE-KRAB-ZNF interactions may influence susceptibility to neurodegenerative disease [14]. This work exemplifies how EAL thinking helps conceptualize the evolutionary trajectories of regulatory networks and their disease implications.

Table 2: Experimental Findings from EAL-Informed Evolutionary Studies

Biological System	Key Finding	Methodological Approach	Evolutionary Significance
Pristionchus mouth plasticity	Two-environmentally sensitive switch genes act as sequential checkpoints [12]	Developmental transcriptomics across environments and mutants	Evolution of plasticity through regulatory changes in switch genes
Dispersal evolution	GRN architecture maintains higher adaptive potential during range expansion [13]	Individual-based metapopulation modeling with GRN traits	Genetic architecture influences contemporary evolution
Human brain evolution	Increased KRAB-ZNF/TE interactions in humans versus other primates [14]	Cross-species correlation networks from RNA-seq data	Evolutionary arms race shapes human-specific features
Stem cell differentiation	Transient states have higher Hopfield energy than stable states [10]	Hopfield network modeling from time-course expression data	Developmental trajectories follow energy minimization

Methodological Guide: Experimental and Computational Protocols

Hopfield Network Analysis of Developmental Time-Course Data

The Hopfield network approach provides a practical methodology for deriving EALs from empirical gene expression data. The following protocol outlines the key steps:

1. Data Collection and Preprocessing:

- Collect **time-course gene expression data** across developmental stages using RNA-seq or single-cell RNA-seq [10].
- Perform **feature selection** to identify genes with significant expression variation (e.g., 3,753 from an initial 48,687 probes) [10].
- Normalize expression data and optionally discretize values (-1, +1) for classical HN implementation [10].

2. Network Construction:

- Compute **co-expression relationships** between genes to establish connection weights [10].
- Construct **weight matrix (W)** storing interactions among all nodes (genes/TFs) in the network [10].
- The weight matrix represents the "wiring diagram" of the underlying GRN that shapes the landscape topography [10].

3. Energy Calculation:

- For each developmental stage (expression pattern), compute the **Hopfield energy** using the formula derived from the network weights [10].
- Associate each developmental stage with its specific energy value (E) [10].
- Validate that stable phenotypic states have lower energy than transitional states [10].

4. Perturbation Analysis:

- Progressively perturb gene expression values (5%-50% of genes) for each stage [10].
- Compute energies of perturbed networks to assess **attractor stability** [10].
- Stable states (attractors) should maintain significantly different energies from randomly perturbed states [10].

5. Identification of Driver Genes:

- Compare discretized expression values between successive stages [10].
- Identify **TFs and genes** that switch activity states (-1 to +1 or vice versa) at transitions [10].
- These "switch genes" represent potential drivers of cell-fate transitions [10].

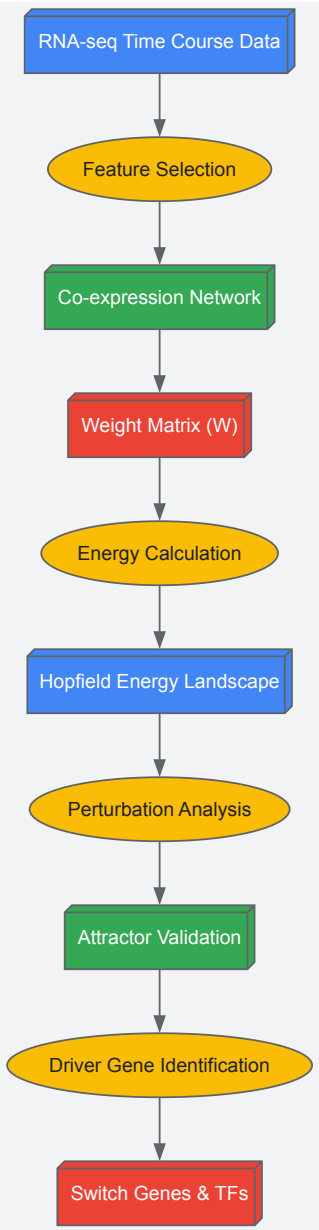


Figure 2: Computational workflow for deriving epigenetic landscapes from gene expression data using Hopfield networks.

Gene Regulatory Network Perturbation Experiments

Understanding causal relationships in GRNs requires carefully designed perturbation experiments:

1. Genetic Perturbations:

- Use **CRISPR/Cas9** or RNAi to systematically knock out or knock down candidate regulator genes [12].
- Employ **inducible systems** for temporal control of perturbations during critical developmental windows [12].
- In the *Pristionchus* system, mutagenesis screens identified >30 genes affecting mouth form [12].

2. Expression Profiling:

- Perform **RNA sequencing** across multiple time points following perturbations [12].
- Include multiple biological replicates and control for batch effects [12].
- Analyze differential expression relative to unperturbed controls [12].

3. Network Inference:

- Construct **regulatory networks** from expression data using information-theoretic or correlation-based approaches [15].
- Validate predicted interactions through **chromatin immunoprecipitation (ChIP-seq)** or reporter assays [15].
- Identify **network motifs** (e.g., feed-forward loops) that confer specific dynamical properties [15].

4. Landscape Mapping:

- Apply Hopfield or Boolean network models to derive **energy landscapes** from expression data [10].
- Characterize how perturbations alter landscape topography (attractor depths, positions, barriers) [10].
- Correlate landscape changes with phenotypic outcomes [10].

Table 3: Essential Research Reagents and Computational Tools for EAL Studies

Resource Category	Specific Tool/Reagent	Function/Application	Key Features
Computational Tools	TEKRABber [14]	Cross-species analysis of TE and gene co-expression	Handles evolutionary comparisons; correlation networks
Computational Tools	Hopfield Network implementation [10]	Landscape modeling from expression data	Energy minimization; attractor identification
Biological Models	*Pristionchus pacificus* [12]	Plasticity GRN studies	Inducible mouth forms; genetic tractability
Experimental Methods	Single-cell RNA-seq [15]	High-resolution expression profiling	Cellular heterogeneity; developmental trajectories
Experimental Methods	CRISPR/Cas9 [12]	Targeted gene perturbation	Precise genome editing; conditional mutations
Analysis Frameworks	Boolean network models [9]	Logical GRN modeling	Handles network complexity; captures essential dynamics
Analysis Frameworks	Ordinary Differential Equations [10]	Quantitative dynamical modeling	Detailed kinetic predictions; continuous dynamics

Future Perspectives: Challenges and Opportunities

While significant progress has been made in quantifying Waddington's epigenetic landscape, several challenges remain. A primary limitation is the **high dimensionality** of biological state spaces, with realistic GRNs involving hundreds to thousands of genes [9]. Current modeling approaches necessarily simplify this complexity, potentially missing important aspects of the true landscape topography [9]. Future work must develop more scalable computational frameworks that can handle biological complexity while remaining computationally tractable.

An exciting frontier lies in integrating **tissue mechanics and self-organization** with GRN dynamics [16]. Rather than conflicting models of development, genetic programs and physical processes appear to play complementary causal roles at different spatial scales [16]. This integration may resolve how macroscopic patterns emerge from molecular regulations, bridging a fundamental gap in evo-devo theory [16].

The increasing availability of **single-cell multi-omics data** provides unprecedented resolution for mapping developmental landscapes [15]. However, this data richness introduces new challenges in distinguishing technical noise from biologically meaningful heterogeneity [15]. Future methodological developments must robustly extract landscape features from sparse, high-dimensional single-cell data while accounting for measurement uncertainty.

Finally, applying EAL thinking to **evolutionary innovation** represents a promising research direction. By comparing landscapes across species and identifying changes in attractor positions and transition barriers, we may decipher how novel cell types and developmental trajectories evolve [14]. Such comparative landscape studies could reveal universal principles governing the evolvability of developmental systems.

The evo-devo toolkit has evolved from Waddington's qualitative metaphor to increasingly sophisticated quantitative models that bridge genes, development, and evolution. As these approaches continue to mature, they promise deeper insights into one of biology's most fundamental questions: how complex forms evolve through changes in developmental processes.

Gene regulatory networks (GRNs) provide a powerful mechanistic model for understanding how conserved developmental processes, such as gastrulation, can evolve across species. In evolutionary developmental biology (evo-devo), GRNs represent the molecular structure of developmental programs—reticulated webs of regulatory interactions that transform single-celled embryos into adult organisms [17]. The **modular nature** of these networks, where discrete subcircuits control specific developmental functions, enables parts of the network to evolve independently while preserving essential functions [18] [19]. This modularity facilitates evolutionary change through mechanisms such as co-option of existing modules or rewiring of regulatory connections [18] [17].

The phylum Cnidaria, particularly reef-building corals of the genus *Acropora*, serves as a pivotal model for studying the evolution of developmental mechanisms. As diploblastic metazoans and the sister group to bilaterians, cnidarians occupy a **basal phylogenetic position** that provides insights into ancestral developmental mechanisms [5] [3]. Within this framework, gastrulation represents a particularly informative process—while morphologically conserved across animals, its underlying cellular mechanisms exhibit remarkable variability [5] [3] [20]. This contrast between morphological conservation and molecular divergence presents an ideal opportunity to investigate how GRN architecture influences evolutionary trajectories.

Experimental Design and Comparative Transcriptomics Approach

Biological Model and Study System

This case study examines two coral species, *Acropora digitifera* and *Acropora tenuis*, which diverged approximately 50 million years ago [5] [3]. These species share similar developmental environments in the plankton but exhibit different spawning times, settling depth preferences, and polyp-stage morphologies [3]. Both species undergo a characteristic early development sequence: from a flattened blastula stage (prawn chip or PC), through gastrulation (G), to a spherical larval stage (sphere or S), and eventually to a planula larva [5].

Transcriptome Processing and Sequencing Methodology

The experimental design incorporated triplicate sampling for each of three early developmental stages (PC, G, S) in both *A. digitifera* and *A. tenuis*, resulting in nine libraries per species [5]. The following standardized RNA-seq workflow was implemented:

- **RNA Extraction and Library Preparation**: Total RNA was extracted from pooled embryos at each developmental stage. Sequencing libraries were prepared using standard protocols.
- **High-Throughput Sequencing**: Libraries were sequenced to generate raw read data, yielding approximately 30.5 million reads for *A. digitifera* and 22.9 million reads for *A. tenuis* after quality filtering [5].
- **Read Alignment and Processing**: Filtered reads were aligned to reference genomes (assembly accessions: GCA*014634065.1 for *A. digitifera*, GCA*014633955.1 for *A. tenuis*), achieving mapping rates of 68.1–89.6% for *A. digitifera* and 67.51–73.74% for *A. tenuis* [5].
- **Transcript Assembly and Quantification**: Aligned reads were assembled into transcript models, resulting in 38,110 merged transcripts for *A. digitifera* and 28,284 for *A. tenuis* [5]. The difference in transcript number was potentially attributed to greater sequencing depth in *A. digitifera*.

Table 1: Transcriptome Sequencing and Assembly Metrics

Parameter	*A. digitifera*	*A. tenuis*
Total reads after filtering	~30.5 million	~22.9 million
Read mapping rate	68.1–89.6%	67.51–73.74%
Merged transcripts	38,110	28,284
Developmental stages analyzed	PC, G, S (triplicates)	PC, G, S (triplicates)

Computational and Functional Analyses

Downstream bioinformatic analyses included:

- **Differential Gene Expression**: Identification of significantly up- and down-regulated genes across developmental stages within and between species.
- **Orthology Assignment**: Mapping of orthologous gene pairs between the two species to distinguish conserved versus divergent expression patterns.
- **Paralog Identification**: Detection of species-specific gene duplicates and analysis of their expression patterns.
- **Alternative Splicing Analysis**: Identification of differentially spliced transcripts across development.
- **Functional Enrichment**: Gene ontology and pathway analysis to identify biological processes enriched at specific developmental stages.

Results: Conserved Kernels and Divergent Regulation

Developmental System Drift in GRN Architecture

The comparative transcriptomic analysis revealed striking divergence in the gene regulatory programs underlying gastrulation in *A. digitifera* and *A. tenuis*. Despite **morphological similarity** during gastrulation, orthologous genes showed significant **temporal and modular expression divergence**, indicating GRN diversification rather than conservation [5] [3] [20]. This phenomenon, known as **developmental system drift**,

demonstrates that natural selection can maintain conserved morphological outcomes despite extensive rewiring of underlying genetic programs [5] [3].

The study identified a core set of 370 differentially expressed genes that were consistently up-regulated at the gastrula stage in both species [5] [3] [20]. This conserved regulatory "**kernel**" contained genes with roles in **axis specification, endoderm formation, and neurogenesis**, suggesting these processes represent essential, constrained components of gastrulation [3] [20]. The persistence of this kernel amidst broader GRN divergence highlights the modular architecture of developmental networks, where core functions can be preserved while peripheral components diverge.

Species-Specific Paralog Usage and Expression

A key finding was the differential utilization of paralogous genes between the two coral species. *A. digitifera* exhibited **greater paralog divergence**, with expression patterns consistent with neofunctionalization—where duplicated genes acquire new functions [5] [3]. In contrast, *A. tenuis* showed more **redundant expression** between paralogs, suggesting greater regulatory robustness in its developmental programs [5] [3].

Table 2: Patterns of Paralog Divergence in Acropora Species

Characteristic	*A. digitifera*	*A. tenuis*
Overall paralog divergence	High	Low
Primary evolutionary pattern	Neofunctionalization	Redundant expression
Regulatory robustness	Lower	Higher
Paralog usage	Divergent, specialized	Redundant, overlapping

These differences in paralog usage represent complementary evolutionary solutions to the challenge of maintaining developmental stability. Neofunctionalization in *A. digitifera* may enable **functional specialization**, while retained redundancy in *A. tenuis* provides **backup capacity** through paralogous genes that can compensate for each other's loss [21]. This functional redundancy between paralogs has been documented in other systems, where paralogous genes maintain the ability to replace each other despite having divergent expression patterns and functions under normal conditions [21].

Alternative Splicing Contributions to GRN Diversification

Beyond paralog divergence, species-specific differences in **alternative splicing** patterns further contributed to GRN diversification [5] [3]. Alternative splicing increases proteomic complexity without requiring genome expansion, allowing a single gene to generate multiple proteins with distinct functions [5]. The study found that differential isoform usage between species represented another mechanism for peripheral rewiring of the conserved gastrulation kernel, potentially contributing to **morphological innovation** and species-specific adaptations [5].

The Scientist's Toolkit: Research Reagent Solutions

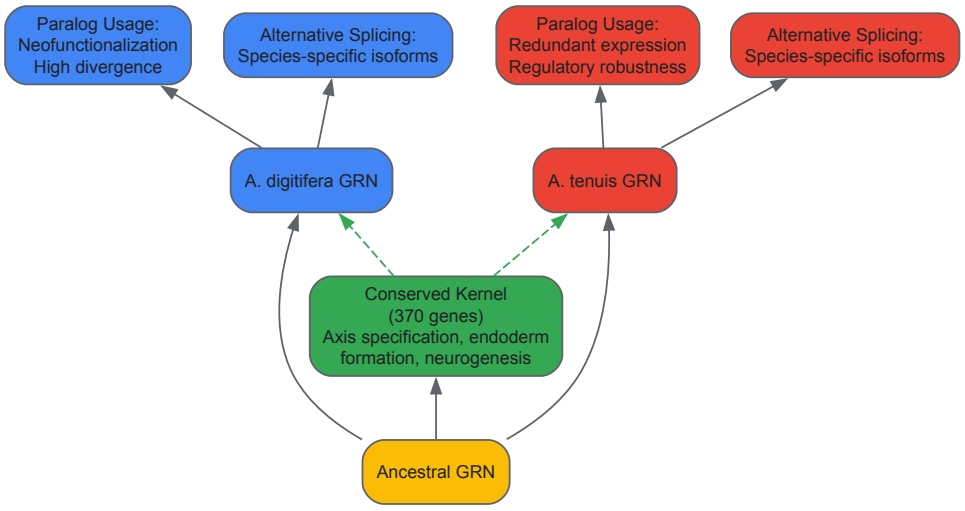
Table 3: Essential Research Reagents and Resources for GRN Studies in Non-Model Systems

Reagent/Resource	Function and Application	Considerations for Evolutionary Studies
Reference Genomes	Foundation for read alignment and transcript assembly; enables orthology mapping	Requires adequate assembly and annotation quality; crucial for cross-species comparisons [17]
Orthology Databases	Identification of conserved genes across species; distinguishes orthologs from paralogs	Challenging for recently duplicated genes; impacts functional inferences [22]
Species-Specific Molecular Probes	Detection of gene expression patterns via in situ hybridization or other methods	Must be designed to avoid cross-reactivity with paralogs; requires validation [22]
Genome Editing Tools (CRISPR/Cas9)	Functional validation through targeted gene knockout or modification	Efficiency varies across systems; requires optimization for each organism [17]

Conceptual Framework and Visualization

Conserved Kernels and Peripheral Rewiring in GRN Evolution

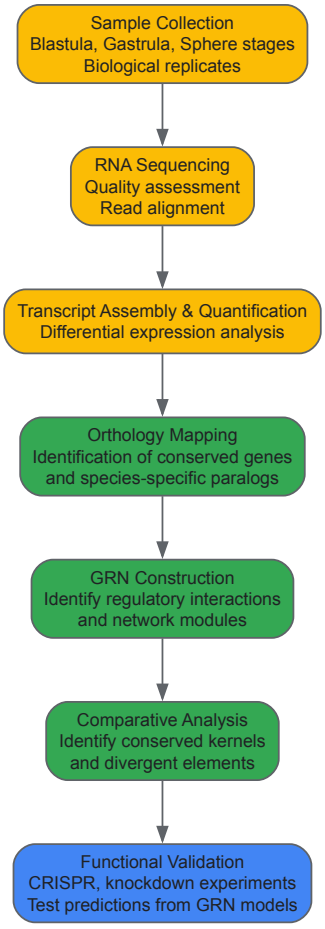
The Acropora study exemplifies how GRN evolution follows a pattern of conserved kernels with peripheral rewiring. The diagram below illustrates this conceptual framework:



This framework demonstrates how evolution maintains a **core regulatory kernel** while allowing **peripheral components** such as paralog usage and alternative splicing patterns to diverge through species-specific evolutionary trajectories [5] [3].

Experimental Workflow for Comparative GRN Analysis

The methodology for constructing and comparing GRNs across species involves an integrated computational and experimental pipeline:



This workflow highlights the sequence from data generation through computational analysis to functional validation, illustrating how modern evolutionary developmental biology integrates high-throughput data with functional experiments [17].

Discussion and Implications

Mechanisms of GRN Evolvability

The Acropora case study demonstrates that GRN evolvability—the capacity to generate phenotypic variation—stems from several structural and regulatory features. **Modular architecture** allows portions of the network to change without disrupting essential functions [19] [23]. This modularity enables developmental system drift, where species can arrive at similar morphological outcomes through different genetic paths [5] [3].

Paralog divergence represents a particularly powerful mechanism for evolutionary innovation. Gene duplicates provide genetic material that can acquire new functions without losing ancestral functions [21] [22]. The observed differences in paralog usage between *A. digitifera* and *A. tenuis* illustrate how the same initial genetic material can be shaped through different evolutionary trajectories—either toward specialization (neofunctionalization) or backup capacity (redundancy) [5] [21].

Relevance for Biomedical and Pharmaceutical Research

Understanding paralog divergence and functional compensation has direct relevance for drug development. The **target specificity** of pharmaceutical compounds can be compromised when drugs bind unintended paralogous proteins with similar structures and binding pockets [24]. Analyzing type-II amino acids with radical shifts in physicochemical properties between paralogs can help identify binding sites that maximize target specificity and reduce off-target effects [24].

Furthermore, the principles of GRN robustness revealed in this study—particularly the backup capacity provided by redundant paralogs in *A. tenuis*—inform our understanding of **genetic resilience** in biological systems. This has implications for understanding why some genetic perturbations cause disease while others are compensated, potentially guiding therapeutic strategies that exploit or modulate backup systems [21].

This case study of modular GRNs and divergent paralog usage in Acropora coral gastrulation illustrates how contemporary evolutionary developmental biology investigates the relationship between genotype and phenotype. By comparing GRN architecture across species, researchers can distinguish conserved regulatory kernels from diverged peripheral elements, revealing the modular structure that enables developmental stability alongside evolutionary innovation.

The findings demonstrate that **developmental system drift** is a widespread evolutionary phenomenon, with species employing distinct molecular programs to achieve conserved morphological outcomes. The differential utilization of paralogs and alternative splicing patterns between *A. digitifera* and *A. tenuis* highlights how genetic redundancy provides raw material for evolutionary innovation while maintaining developmental robustness.

These insights from coral development not only advance our fundamental understanding of evolutionary mechanisms but also provide conceptual frameworks relevant to biomedical research, particularly in understanding genetic redundancy, compensatory mechanisms, and the challenges of target specificity in drug development. As genomic technologies continue to advance, the GRN concept will remain an essential framework for unraveling the molecular basis of phenotypic diversity across the tree of life.

Advanced Methods for Reconstructing and Analyzing Evolutionary GRNs

Leveraging Single-Cell Multi-Omics for Cell-Type-Specific GRN Inference

Gene Regulatory Networks (GRNs) are complex circuits of molecular interactions that dictate cellular identity and function by controlling gene expression. A comprehensive understanding of GRNs is fundamental to explaining how cells perform diverse functions, respond to environmental changes, and how noncoding genetic variants cause disease [25]. Historically, GRN inference relied on bulk genomic data, which averaged signals across mixed cell populations, obscuring cell-type-specific regulatory mechanisms. The advent of single-cell multi-omics technologies now enables the simultaneous measurement of gene expression and chromatin accessibility within the same single cell, providing an unprecedented opportunity to decipher GRNs at the resolution of individual cell types within complex tissues [26].

In the context of evolutionary development research, understanding the dynamics of GRNs across different cell types and lineages is paramount. Cell type-specific gene expression patterns are the direct outputs of transcriptional GRNs that connect transcription factors and signaling proteins to their target genes [27]. These networks reconfigure during dynamic processes such as embryonic development, cellular differentiation, and disease progression. Single-cell technologies like single-cell RNA-sequencing (scRNA-seq) and single-cell Assay for Transposase-Accessible Chromatin using sequencing (scATAC-seq) can examine cell-type-specific gene regulation at unprecedented detail, offering new insights into the regulatory logic that governs evolutionary processes [27] [28].

Computational Methodologies for GRN Inference

The integration of scRNA-seq and scATAC-seq data presents both opportunities and challenges for GRN inference. Several sophisticated computational frameworks have been developed to leverage these multi-modal data, each employing distinct strategies to overcome the limitations of single-omics approaches.

Key Methodological Frameworks

- **LINGER (Lifelong neural network for gene regulation)**: This machine-learning method incorporates atlas-scale external bulk data across diverse cellular contexts and prior knowledge of transcription factor motifs as a manifold regularization. Its lifelong learning approach leverages knowledge from previous tasks (bulk data) to learn new tasks (single-cell data) more efficiently with limited data. LINGER uses a neural network model to fit the expression of target genes, taking as input TF expression and the accessibility of regulatory elements, with a second layer consisting of weighted sums of TFs and REs forming regulatory modules guided by TF-RE motif matching [25].
- **scMTNI (single-cell Multi-Task Network Inference)**: This framework integrates cell lineage structure with scRNA-seq and scATAC-seq measurements to jointly infer cell type-specific GRNs. It uses a probabilistic lineage tree prior that models GRN change from a start state (e.g., progenitor cell) to an end state (e.g., differentiated cell) as a series of individual edge-level probabilistic transitions. scMTNI can incorporate both linear and branching lineage structures, making it particularly valuable for developmental studies [27] [28].
- **PRISM-GRN**: A Bayesian model that incorporates known GRNs along with scRNA-seq and scATAC-seq data into a probabilistic framework. It employs a biologically interpretable architecture rooted in the established gene regulatory mechanism that gene expression is influenced by TF expression levels and gene chromatin accessibility through GRNs. PRISM-GRN decomposes observable data into biologically meaningful latent variables through a mechanism-informed generation process [29].
- **scSAGRN**: This method uses neighborhood information obtained by weighted nearest neighbor (WNN) analysis and combines it with spatial association to compute the relationship between gene expression and chromatin accessibility correlation. It links distal cis-regulatory elements to genes, identifies key TFs, and infers GRNs by specifying whether TFs activate or repress gene expression, addressing a limitation of many existing methods [30].

Table 1: Comparison of Major GRN Inference Methods

Method	Core Approach	Data Integration	Key Innovation	Reported Performance Gain
LINGER	Lifelong machine learning	scMultiome + external bulk data + TF motifs	Manifold regularization with external knowledge	4-7x relative increase in accuracy [25]
scMTNI	Multi-task graph learning	scRNA-seq + scATAC-seq + lineage structure	Lineage tree prior for network dynamics	Superior to single-task algorithms in simulation [27]
PRISM-GRN	Bayesian modeling	scRNA-seq + scATAC-seq + prior GRNs	Biologically interpretable latent variables	Higher precision in imbalanced scenarios [29]
scSAGRN	Spatial association	Paired scRNA-seq + scATAC-seq	WNN-based neighborhood information	Better TF-gene linkage prediction [30]

Experimental Design and Workflow

A standardized workflow for GRN inference from single-cell multi-omics data typically involves several critical stages. The following diagram illustrates the core logical process and data flow:

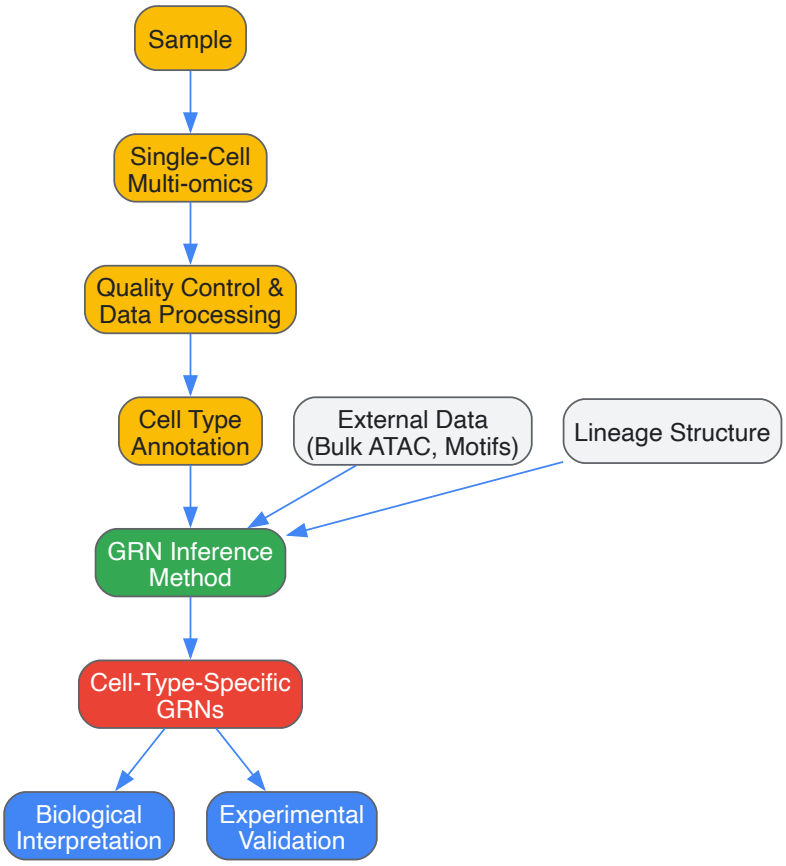


Diagram 1: Core Workflow for GRN Inference. The process begins with sample preparation and progresses through data generation, processing, and computational analysis to generate testable biological hypotheses.

Technical Protocols and Validation Frameworks

Benchmarking and Validation Standards

Rigorous validation of inferred GRNs remains challenging due to the lack of comprehensive ground truth datasets. However, several benchmarking approaches have emerged as standards in the field:

- **ChIP-seq validation**: For trans-regulatory relationships, putative targets of TFs from chromatin immunoprecipitation followed by sequencing (ChIP-seq) data are collected as ground truth. Performance is evaluated using area under the receiver operating characteristic curve (AUC) and area under the precision-recall curve (AUPR) ratio by sliding the trans-regulatory predictions [25].
- **eQTL consistency**: For cis-regulatory inference, the consistency of cis-regulatory coefficients with expression quantitative trait loci (eQTL) studies that link genotype variants to their target genes serves as validation. RE-TG pairs are divided into different distance groups, and AUC/AUPR metrics are calculated for each group [25].
- **Simulation frameworks**: For methods like scMTNI, in silico datasets are created with known ground truth networks. This involves creating a cell lineage, generating synthetic networks and corresponding single-cell expression datasets for each cell type on the lineage, then applying algorithms to recover the known networks [27].

Table 2: Experimental Validation Approaches for GRN Inference

Validation Method	Basis of Validation	Application	Key Metrics
ChIP-seq comparison	Direct TF-binding evidence	Trans-regulatory edges	AUC, AUPR ratio [25]
eQTL consistency	Genetic variant-gene links	Cis-regulatory edges	Distance-stratified AUC [25]
In silico benchmarking	Simulated networks with known truth	Method comparison	AUPR, F-score of top k edges [27]
TF recovery	Known TF-target relationships	Overall network quality	Precision, recall [30]

LINGER Protocol Implementation

The LINGER methodology implements a specific multi-step protocol for GRN inference:

- **Data Input**: Count matrices of gene expression and chromatin accessibility along with cell type annotation.
- **External Data Incorporation**: Pre-training using external bulk data from resources like the ENCODE project, which contains hundreds of samples covering diverse cellular contexts (BulkNN).
- **Single-Cata Refinement**: Application of elastic weight consolidation (EWC) loss, using bulk data parameters as a prior. The magnitude of parameter deviation is determined by the Fisher information.
- **Regulatory Strength Inference**: Using Shapley value to estimate the contribution of each feature for each gene after training the neural network model on single-cell data.
- **TF-RE Binding Assessment**: Generation of TF-RE binding strength by correlation of TF and RE parameters learned in the second layer.
- **Network Construction**: Building cell type-specific and cell-level GRNs based on the general GRN and cell type-specific profiles [25].

The following diagram illustrates LINGER's specific computational architecture:

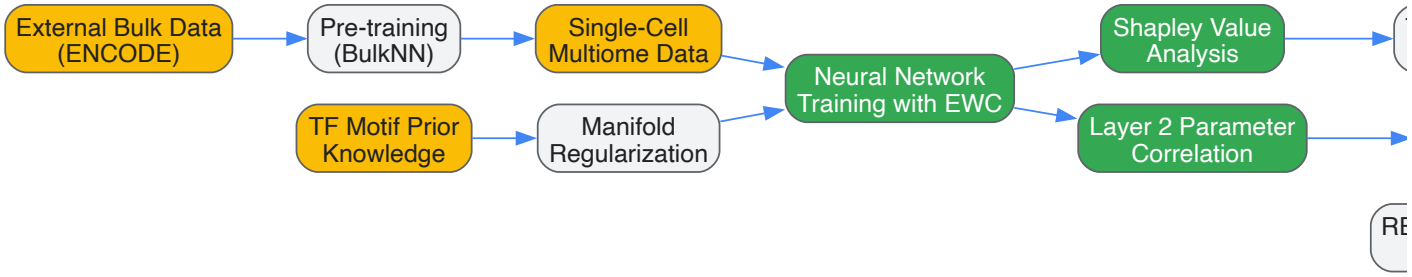


Diagram 2: LINGER Computational Architecture. The method integrates external bulk data and motif knowledge through lifelong learning and specialized regularization techniques.

Successful implementation of single-cell multi-omics GRN studies requires specific experimental and computational resources. The following table details essential components of the research toolkit:

Table 3: Research Reagent Solutions for Single-Cell Multi-Omics GRN Studies

Resource Category	Specific Examples	Function/Application	Technical Considerations
Single-Cell Platforms	10x Genomics Multiome, SNARE-seq, SHARE-seq	Simultaneous measurement of gene expression and chromatin accessibility	Choice affects cell throughput, data sparsity, and integration complexity [30] [26]
Reference Data	ENCODE cCREs, brainSCOPE, PsychENCODE	Provide prior knowledge for regulatory elements and gene regulation	Tissue-specific references improve accuracy; brainSCOPE contains >550K cell-type-specific regulatory elements [31]
Motif Databases	JASPAR, CIS-BP, HOCOMOCO	TF binding specificity models for linking REs to potential regulators	Quality and coverage affect prior network construction [27]
Validation Resources	ChIP-seq datasets, GTEx eQTLs, eQTLGen	Experimental validation of predicted regulatory interactions	Tissue/cell-type matching between prediction and validation is critical [25]
Computational Frameworks	LINGER, scMTNI, PRISM-GRN, scSAGRN	Core algorithms for GRN inference from multi-omics data	Method choice depends on data type, lineage information, and prior knowledge availability [25] [27] [29]

Applications in Evolutionary and Developmental Biology

Single-cell multi-omics approaches have revealed fundamental insights into developmental processes by mapping GRN dynamics across cell lineages. The scMTNI framework, for instance, has been applied to reconstruct GRN dynamics during cellular reprogramming and differentiation, identifying key regulators of fate transitions [27]. These approaches are particularly powerful when applied to model systems of evolutionary development.

In mouse preimplantation embryos, single-cell multi-omics has been used to analyze replication timing and gene expression, revealing that RT is established at the 1-cell stage prior to zygotic gene activation (ZGA). Surprisingly, the coordinated RT and gene expression control differs in early totipotent embryos compared to somatic cells, with late replicating regions correlating with higher gene expression and open chromatin - opposite to patterns observed in somatic cells [32].

The creation of atlas-scale resources like brainSCOPE, which comprises >2.8 million nuclei from 388 human prefrontal cortices, enables the identification of cell-type-specific regulatory elements and expression quantitative trait loci (eQTLs) across 28 brain cell types. This resource has been used to build cell-type regulatory and cell-to-cell communication networks that manifest cellular changes in aging and neuropsychiatric disorders [31].

The integration of single-cell multi-omics data for GRN inference represents a transformative advancement in our ability to understand gene regulation at cellular resolution. Methods like LINGER, scMTNI, PRISM-GRN, and scSAGRN demonstrate that leveraging external data, lineage information, and innovative machine learning approaches can significantly improve the accuracy of inferred networks. As these technologies mature, they will increasingly enable researchers to build dynamic models of regulatory network evolution across developmental lineages, providing crucial insights into the mechanisms driving evolutionary developmental processes.

The field continues to face challenges, including the computational complexity of methods, the need for comprehensive benchmarking datasets, and the integration of additional data modalities such as protein expression and spatial information. However, the rapid pace of methodological development suggests that single-cell multi-omics will soon become the standard approach for mapping gene regulatory networks across the tree of life, finally enabling a comprehensive understanding of how regulatory evolution shapes biological diversity.

Role-Based Gene Embedding (Gene2role) for Comparative Topological Analysis

In evolutionary developmental biology, the alteration of the functional organization of **gene regulatory networks (GRNs)** is recognized as a fundamental mechanism driving evolutionary change in animal morphology [33]. These networks, which control the development of the body plan, possess a hierarchical structure where specifically expressed genes encode transcription factors that direct developmental events through hardwired functional linkages [33]. The evolution of body plans occurs primarily through derived changes in GRN operation, which stem from structural alterations in the GRNs themselves compared to their ancestral states [33]. A major mechanism of this evolutionary change occurs through **cis-regulatory modifications** that determine regulatory gene expression, ultimately rewiring network connections [33] [34].

However, analyzing these evolutionary changes has presented significant challenges. Traditional comparative methods often focus on simple topological information or direct connections, overlooking deeper structural relationships in GRNs [35]. This limitation hinders our ability to fully capture the similarities and differences between complex GRNs across species or cellular states. As the field moves toward understanding GRN dynamics at a systems level, novel computational approaches are required to quantify and compare the intricate topological roles of genes within these networks. The emergence of **role-based embedding methods** represents a promising frontier for addressing these challenges in evolutionary developmental research.

Understanding Gene2role: Core Concepts and Methodology

Problem Definition and Biological Rationale

Gene2role addresses a critical gap in comparative GRN analysis by moving beyond simple topological metrics to capture **multi-hop topological information** from genes within signed GRNs [35]. While existing methods often focus solely on direct topological information of genes, they overlook deeper structural connections (e.g., 1-hop and 2-hop neighbors), resulting in a shallow understanding of GRN complexity [35]. This limitation is particularly problematic in evolutionary studies where network rewiring may preserve functional roles despite changes in immediate connectivity.

The method specifically operates on **signed GRNs**, represented as $G = (V, E+, E-)$, where $V = \{v_1, v_2, ..., v_n\}$ denotes the set of genes, $E+$ represents positive (activating) interactions, and $E-$ represents negative (inhibitory) interactions [35]. This signing is crucial for biological accuracy, as the nature of regulatory relationships fundamentally impacts gene function and network dynamics.

Mathematical Foundation

Signed-Degree Representation

Gene2role introduces the concept of the **signed-degree vector** to initially capture topological nuances of each gene [35]. For a gene in the network, this 2-dimensional vector is defined as:

$$\mathbf{d} = [d+, d-]$$

where $d+$ and $d-$ represent the positive and negative degrees, respectively [35]. This representation maps each gene from the signed GRNs to a point on a plane, providing a foundational topological signature.

Topological Similarity Calculation

To quantify topological similarity between genes, Gene2role employs the **Exponential Biased Euclidean Distance (EBED)** function, which evaluates the zero-hop distance (D_0) between the signed-degree vectors of two genes [35]. The formula is defined as:

$$D_O(u,v) = EBED(d_u, d_v) = \exp(\sqrt{(\log(d_{u+} + 1)/(d_{v+} + 1))^2 + (\log(d_{u-} + 1)/(d_{v-} + 1))^2})$$

The EBED function addresses the scale-free nature of GRNs, where gene degrees often follow a power-law distribution [35]. The logarithmic transformation mitigates the effects of this distribution, followed by Euclidean distance calculation and exponential transformation to preserve original distance proportionality.

Multi-Hop Neighborhood Analysis

Gene2role extends beyond direct connections by defining $R_k(u)$ as the sorted sequence of degrees for genes that are k hops away from gene u [35]. This multi-hop perspective enables the method to capture the broader topological context that influences a gene's regulatory role, essential for understanding conserved functional modules in evolutionary comparisons.

Algorithmic Workflow and Implementation

The following diagram illustrates the complete Gene2role analytical workflow, from network preparation through to downstream evolutionary analysis:

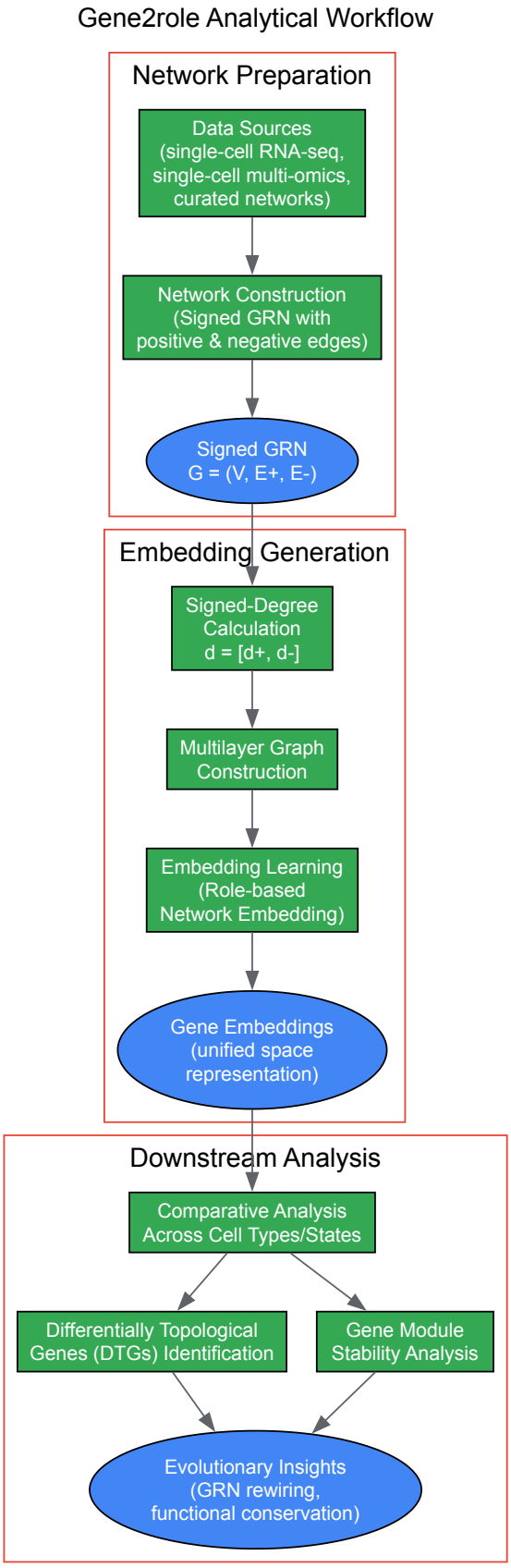


Figure 1: Gene2role implements a three-stage workflow for comparative GRN analysis, transforming raw network data into evolutionary insights through role-based embedding.

Experimental Protocols and Validation Frameworks

Gene2role validation encompasses multiple network types to ensure methodological robustness [35]:

- **Simulated Networks**: A simple simulated network comprising 31 genes mimics the scale-free characteristics of GRNs, providing a controlled validation environment [35].
- **Curated Biological Networks**: Four manually curated networks—hematopoietic stem cell (HSC), mammalian cortical area development (mCAD), ventral spinal cord (VSC), and gonadal sex determination (GSD)—containing between 5 and 19 genes were downloaded from BEELINE [35].
- **Single-cell RNA-seq Networks**: Utilizing count matrices and cell type annotation data from published studies, including:
  - Human glioblastoma at glioblastoma stem-like cells (0-h) and serum-induced differentiated (12-h) stages [35]
  - Human bone marrow mononuclear cells (BMMC) and human peripheral blood mononuclear cells (PBMC) [35]
  - Construction of cell type-specific GRNs using EEISP and Spearman correlation on 2000 highly variable genes [35]
- **Single-cell Multi-omics Networks**: Networks obtained from CellOracle, integrating scRNA-seq and sci-ATAC-seq data from differentiating mouse myeloid progenitors [35]. These networks encompass 24 cell states during myeloid differentiation, with only connections exhibiting p-value < 0.01 maintained, and the top 2000 edges selected based on highest absolute coefficient values [35].

Embedding Generation Protocol

The core embedding process follows these methodological steps:

- **Signed-Degree Calculation**: For each gene in the signed GRN, compute the signed-degree vector d = [d+, d-] [35].
- **Topological Similarity Computation**: Calculate pairwise distances between genes using the EBED function to establish initial similarity metrics [35].
- **Multilayer Graph Construction**: Build a multi-layer weighted graph that reflects structural similarities among nodes at various depths, following the struc2vec framework [35].
- **Embedding Learning**: Apply role-based network embedding techniques to project genes from potentially separate networks into closely positioned spaces for comparative analysis [35].

Validation Metrics and Performance Assessment

Gene2role employs multiple evaluation strategies to assess embedding quality and biological relevance:

- **Topological Capture Validation**: Demonstrate the method's ability to capture intricate topological nuances of genes using GRNs inferred from distinct data sources [35].
- **Biological Significance Testing**: Apply the embeddings to identify genes with significant topological changes across cell types or states, providing perspectives beyond traditional differential gene expression analyses [35].
- **Module Stability Assessment**: Quantify the stability of gene modules between cellular states by measuring changes in gene embeddings within these modules [35].

Quantitative Results and Comparative Analysis

Methodological Performance Benchmarks

Table 1: Performance comparison of Gene2role against traditional topological analysis methods

Analysis Type	Traditional Methods	Gene2role Approach	Biological Insights Gained
Similarity Measurement	Direct topological information (degree centrality)	Multi-hop topological neighborhoods	Captures functional roles beyond immediate connectivity
Cross-Network Comparison	Limited by network-specific metrics	Unified embedding space	Enables direct comparison of genes across different cellular states
Evolutionary Analysis	Focus on sequence conservation	Topological role conservation	Identifies functional conservation despite sequence divergence
Module Detection	Based on immediate connectivity	Role-based community detection	Reveals functionally coherent modules with stable topological signatures

Application to Evolutionary Developmental Questions

Table 2: Gene2role applications in evolutionary developmental biology research

Research Question	Experimental Approach	Gene2role Analysis	Evolutionary Implications
GRN Rewiring Events	Comparative functional genetics across species	Quantify topological role changes in orthologous genes	Identify functional compensation events (e.g., amphioxus Gdf1/3-like takeover [34])
Cis-Regulatory Evolution	Enhancer validation through transgenic assays	Track topological impact of regulatory changes	Connect sequence changes to network-level functional alterations [33]
Developmental System Drift	Expression pattern comparison across lineages	Quantify topological role conservation despite expression divergence	Reveal hidden functional conservation maintaining phenotype stability
Gene Duplication Fate	Functional characterization of duplicates	Compare topological roles of paralogs	Predict subfunctionalization vs. neofunctionalization outcomes

The Scientist's Toolkit: Research Reagent Solutions

Table 3: Essential research reagents and computational tools for GRN evolutionary analysis

Reagent/Tool	Type	Function in Analysis	Implementation Notes
Single-cell Multi-omics Data	Biological Data	GRN inference across cell states	Integrates scRNA-seq and scATAC-seq (e.g., CellOracle [35])
Curated Network Databases	Reference Data	Benchmarking and validation	BEELINE benchmark networks [35]
Struc2Vec Framework	Algorithm	Role-based network embedding	Adapted for signed GRNs in Gene2role [35]
SignedS2V	Algorithm	Signed network embedding	Handles positive/negative edges in GRNs [35]
EEISP	Software	GRN construction from scRNA-seq	Identifies co-dependency and mutual exclusivity [35]
CellOracle	Software	GRN inference from multi-omics data	Integrates TF binding motifs and co-expression [35]

Visualization Framework for GRN Topological Relationships

Signed GRN Representation

The following diagram illustrates the concept of signed-degree calculation and multi-hop neighborhood analysis in a signed gene regulatory network:

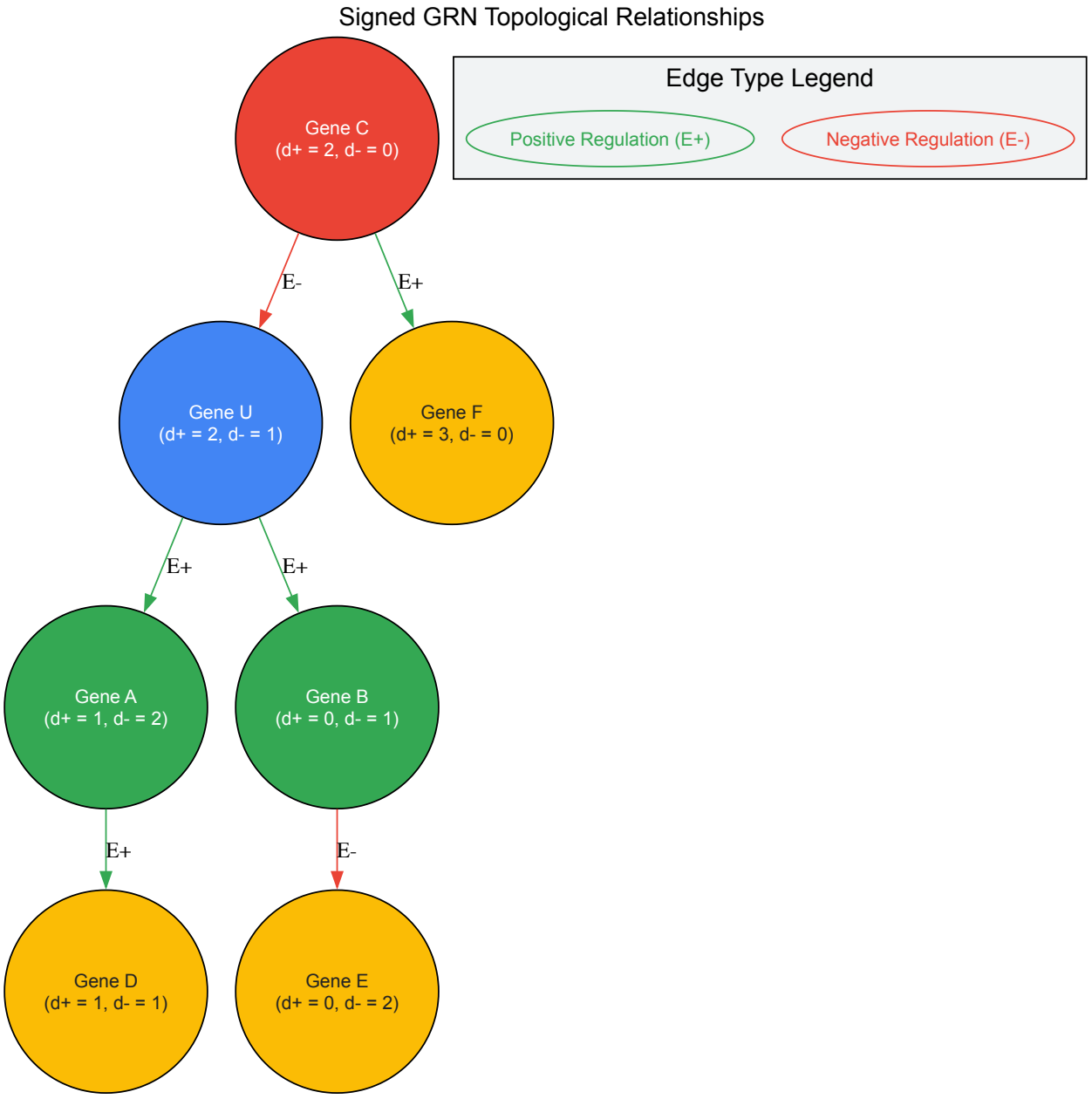


Figure 2: Signed GRN representation showing positive (green) and negative (red) regulatory relationships, with signed-degree calculations for each gene and multi-hop neighborhood connections.

Multilayer Graph Construction for Role-Based Embedding

The following diagram visualizes the multilayer graph construction process that enables role-based similarity measurement in Gene2role:

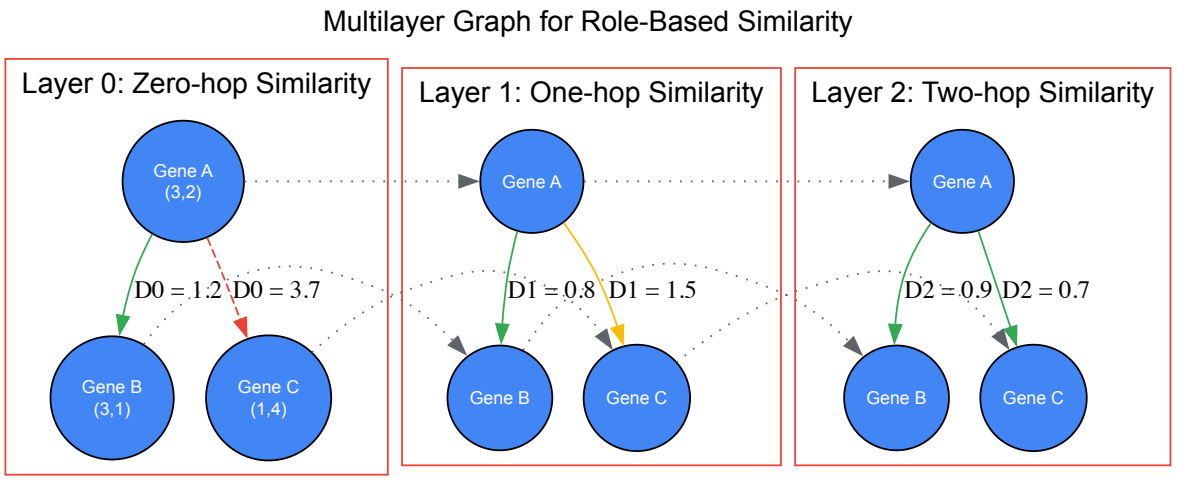


Figure 3: Multilayer graph construction in Gene2role, where each layer captures topological similarities at different neighborhood depths (zero-hop, one-hop, two-hop), enabling comprehensive role-based embedding.

Discussion: Implications for Evolutionary Developmental Biology

The Gene2role framework provides a powerful methodological advancement for evolutionary developmental biology by enabling **quantitative comparison of GRN topological structures** across species and developmental stages. This approach addresses a fundamental challenge in evo-devo: distinguishing between mere sequence conservation and functional conservation of regulatory roles within networks.

The method's capacity to identify **differentially topological genes (DTGs)** offers a novel perspective beyond traditional differential expression analysis, potentially revealing genes that have undergone significant functional role changes during evolution [35]. Similarly, the quantification of **gene module stability** through embedding space analysis provides a systematic approach to studying developmental system drift—the phenomenon where developmental processes diverge while producing conserved morphological outcomes [35] [34].

Case studies like the rewiring of the Nodal signaling pathway in amphioxus, where Gdf1/3-like functionally replaced Gdf1/3 in body axis formation possibly through enhancer hijacking, demonstrate the type of evolutionary events that Gene2role can help identify and characterize systematically [34]. By providing a framework to quantify such topological role changes, Gene2role enables researchers to move beyond qualitative descriptions of network evolution toward predictive models of GRN evolvability.

The integration of role-based embedding with single-cell multi-omics data creates opportunities to study GRN evolution at unprecedented resolution, potentially revealing conserved topological motifs that underlie fundamental developmental processes across diverse lineages. This methodological advancement represents a significant step toward deciphering the evolutionary patterns of gene regulatory networks that control embryonic development and the mechanisms underlying GRN evolution.

AI and Large Language Models in Mining Literature and Predicting Regulatory Interactions

Gene regulatory networks (GRNs) represent the causal relationships between transcription factors (TFs) and their target genes, forming the fundamental architecture that controls developmental processes, cellular identity, and disease mechanisms [36] [15]. Understanding these networks is crucial for unraveling the complexities of evolutionary development and identifying therapeutic targets. However, the exponential growth of biological literature has made manual curation of regulatory interactions increasingly impractical [37].

The emergence of large language models (LLMs) trained on vast text corpora offers a transformative approach to this challenge. These models contain rich biological information that can be mined as a comprehensive knowledge graph, enabling researchers to extract gene-gene interactions and pathway knowledge at unprecedented scale and speed [37]. This technical guide examines the current state of LLM applications in GRN inference, providing researchers with methodological frameworks and practical tools for leveraging artificial intelligence in evolutionary developmental biology research.

LLM Performance Benchmarking in Gene Relation Mining

Quantitative Evaluation of LLM Capabilities

Recent comprehensive evaluations have assessed the capacity of various LLMs to mine biological relationships from literature. These studies typically focus on predicting gene regulatory relations (including activation, inhibition, and phosphorylation) and reconstructing known pathway components from databases like the Kyoto Encyclopedia of Genes and Genomes (KEGG) [37].

Table 1: Performance Comparison of Large Language Models in Gene Relation Mining

Model Type	Model Name	Gene Relation F1 Score	KEGG Pathway Jaccard Similarity	Key Strengths
API-based Models	GPT-4	0.4448	0.2778	Superior knowledge integration, contextual understanding
	Claude-Pro	0.4386	0.2657	Strong analytical capabilities, precise extraction
Open-source Models	Falcon-180b	0.2787	0.2237	Best-performing open-source alternative
	Llama2-7b	0.1923	0.2207	Efficient despite smaller size
	Other Open-source	<0.19	<0.22	Variable performance across tasks

The evaluation results indicate a significant performance disparity between API-based and open-source models, with GPT-4 and Claude-Pro demonstrating superior capability in both gene regulatory relation prediction and pathway mapping tasks [37]. This performance gap highlights the importance of model selection for research applications requiring high precision in relationship extraction.

Evaluation Methodologies

The benchmark assessments typically employ structured evaluation frameworks that compare LLM-extracted relationships against gold-standard databases. Key evaluation metrics include:

- **F1 Scores**: Balance precision and recall in identifying specific regulatory relationships (activation, inhibition, phosphorylation)
- **Jaccard Similarity Index**: Measures overlap between predicted pathway components and established KEGG pathway annotations
- **Task-Based Evaluation**: Addresses the absence of ground truth causal graphs by using LLM-suggested GRNs to guide synthetic data generation, then comparing statistical similarity and biological plausibility against original datasets [36]

LLM Applications in Gene Regulatory Network Inference

Methodological Frameworks

The LLM4GRN framework demonstrates how large language models can be leveraged for GRN discovery through multiple approaches:

1. Direct GRN Generation: LLMs synthesize complete causal graphs between transcription factors and target genes using their embedded biological knowledge [36].

2. Prior Knowledge Integration: LLMs provide curated lists of potential transcription factors and regulatory relationships that enhance traditional statistical causal discovery algorithms [36].

3. Hybrid Statistical-LLM Approaches: Combining LLM-generated priors with causal discovery algorithms like PC, FGES, and DAGMA has shown improved performance over either method alone [36].

These methodologies address critical limitations in conventional GRN inference, which often relies on curated databases (TRANSFAC, RegNetwork, ENCODE) that lack essential contextual information about specific cell types or conditions [36].

Experimental Protocols for LLM-Enhanced GRN Discovery

Protocol 1: Direct GRN Generation Using LLMs

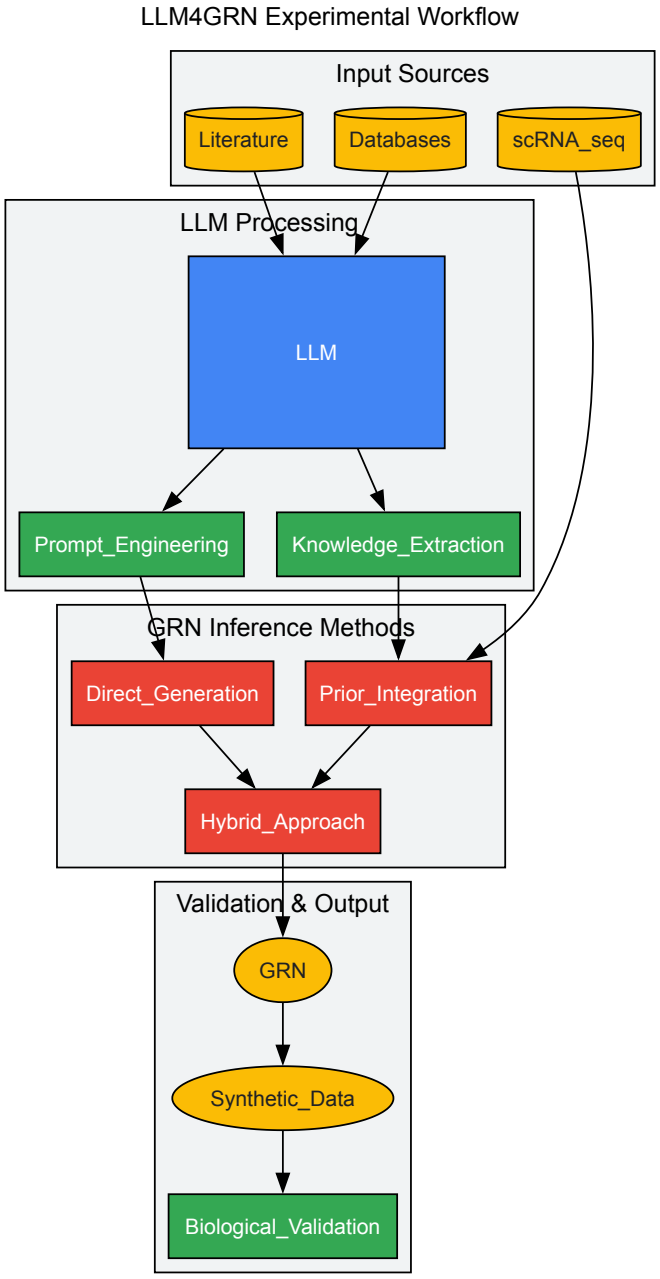
- **Input Preparation**: Compile a list of candidate genes and transcription factors relevant to your developmental system
- **Prompt Engineering**: Design structured prompts requesting causal regulatory relationships between transcription factors and target genes
- **Graph Construction**: Parse LLM responses to extract directed edges representing activation/inhibition relationships
- **Validation**: Assess network plausibility through synthetic data generation and comparison with experimental data [36]

Protocol 2: LLM-Augmented Statistical Causal Discovery

- **Prior Knowledge Extraction**: Use LLMs to generate ranked lists of potential transcription factors for specific cell types or developmental contexts
- **Data Integration**: Combine LLM-generated priors with single-cell RNA sequencing data
- **Network Inference**: Apply statistical causal discovery algorithms (e.g., NOTEARS, DCDI) constrained by LLM-provided priors
- **Downstream Validation**: Generate synthetic data from inferred networks and assess biological plausibility through statistical and functional metrics [36]

Protocol 3: Causal Synthetic Data Generation for Validation

- **Network Embedding**: Incorporate the inferred GRN as a causal acyclic graph guiding data generation
- **Generator Training**: Train causal generative adversarial networks (GANs) using the graph structure and original scRNA-seq data
- **Data Synthesis**: Generate synthetic single-cell expression data preserving the causal relationships of the input GRN
- **Evaluation**: Compare synthetic data with original dataset using statistical metrics and biological knowledge [36]



GRN Analysis in Evolutionary Developmental Biology

Case Study: Developmental System Drift in Acropora Corals

Research on phylogenetically distant Acropora species (A. digitifera and A. tenuis) provides compelling insights into GRN evolution during early development. Despite diverging approximately 50 million years ago, these species maintain conserved gastrulation morphology while exhibiting significant GRN diversification—a phenomenon known as "developmental system drift" [3].

Key Findings from Comparative Transcriptomics:

- **Conserved Regulatory Kernel**: Identification of 370 differentially expressed genes consistently up-regulated during gastrulation in both species, functioning in axis specification, endoderm formation, and neurogenesis
- **Peripheral Network Rewiring**: Species-specific differences in paralog usage and alternative splicing patterns modifying network connectivity

- **Divergent Evolutionary Strategies**: A. digitifera exhibits greater paralog divergence (neofunctionalization), while A. tenuis shows more redundant expression (regulatory robustness) [3]

Experimental Framework for Evolutionary GRN Analysis

Protocol 4: Comparative GRN Analysis Across Species

- **Sample Collection**: Obtain embryos at equivalent developmental stages from multiple species (e.g., blastula, gastrula, sphere stages)
- **Transcriptomic Profiling**: Perform RNA sequencing with sufficient biological replicates for statistical power
- **Orthology Mapping**: Identify orthologous genes across species using genome annotations and phylogenetic analysis
- **Network Inference**: Reconstruct GRNs for each species using combined statistical and LLM-based approaches
- **Conservation Assessment**: Identify conserved network kernels versus diverged peripheral connections
- **Functional Validation**: Test conserved regulatory relationships through experimental perturbation [3]

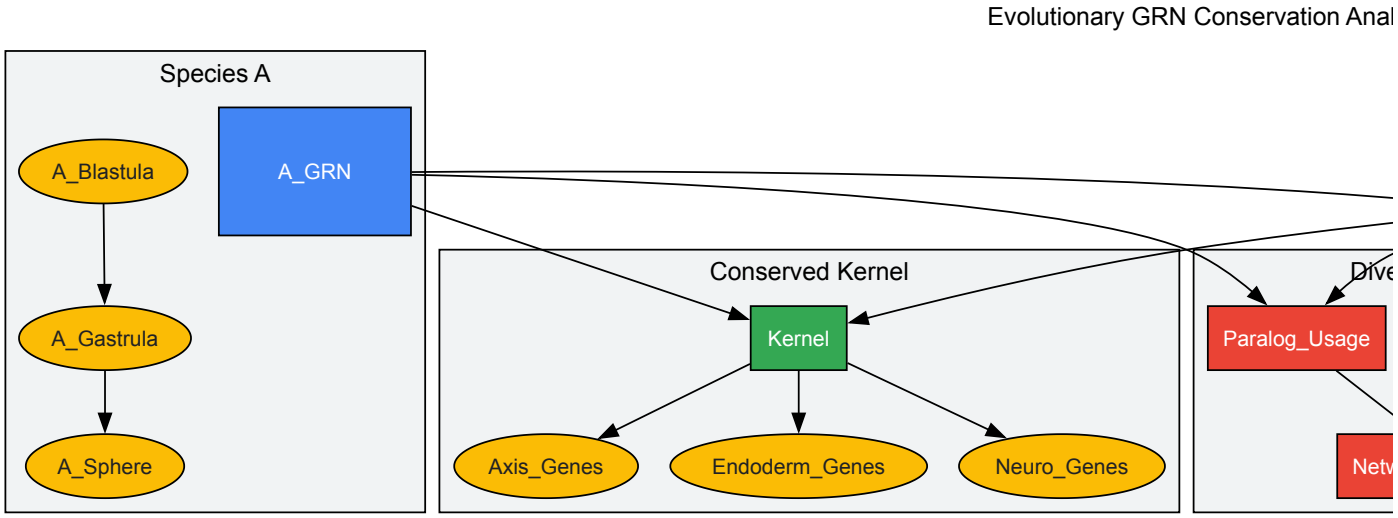


Table 2: Key Research Reagent Solutions for LLM-Guided GRN Analysis

Category	Specific Resource	Function/Application	Key Features
LLM Platforms	GPT-4 API	Gene relation extraction, pathway mapping	Highest performance in biological relation mining [37]
	Claude-Pro	Regulatory relationship inference	Strong analytical capabilities for complex biological texts [37]
	Falcon-180B	Open-source alternative for GRN inference	Best-performing open-source model for gene relations [37]
Data Resources	Single-cell RNA-seq Data	GRN inference input, expression validation	Enables cell-type specific network reconstruction [36] [3]
	KEGG Pathway Database	Validation benchmark for pathway predictions	Gold-standard pathway knowledge [37]
	TRANSFAC/RegNetwork	Prior knowledge for statistical methods	Curated transcription factor databases [36]
Analysis Tools	Causal GAN Frameworks	Synthetic data generation from GRNs	Validates biological plausibility of inferred networks [36]
	BioTapestry	GRN visualization and analysis	Specialized for developmental gene networks [38]
	Statistical Causal Discovery (NOTEARS, DCDI)	Network inference from expression data	Infers causal relationships from observational data [36]
Experimental Validation	CRISPR/Cas9 Systems	Functional validation of regulatory relationships	Tests necessity of predicted interactions [3]

Category	Specific Resource	Function/Application	Key Features
	Single-cell ATAC-seq	cis-regulatory element mapping	Identifies potential regulatory regions [15]

Integrated Workflow for LLM-Enhanced GRN Research

The most effective approach to GRN analysis in evolutionary developmental biology combines LLM-based knowledge extraction with statistical causal discovery and experimental validation.

Protocol 5: Integrated LLM-Statistical-Experimental Pipeline

- **Literature Mining**: Deploy best-performing LLMs (GPT-4, Claude-Pro) to extract potential regulatory relationships from entire corpora of biological literature
- **Prior Knowledge Curation**: Filter and validate LLM-generated relationships against established databases and experimental evidence
- **Multi-Omics Data Integration**: Combine scRNA-seq, scATAC-seq, and epigenetic data to provide comprehensive input for network inference
- **Hybrid GRN Inference**: Implement statistical causal discovery algorithms constrained by LLM-generated priors
- **Evolutionary Comparison**: Apply conserved kernel identification across multiple species to distinguish core versus lineage-specific network components
- **Synthetic Validation**: Generate causal synthetic data to assess network plausibility before resource-intensive experimental validation
- **Functional Testing**: Use CRISPR-based perturbations to validate critical predicted interactions in model systems [36] [37] [3]

This integrated framework leverages the respective strengths of each methodology: LLMs for comprehensive knowledge synthesis, statistical methods for robust inference from empirical data, and experimental approaches for biological validation. The resulting GRNs provide unprecedented insights into the evolutionary principles governing developmental processes, with applications ranging from basic evolutionary biology to pharmaceutical development targeting regulatory network dysfunctions.

Gene regulatory networks (GRNs) form the computational core of biological development, translating genotype into phenotype through complex, dynamic interactions. In evolutionary developmental biology (evo-devo), understanding how these networks evolve to produce diverse morphological outcomes is a central pursuit. Dynamic modeling provides the essential mathematical framework to formalize this understanding, moving from static network diagrams to predictive, quantitative models of temporal behavior. The journey from simple Boolean networks to continuous ordinary differential equations (ODEs) and ultimately to stochastic formulations using Fokker-Planck equations represents increasing sophistication in capturing biological reality. Boolean models offer intuitive, discrete representations of gene activity states, while ODEs provide continuous quantification of concentration changes. Fokker-Planck equations extend this framework to account for intrinsic stochasticity, enabling researchers to model the probabilistic evolution of gene expression states and formalize concepts like Waddington's epigenetic landscape [39] [40]. This technical guide explores these interconnected modeling paradigms within the context of evo-devo research, providing researchers with the mathematical foundations and practical methodologies to implement these approaches for studying the evolutionary dynamics of developmental systems.

Mathematical Foundations of Gene Regulatory Network Modeling

Boolean Networks: Discrete Dynamics of Gene Regulation

Boolean networks (BNs) represent the simplest dynamic modeling approach for GRNs, conceptualizing genes as binary switches that are either ON (expressed, 1) or OFF (not expressed, 0). Formally, a Boolean network is defined on a set of n binary-valued nodes (genes) $V = \{x_1, x_2, ..., x_n\}$ where $x_i \in \{0,1\}$. Each node x_i has k_i parent nodes (regulators) selected from V, and its value at time t+1 is determined by its parent nodes at time t through a Boolean function f_i [41]:

$$x_i(t+1) = f_i(x_{i1}(t), x_{i2}(t), ..., x_{ik_i}(t))$$

The network function $f = (f_1, f_2, ..., f_n)$ governs the state transitions of the system, where the network state at time t is $x(t) = (x_1(t), x_2(t), ..., x_n(t))$, and the state transition is written as $x(t+1) = f(x(t))$ [41]. The deterministic nature of Boolean networks makes them particularly valuable for identifying fundamental logical principles of developmental regulation, where specific gene combinations necessarily lead to particular developmental outcomes.

Table 1: Key Concepts in Boolean Network Modeling

Concept	Mathematical Representation	Biological Interpretation
State	$x(t) = (x_1(t), x_2(t), ..., x_n(t))$	Gene expression profile at time t
State Space	2^n possible states	All possible gene expression profiles
Regulatory Function	$f_i: \{0,1\}^{k_i} \rightarrow \{0,1\}$	Logic rule determining gene i's expression

Concept	Mathematical Representation	Biological Interpretation
Attractor	$x(t)$ such that $x(t+1) = x(t)$	Stable phenotypic state (e.g., cell type)
Basin of Attraction	Set of states leading to an attractor	Developmental pathway potential

To address biological stochasticity, probabilistic Boolean networks (PBNs) extend this framework by incorporating multiple possible regulatory functions for each gene with associated selection probabilities. A PBN is defined as $G(V, F, c, p)$, where $F = \{f^1, f^2, ..., f^r\}$ is a set of r network functions, $c = \{c_1, c_2, ..., c_r\}$ are network selection probabilities, and p is the random gene perturbation probability [41]. This formulation enables modeling of the inherent noise and contextual variability in real biological systems.

Continuous ODE Models: Quantifying Gene Expression Dynamics

Continuous models using ordinary differential equations (ODEs) provide a more quantitative framework for modeling GRNs by describing the temporal evolution of molecular concentrations. The core mathematical structure for a GRN with n genes can be represented as:

$$dX_i/dt = F_i(X_1, X_2, ..., X_n) - \gamma_i X_i$$

where X_i represents the concentration of gene product i, F_i is a function summarizing the regulatory inputs controlling the production of i, and γ_i is the degradation rate constant [39] [42]. The regulatory function F_i often takes the form of Hill kinetics to capture cooperative binding effects:

$$F_i(X) = \alpha_i \prod_{j \in \text{Activators}} (X_j^{n_{ij}})/(K_{ij}^{n_{ij}} + X_j^{n_{ij}}) \prod_{k \in \text{Repressors}} 1/(K_{ik}^{n_{ik}} + X_{ik}^{n_{ik}})$$

where α_i is the maximum production rate, K_{ij} represents the activation/repression threshold, and n_{ij} is the Hill coefficient quantifying cooperativity [42]. This formulation enables precise modeling of dose-response relationships in gene regulation, which is essential for understanding how quantitative variation in gene expression leads to phenotypic differences in evolutionary contexts.

Table 2: Common Rate Laws in ODE Models of GRNs

Rate Law	Equation	Application Context
Mass Action	$dX/dt = k\prod[\text{Reactants}]$	Elementary biochemical reactions
Michaelis-Menten	$v = V_{max}[S]/(K_m + [S])$	Enzyme-catalyzed reactions
Hill Equation	$v = V_{max}[X]^n/(K^n + [X]^n)$	Cooperative binding (transcription)
Goldbeter-Koshland	$v = V_1(1 - Y) - V_2Y$	Ultrasensitive switches

Fokker-Planck Equations: Capturing Stochasticity in Gene Regulation

The Fokker-Planck equation (FPE) provides a complete statistical description of stochastic dynamical systems by modeling the temporal evolution of the probability density function (PDF) of system states. For a GRN described by the stochastic differential equation (SDE):

$$dX_t = v(X_t)dt + \sqrt{\varepsilon}\, dB_t$$

where $v(X_t)$ is the drift term representing deterministic dynamics, ε is the noise intensity, and B_t is Brownian motion, the corresponding Fokker-Planck equation for the probability density $p(x,t)$ is [43] [40]:

$$\partial p(x,t)/\partial t = -\nabla\cdot[v(x)p(x,t)] + (\varepsilon/2)\Delta p(x,t)$$

This partial differential equation describes how the probability distribution of possible gene expression states evolves over time [39] [40]. The stationary solution ($\partial p/\partial t = 0$) provides access to the epigenetic landscape concept through the relationship $F(x) = -\log p_{ss}(x)$, where $p_{ss}(x)$ is the stationary distribution and $F(x)$ represents the potential function that characterizes the landscape [39]. This formulation enables researchers to quantify the relative stability of different phenotypic states and the probabilities of transitions between them—key considerations in evolutionary developmental biology.

Methodological Implementation

Protocol 1: Constructing a Boolean Network Model from Expression Data

Objective: To create a discrete dynamic model of a GRN from qualitative gene expression patterns.

Materials and Reagents:

- Binary gene expression data (e.g., from RNA in situ hybridization)

- Network inference algorithms (e.g., mutual information, correlation)
- Software: BioTapestry, CellCollective, BoolNet

Procedure:

- **Define Network Components:** Identify key genes and their regulatory interactions based on literature and experimental data. For the Arabidopsis thaliana flower morphogenesis network, this includes genes such as EMF1, TFL1, LFY, AP1, and AG [39].
- **Construct Interaction Matrix:** Create a weight matrix $W = [w_{ij}]$ where w_{ij} represents the influence of gene j on gene i (positive for activation, negative for repression). For example, the Arabidopsis network was defined with integer weights [39]:
- **Define Update Rules:** Implement Boolean functions for each gene. For instance:
  - $x_i(t+1) = H(\sum_j w_{ij} x_j(t) - \theta_i)$
  - where H is the Heaviside step function: $H(x) = 1$ if $x > 0$, else 0 [39]
- **Identify Attractors:** Enumerate all 2^n possible states and compute their trajectories to identify stable steady states (attractors) and their basins of attraction.
- **Validate with Biological Phenotypes:** Compare identified attractors with known phenotypic states. The Arabidopsis model successfully identified four stationary states corresponding to phenotypic stages in flower development [39].

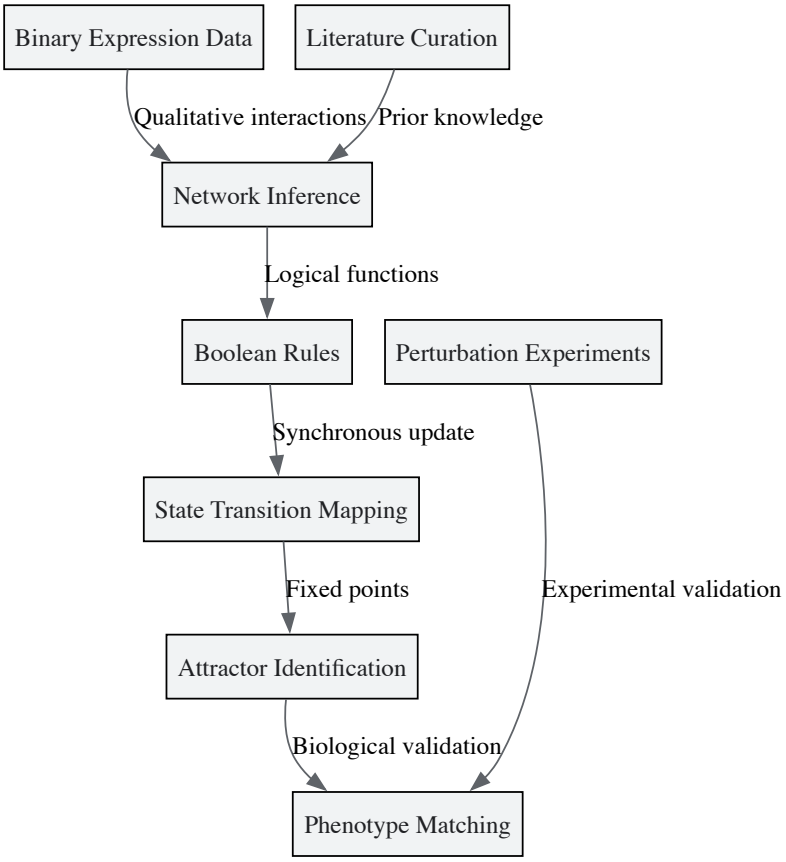


Figure 1: Workflow for Boolean Network Construction

Protocol 2: Developing Continuous ODE Models from Boolean Networks

Objective: To convert a discrete Boolean model into a continuous ODE framework for quantitative analysis.

Materials and Reagents:

- Established Boolean network model
- Quantitative expression data (e.g., RNA-seq, protein concentrations)
- Parameter estimation algorithms
- Software: COPASI, MATLAB, Tellurium, SciPy

Procedure:

- **Define Biochemical Reactions:** Translate logical interactions into explicit biochemical events. For each gene, specify:
  - Transcription: $Ga + n_a bPb \rightarrow Kab + Ga^*$ (activation)
  - Translation: $mRNAa \rightarrow \beta a\ mRNAa + Pa$
  - Degradation: $Pa \rightarrow \delta a\ \varnothing$ [39]

- **Formulate Rate Equations:** Derive ODEs from reaction schemes. For mRNA and protein concentrations:

  - $dm_a/dt = \alpha_a b(k_a bp_a b^a b)/(1 + k_a bp_a b^a b) - \gamma_a m_a$
  - $dp_a/dt = \beta_a m_a - \delta_a p_a$ [39]

- **Parameter Estimation:** Use optimization algorithms to fit model parameters to quantitative expression data:

  - Apply global optimization methods (genetic algorithms) to avoid local minima
  - Use least squares minimization for deterministic models
  - Employ maximum likelihood estimation for stochastic models [42]

- **Model Validation:** Compare simulation results with independent datasets not used for parameterization:

  - Perform cross-validation using temporal expression profiles
  - Assess model performance using statistical measures (R-squared, AIC)
  - Conduct sensitivity analysis to identify critical parameters [42]

- **Bifurcation Analysis:** Explore parameter space to identify regions of multistability corresponding to different cell fates.

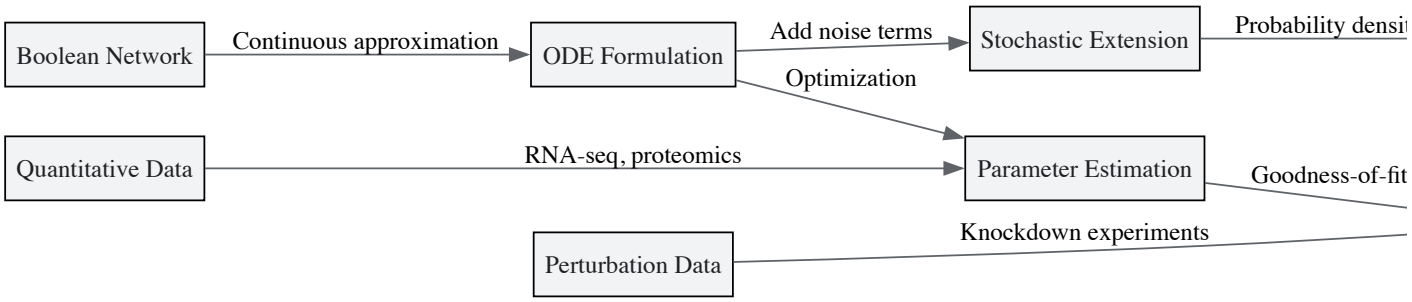


Figure 2: From Boolean to ODE Modeling Pipeline

Protocol 3: Solving Fokker-Planck Equations for Epigenetic Landscape Analysis

Objective: To compute the stationary probability distribution of gene expression states and reconstruct the epigenetic landscape.

Materials and Reagents:

- Validated ODE model of GRN
- High-performance computing resources
- Numerical PDE solvers
- Software: DSN-PINNs, custom MATLAB/Python implementations

Procedure:

- **Formulate Stochastic Differential Equations:** Start with the deterministic ODE model and add noise terms:
  - $dX_t = v(X_t)dt + \sqrt{\varepsilon}\, dB_t$
  - where $v(X_t)$ represents the deterministic drift from ODEs [43]

- **Derive Fokker-Planck Equation:** Convert the SDE to its corresponding FPE:

  - $\partial p(x,t)/\partial t = -\nabla\cdot[v(x)p(x,t)] + (\varepsilon/2)\Delta p(x,t)$ [39]

- **Implement Numerical Solution Method:** Apply the Distribution Self-adaptive Normalized Physics-Informed Neural Network (DSN-PINNs) approach:

  - **Pretraining Phase:** Use normalization-enhanced PINN to establish solution's global structure
  - **Adaptive Resampling:** Dynamically reallocate training points via weighted kernel density estimation
  - **Convergence Check:** Monitor loss function until stabilization [43]

- **Compute Stationary Distribution:** Solve for $p_{ss}(x)$ satisfying $\partial p/\partial t = 0$:

- $0 = -\nabla\cdot[v(x)p_{ss}(x)] + (\varepsilon/2)\Delta p_{ss}(x)$ [39]

- **Reconstruct Epigenetic Landscape:** Calculate the potential function:
  - $F(x) = -\log p_{ss}(x)$
  - This represents Waddington's epigenetic landscape [39]

- **Validate with Experimental Data:** Compare model predictions with gene coexpression matrices from experimental data (e.g., microarray) [39].

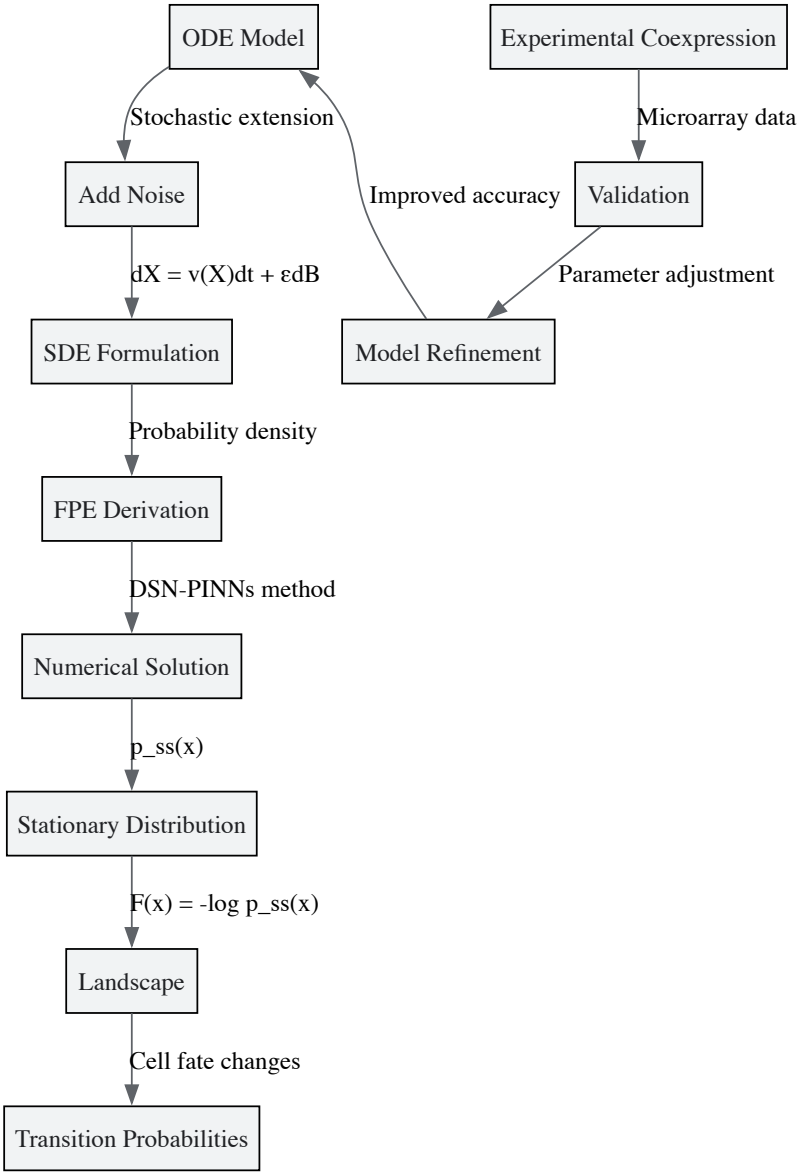


Figure 3: Fokker-Planck Equation Solution Workflow

Case Studies in Evolutionary Developmental Biology

Flower Morphogenesis in Arabidopsis thaliana

The GRN controlling Arabidopsis thaliana flower development provides a compelling case study for dynamic modeling approaches. The network comprises 12 key genes (EMF1, TFL1, LFY, AP1, CAL, LUG, UFO, BFU, AG, AP3, PI, SUP) with well-characterized interactions [39]. Researchers first developed a Boolean model that successfully captured four stable attractors corresponding to different floral organ identities. This discrete model was subsequently converted to a continuous ODE framework, incorporating realistic biochemical kinetics. Finally, the Fokker-Planck equation was solved numerically using a gamma mixture model, enabling quantification of the epigenetic landscape and prediction of transition probabilities between floral organ states [39]. The model's predictions showed good agreement with experimental coexpression matrices, validating its biological relevance and demonstrating how dynamic modeling can bridge qualitative genetic knowledge with quantitative phenotypic outcomes.

Table 3: Arabidopsis Flower Development GRN Components

Gene	Position	Primary Function	Regulatory Role
EMF1	1	Represses flowering	Repressor
TFL1	2	Maintains inflorescence	Repressor
LFY	3	Flower meristem identity	Activator
AP1	4	Sepal/petal identity	Activator
CAL	5	Redundant with AP1	Activator
LUG	6	Floral organ development	Repressor
UFO	7	FlorAL organ identity	Activator
BFU	8	Floral regulation	Activator
AG	9	Stamen/carpel identity	Activator
AP3	10	Petal/stamen identity	Activator
PI	11	Petal/stamen identity	Activator
SUP	12	Stamen development	Repressor

Gastrulation in Acropora Corals: Evolutionary Conservation and Divergence

Comparative analysis of gastrulation in Acropora digitifera and Acropora tenuis reveals how dynamic modeling can illuminate evolutionary processes. Despite diverging approximately 50 million years ago, these coral species exhibit remarkably conserved gastrulation morphology while employing divergent GRN architectures [3]. Transcriptomic profiling across three developmental stages (blastula, gastrula, sphere) identified a conserved regulatory "kernel" of 370 differentially expressed genes involved in axis specification, endoderm formation, and neurogenesis. However, significant temporal and modular expression divergence was observed in orthologous genes, illustrating developmental system drift—where conserved phenotypes are maintained despite genetic network rewiring [3]. This case study highlights how dynamic modeling approaches can distinguish between conserved and divergent regulatory elements, revealing the evolutionary flexibility of developmental programs. Boolean modeling of the core kernel identified stable attractors corresponding to pre- and post-gastrulation states, while ODE models captured quantitative expression differences between species, providing insights into how evolutionary tinkering with regulatory parameters shapes developmental outcomes.

Mouth-Form Plasticity in Pristionchus pacificus

The nematode Pristionchus pacificus exhibits a remarkable plasticity in mouth-form development, producing either a predatory or grazing morphology in response to environmental cues. Dynamic modeling of this plasticity GRN revealed a sequential checkpoint mechanism, with environmental sensitivity concentrated in two key genes (eud-1 and seud-1/sult-1) that act at different time points during the critical window of plasticity [12]. The network architecture incorporates positive feedback loops that lock in the developmental decision, creating bistability that can be modeled using both Boolean frameworks (discrete states) and ODE approaches with bifurcation analysis. This system exemplifies how dynamic modeling can elucidate the mechanistic basis of phenotypic plasticity—an important evolutionary capability that enables organisms to produce alternative phenotypes matched to environmental conditions. Parameter sensitivity analysis of the ODE model identified specific regulatory interactions that control the threshold for phenotype switching, suggesting potential evolutionary modules for adaptation to different ecological niches [12].

Table 4: Research Reagent Solutions for GRN Dynamic Modeling

Reagent/Resource	Function	Application Context
Boolean Network Software (BoolNet, CellCollective)	Discrete dynamics simulation	Initial network modeling, attractor identification
ODE Solvers (COPASI, Tellurium, MATLAB)	Continuous dynamics simulation	Quantitative modeling, parameter estimation
FPE Solvers (DSN-PINNs, custom code)	Stochastic dynamics solution	Epigenetic landscape reconstruction
RNA-seq Time Series Data	Expression quantification	Parameter estimation, model validation
Perturbation Tools (CRISPR, RNAi)	Network intervention	Model testing, causal validation
BioTapestry	Network visualization	Circuit representation, interaction mapping

Reagent/Resource	Function	Application Context
Single-Cell RNA-seq	Cellular heterogeneity measurement	Stochastic parameterization, state diversity

Future Perspectives and Concluding Remarks

Dynamic modeling of gene regulatory networks has evolved from simple Boolean abstractions to sophisticated stochastic frameworks that capture the probabilistic nature of developmental processes. The integration of these computational approaches with experimental evo-devo research has created powerful synergies, enabling researchers to move beyond descriptive network diagrams to predictive, quantitative models of developmental dynamics. The Fokker-Planck framework, in particular, provides a mathematical formalization of the epigenetic landscape concept, offering insights into the stability of cell states and the probabilities of transitions between them—fundamental considerations in evolutionary developmental biology.

Future directions in this field will likely focus on several key challenges: improving the scalability of FPE solutions for high-dimensional networks, integrating multi-scale models that connect gene regulation to tissue-level morphogenesis, and developing more efficient parameter estimation methods for complex stochastic models. Additionally, as single-cell technologies provide increasingly detailed views of cellular heterogeneity, dynamic models will need to incorporate this variability explicitly, further emphasizing the importance of stochastic approaches. The continued dialogue between mathematical modeling and experimental biology promises to deepen our understanding of how evolutionary changes in regulatory networks generate morphological diversity, ultimately illuminating the fundamental principles that shape biological form across phylogeny.

Challenges and Solutions in GRN Inference and Modeling

Overcoming High Dimensionality and Stochasticity in Dynamic Models

Gene regulatory networks (GRNs) form the fundamental control systems governing developmental processes, cellular differentiation, and evolutionary adaptation. Understanding their dynamics is crucial for unraveling the mechanisms behind phenotypic plasticity, morphological evolution, and disease pathogenesis. However, a significant challenge persists: the **high dimensionality and inherent stochasticity** of GRN dynamics create substantial computational barriers for accurate modeling and prediction. These networks operate through intricate interactions where genes encode transcription factors that regulate other genes or themselves, forming complex directed graphs of interactions [44].

The most biologically realistic descriptions of these networks utilize the **chemical master equation (CME)** framework, which predicts not only mean expression levels but also the distributions of discrete numbers of mRNAs and proteins across cell populations [44]. This stochasticity arises from multiple sources, including biological intrinsic and extrinsic noise, leading to the substantial variation in gene expression observed experimentally from cell to cell [44]. Unfortunately, with increasing sophistication comes prohibitive computational cost. The state space of a GRN is typically infinite, rendering direct solution of CMEs impossible, while finite-state projection (FSP) algorithms remain limited to very small networks with only one or two interacting genes [44]. For larger networks, Monte Carlo simulations using the stochastic simulation algorithm (SSA) become necessary but require substantial sampling to obtain smooth distributions, resulting in considerable computational burden [44]. This review examines innovative methodologies that overcome these challenges, with particular emphasis on applications within evolutionary developmental biology.

Methodological Approaches: From Traditional Techniques to Modern Innovations

Conventional Simulation Methods and Their Limitations

Traditional approaches to GRN modeling encompass Boolean networks, ordinary differential equations (ODEs), and chemical master equations (CMEs), each with distinct advantages and limitations [44]. While Boolean networks offer computational efficiency through binary representation of gene expression, they lack quantitative precision. ODEs provide more refined descriptions of time-dependent concentrations but fail to capture system stochasticity. The CME approach offers the most comprehensive framework by predicting complete probability distributions over molecular counts, but its computational demands have historically constrained applications.

The **finite-state projection (FSP)** algorithm addresses the infinite state space problem by truncation to a finite domain, enabling numerical solutions for small networks [44]. However, for networks with multiple interacting genes, the state space expands exponentially, severely limiting FSP's practical application. The **stochastic simulation algorithm (SSA)** generates statistically correct trajectories by calculating the timing and identity of reaction events using random numbers [44]. While more practical for larger networks, SSA requires extensive sampling to achieve smooth distributions, resulting in substantial computational overhead that hinders parameter exploration and sensitivity analysis.

The Holimap Framework: Principles and Implementation

A recently developed approach called **Holimap** (high-order linear-mapping approximation) addresses these computational challenges through an innovative mapping strategy [44]. The core concept involves approximating the protein or mRNA number distributions of a complex gene regulatory network by the distributions of a significantly simpler reaction system. Specifically, Holimap transforms the dynamics of a complex network with

second or higher-order interactions (nonlinear propensities) to the dynamics of a simpler system where all reactions are first-order (linear network) [44].

The reaction rates in this simplified linear system are typically time-dependent complex functions of the original network's reaction rates, determined through **conditional moment-matching** [44]. For example, in an autoregulatory feedback loop where protein expressed from a gene regulates its own transcription through cooperative binding, the nonlinear reactions G + hP ⇌ G* are replaced by linear reactions G ⇌ G* [44]. The effective binding rate $\hat{\sigma}_b$ in the linear-mapping approximation is calculated as $\hat{\sigma}_b = \sigma_b \times \langle n|i=0\rangle = (\sigma_b \times \mu_{1,0})/g_0$, where $\mu_{1,0}$ represents the first moment of protein numbers and g_0 the probability of the gene being in the unbound state [44].

Table 1: Key Notation in Holimap Framework for Autoregulatory Network

Symbol	Biological Meaning	Mathematical Definition
G	Gene in unbound state	Regulatory element available for transcription
G*	Gene in bound state	Regulatory element with transcription factor bound
P	Protein	Gene product acting as transcription factor
σ_b	Protein-gene binding rate	Bimolecular reaction rate constant
σ_u	Protein-gene unbinding rate	First-order dissociation rate constant
ρ_u, ρ_b	Burst frequencies in unbound/bound states	Protein production rates in different gene states
d	Protein degradation rate	First-order degradation/dilution rate constant
B	Mean protein burst size	Average proteins produced per transcription event $\langle k\rangle = p/(1-p)$
$\mu_{m,i}$	m-th factorial moment in state i	$\Sigma_n\, n(n-1)\cdots(n-m+1)p_{i,n}$

This linearization strategy dramatically reduces the system's state space, making previously intractable FSP simulations feasible and enabling efficient computation of smooth protein number distributions [44]. The Holimap approach maintains accuracy across diverse parameter regions, including those exhibiting oscillatory or multistable dynamics, while achieving significant computational speedup compared to conventional SSA.

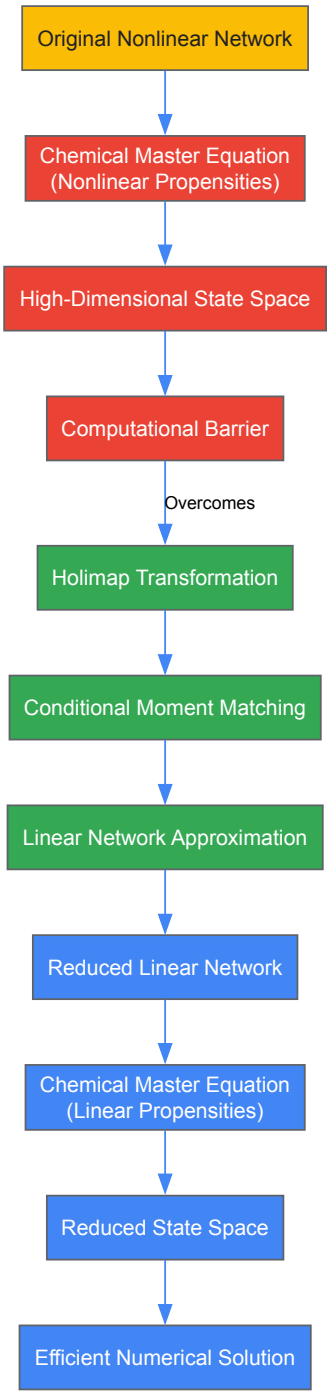


Figure 1: Holimap Methodological Workflow: From complex nonlinear networks to tractable linear approximations through conditional moment matching.

Hybrid Approximation Methods for Stochastic Dynamics

Complementary to Holimap, **hybrid approximation approaches** based on systems of partial differential equations offer another powerful strategy for complexity reduction [45]. These methods assume continuous-deterministic evolution for protein counts while maintaining discrete-stochastic representations for gene states, effectively reducing the computational burden from "one ordinary differential equation for each state to one partial differential equation for each mode of the system" [45].

This hybrid approach demonstrates particular utility for scenarios with sufficiently large molecule counts, where the continuous approximation becomes valid while still capturing essential stochastic effects from low-copy-number genetic elements [45]. For the specific case of self-regulatory genes, analytical steady-state solutions exist for the hybrid model, providing valuable benchmarks for numerical methods and theoretical insights into network behavior [45].

Technical Implementation: Protocols and Reagent Solutions

Computational Implementation of Holimap

The implementation of Holimap for analyzing stochastic GRN dynamics follows a structured protocol:

- **Network Characterization**: Define all species, reactions, and kinetic parameters in the original nonlinear GRN, with special attention to higher-order interactions (e.g., protein-gene binding with cooperativity).
- **Moment Equation Formulation**: Derive the time evolution equations for key moments from the chemical master equation. For an autoregulatory network with states i ∈ {0,1} (unbound/bound), these include probabilities g*i of gene states and factorial moments* μm,i of protein numbers [44].
- **Equation System Construction**: Assemble the moment equations, acknowledging the inherent non-closure property where equations for lower-order moments depend on higher-order moments [44].
- **Linear Mapping Approximation**: Replace nonlinear reactions with effective first-order reactions, calculating transformed rate parameters through conditional moment matching [44].
- **Simplified System Simulation**: Solve the resulting linear network using efficient numerical methods such as FSP, obtaining time-dependent probability distributions for molecular species.
- **Validation and Hybrid Refinement**: Compare results with limited SSA simulations for validation, implementing hybrid approaches where necessary to balance accuracy and computational efficiency [44].

Table 2: Research Reagent Solutions for GRN Modeling

Reagent/Resource	Function	Application Context
Holimap Algorithm	Approximates nonlinear GRN dynamics via linear mapping	Reduction of computational complexity for stochastic simulations
FSP (Finite State Projection)	Numerical solution of truncated CME	Direct computation of probability distributions for small networks
SSA (Stochastic Simulation Algorithm)	Monte Carlo trajectory generation	Exact simulation of discrete stochastic systems
Hybrid PDE Methods	Combines continuous and discrete representations	Efficient analysis for systems with large protein counts
BioTapestry	GRN visualization and analysis	Network architecture representation and validation [38]
Jupyter Notebooks	Interactive computational environment	Implementation of modeling protocols and data analysis [38]

Experimental Validation Frameworks

Validating computational predictions of GRN dynamics requires integration with experimental data. Contemporary approaches include:

- **Parameter Inference from Single-Cell Data**: Utilizing quantitative measurements of protein and mRNA distributions across cell populations to constrain model parameters.
- **Perturbation Response Analysis**: Comparing predicted and observed network responses to genetic perturbations (knockdown, overexpression) or environmental cues.
- **Time-Course Validation**: Assessing the accuracy of predicted temporal dynamics through live-cell imaging and time-lapse measurements of reporter expression.
- **Cross-Species Comparison**: Evaluating model generalizability by testing predictions across evolutionary lineages, particularly in model organisms used in evolutionary developmental biology [38].

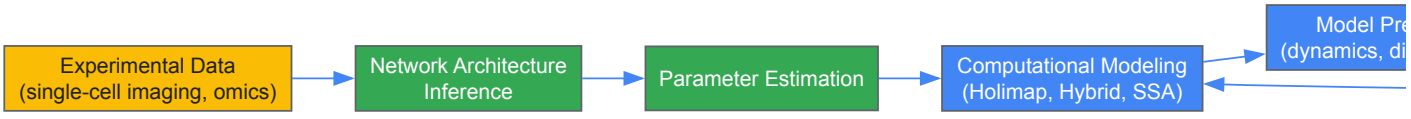


Figure 2: GRN Model Development Workflow: Iterative cycle integrating experimental data, computational modeling, and validation.

Applications in Evolutionary Developmental Biology

Modeling Density-Dependent and Sex-Biased Dispersal Evolution

The power of efficient GRN modeling approaches becomes particularly evident in evolutionary applications, such as understanding the genetic architecture underlying **density-dependent and sex-biased dispersal** during range expansions [13]. Traditional reaction norm (RN) approaches have derived optimal dispersal plastic responses from first principles, but lack insight into genetic and molecular mechanisms [13].

By representing dispersal traits as GRNs that process environmental inputs (population density) and internal cues (sex), researchers can model how **dispersal plasticity** emerges and evolves during range expansions [13]. Under equilibrium metapopulation conditions, GRN models produce emergent density-dependent and sex-biased dispersal response shapes consistent with RN theoretical expectations [13]. However, during range expansion with substantial mutation effects, GRN models demonstrate faster expansion dynamics due to their enhanced capacity to maintain **adaptive potential** [13].

This application highlights a crucial insight: understanding contemporary eco-evolutionary dynamics requires consideration of genetic architecture, not just phenotypic outcomes [13]. The efficiency of Holimap and hybrid methods enables the extensive simulations necessary to explore these evolutionary scenarios across meaningful parameter spaces.

Evolutionary Innovation through GRN Architecture

The structure of GRNs themselves facilitates evolutionary innovation. Computational models reveal how specific network motifs (autoregulatory loops, feedforward loops, mutual inhibition) can constrain or enhance evolutionary trajectories. For example:

- **Reduced Adaptive Potential in RN models** compared to GRN architectures during range expansions [13].
- **Phenotypic plasticity** emergence as a inherent property of specific GRN topologies rather than explicit selection.
- **Cryptic genetic variation** storage and release mechanisms embedded within GRN architecture.
- **Modularity and hierarchy** in GRN organization enabling compartmentalized evolution without global disruption.

Table 3: Performance Comparison of GRN Modeling Approaches

Method	Computational Complexity	Accuracy	Best Application Context
Boolean Networks	Low	Qualitative only	Large networks with binary outputs
Ordinary Differential Equations	Moderate	Deterministic means only	Systems with minimal stochasticity
Chemical Master Equation (FSP)	Very High (unfeasible for large N)	Exact for truncated space	Very small networks (1-2 genes)
Stochastic Simulation Algorithm	High (scales with samples)	Exact but noisy	Medium networks with sufficient compute resources
Hybrid Approximation	Moderate	Accurate for large protein counts	Systems with discrete genes, continuous proteins
Holimap	Low to Moderate	High across parameter space	Medium to large nonlinear networks

The challenges of high dimensionality and stochasticity in dynamic models of gene regulatory networks represent a significant frontier in evolutionary developmental biology. Methods like Holimap and hybrid approximations provide powerful strategies to overcome these barriers by transforming complex nonlinear systems into more tractable representations while preserving essential biological features. Through conditional moment matching and linear mapping, these approaches enable efficient exploration of GRN dynamics across parameter spaces that were previously computationally prohibitive.

The application of these methods to evolutionary questions, such as dispersal evolution during range expansions, demonstrates their potential to bridge gaps between genetic architecture and phenotypic outcomes. As single-cell technologies generate increasingly detailed experimental data on gene expression variability, the integration of efficient computational frameworks with empirical validation will continue to drive advances in understanding how GRNs control developmental processes and evolve new functions. Future methodological developments will likely focus on multi-scale integrations, combining molecular-level stochasticity with tissue-level patterning and evolutionary dynamics across longer timescales.

The reconstruction of **gene regulatory networks (GRNs)** is a cornerstone of modern computational biology, vital for deciphering the complex mechanisms that control cellular processes, development, and disease. In evolutionary developmental biology, comparing GRNs across species provides unparalleled insights into how conserved morphological structures arise from divergent molecular programs, a phenomenon known as **developmental system drift** [3]. However, the field of GRN inference is characterized by a proliferation of computational methods, each with

distinct underlying assumptions, strengths, and limitations. This diversity has led to a significant lack of consensus, where different algorithms applied to the same biological system can yield markedly different network architectures. This inconsistency poses a substantial challenge for biologists and drug development professionals who rely on accurate network models to generate testable hypotheses about gene function, regulatory mechanisms, and potential therapeutic targets [46].

The core of the problem lies in the traditional approach to evaluating these inference methods. For years, the field has relied heavily on **synthetic datasets**—simulated data where the "ground truth" network is known—for benchmarking. While convenient, this approach fails to capture the immense complexity, noise, and biological nuance present in real-world systems. Consequently, methods that perform exceptionally well on synthetic data often do not generalize effectively to experimental biological data, creating a gap between theoretical promise and practical utility [46]. Addressing this lack of consensus requires a paradigm shift towards rigorous, biologically grounded, and standardized benchmarking using real-world data. This whitepaper explores the current challenges in GRN inference benchmarking, outlines the emergence of new frameworks designed to address these challenges, and provides a practical guide for researchers navigating this complex landscape.

The Core Challenge: Why Benchmarking GRN Inference is Inherently Difficult

Inferring GRNs from high-throughput gene expression data, particularly from **single-cell RNA sequencing (scRNA-seq)**, is an inherently underdetermined problem. The high dimensionality of the data (thousands of genes across thousands of cells) and the complex, non-linear nature of gene interactions create a formidable computational challenge. The fundamental obstacle in benchmarking is the general **absence of a known ground truth** for real biological networks. Without a definitive map of all gene-gene interactions in a living cell, it is impossible to definitively assess the accuracy of an inferred network [46].

This lack of ground truth has traditionally been circumvented by using simulated data. However, simulations often incorporate simplifying assumptions that reduce biological fidelity. They may fail to accurately model key aspects such as **dropout events** (common in scRNA-seq), **technical batch effects**, the continuous nature of cellular differentiation, and the complex hierarchical structure of true regulatory networks [47]. As a result, a method may excel at learning the patterns of a specific simulation but fail when confronted with the messy reality of biological data. This issue was starkly highlighted by the CausalBench initiative, which found that the performance of state-of-the-art methods on synthetic benchmarks did not correlate with their performance on large-scale, real-world perturbation data [46].

Furthermore, the field lacks universal agreement on the metrics for success. Different evaluation metrics prioritize different aspects of network performance. Some focus on the **topology** of the network (e.g., edge presence), while others focus on the **dynamics** it can produce (e.g., predicting the outcome of a perturbation). This fragmentation in evaluation criteria further contributes to the lack of consensus, making it difficult to compare methods directly and objectively.

Current Landscape of GRN Inference Methods

The landscape of GRN inference algorithms is diverse, encompassing a wide range of statistical, machine learning, and deep learning approaches. Each class of methods brings a unique set of capabilities and limitations to the problem.

- **Tree-Based Models (e.g., GENIE3, GRNBOOST2):** These are among the most widely used and scalable methods. They operate on an **"One-vs-Rest" (OvR)** formulation, where the expression of each gene is predicted as a function of all other genes using tree models like Random Forest or Gradient Boosting. The importance scores from these models are then interpreted as regulatory strengths. Their key advantages are **scalability** to thousands of genes and inherent **explainability**. However, a major limitation is their inability to distinguish between **activation and inhibition** types of regulation, as importance scores are inherently positive. Additionally, as piecewise continuous functions, they may not perfectly capture the smooth, continuous dynamics of cellular processes [47].

- **Information Theory-Based Models:** These methods, such as those using **Mutual Information (MI)**, search for statistical dependencies between genes without assuming a specific functional form. They are effective at detecting non-linear relationships but can struggle with distinguishing direct from indirect regulation.

- **Differential Equation Models:** These approaches model gene expression changes using systems of ordinary differential equations (ODEs), providing a natural framework for capturing dynamics. However, they often require temporal data and can be computationally intensive for large networks.

- **Deep Learning Models:** Recent years have seen the emergence of various deep learning architectures, including Convolutional Neural Networks (CNNs), Variational Autoencoders (VAEs), and time-series models. While powerful, many supervised deep learning models depend on a list of known regulations for training, which limits their application in novel biological contexts. They also often face challenges in **scalability** and **interpretability** [47].

A notable recent advancement is the **scKAN model**, which employs **Kolmogorov-Arnold Networks (KANs)**. This approach models gene expression as a differentiable function, aligning with the smooth nature of cellular dynamics. A key innovation of scKAN is its use of **explainable AI (XAI)** techniques on the learned function to not only infer regulatory links but also to classify them as activations or inhibitions. Furthermore, it can infer **context-specific GRNs** for different cell lineages or states, moving beyond a one-size-fits-all network for the entire dataset [47].

Table 1: Comparison of Major GRN Inference Method Classes

Method Class	Representative Algorithms	Key Strengths	Inherent Limitations
Tree-Based	GENIE3, GRNBOOST2 [47]	High scalability, model explainability, robust performance	Cannot distinguish regulation type (activation/inhibition); piecewise continuous model
Information Theory	PIDC, MINDy	Detects non-linear relationships; model-agnostic	Can infer indirect correlations; limited directional inference
Differential Equation	SINCERITIES, dynGENIE3	Models dynamic processes; mechanistic interpretation	Requires temporal data; computationally intensive for large networks
Deep Learning	CNNs, VAEs, scKAN [47]	Can capture complex, non-linear patterns; high predictive power (scKAN: signed networks, context-specific)	Often less interpretable (not scKAN); can require known interactions for training (supervised models); data hunger

Emerging Solutions: The Rise of Real-World Benchmarks

The recognition of the limitations of synthetic benchmarks has catalyzed the development of new evaluation frameworks grounded in real-world data. The most promising of these leverage large-scale **perturbation data**, where the effects of experimentally disrupting specific genes are measured across the transcriptome. This provides a causal anchor for evaluating inferred networks.

The CausalBench Framework

A transformative development in the field is the introduction of **CausalBench**, a comprehensive benchmark suite designed specifically to evaluate GRN inference methods on real-world, large-scale single-cell perturbation data [46]. CausalBench is built upon two large-scale perturbational scRNA-seq datasets (from RPE1 and K562 cell lines) that collectively contain over 200,000 interventional data points from CRISPRi-based gene knockdowns.

Since the true, complete causal graph remains unknown, CausalBench employs a dual-strategy evaluation system to assess method performance without relying on a static ground truth:

- **Biology-Driven Evaluation:** This uses an approximation of biological ground truth, likely derived from curated knowledge bases or targeted experimental validations, to calculate metrics like precision and recall.
- **Quantitative Statistical Evaluation:** This leverages the gold-standard interventional data to compute causal metrics, including:
  - **Mean Wasserstein Distance:** Measures the extent to which a model's predicted interactions correspond to strong, empirically verified causal effects.
  - **False Omission Rate (FOR):** Measures the rate at which true causal interactions are missed by the model [46].

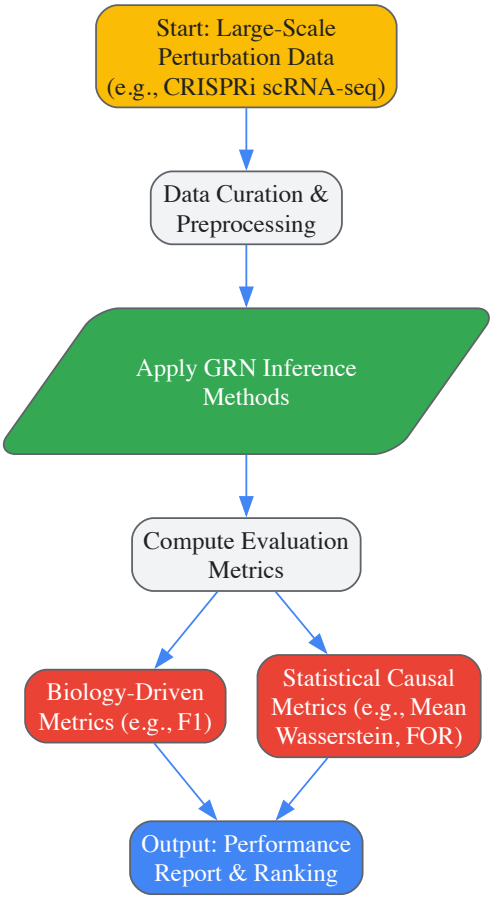
Table 2: Performance of Select GRN Inference Methods on CausalBench [46]

Method	Type	Key Characteristic	Performance on CausalBench
Mean Difference	Interventional	Top method from CausalBench challenge	High performance on statistical evaluation (Mean Wasserstein-FOR trade-off)
Guanlab	Interventional	Top method from CausalBench challenge	High performance on biological evaluation (F1 score)
GRNBoost	Observational	Tree-based (Gradient Boosting)	High recall but low precision; misses many interactions when restricted to TFs
PC	Observational	Constraint-based causal discovery	Low information extraction from data (low recall/precision)
NOTEARS	Observational	Continuous optimization-based	Low information extraction from data (low recall/precision)
Betterboost & SparseRC	Interventional		Good statistical evaluation performance, but poorer biological evaluation performance

The initial application of CausalBench has yielded critical insights. It revealed that **poor scalability** of existing methods is a major factor limiting performance on large, real-world datasets. Perhaps more surprisingly, it demonstrated that methods specifically designed to use **interventional data** did not consistently outperform those using only **observational data** on this real-world benchmark—a finding that contradicts results often seen on synthetic benchmarks [46]. This underscores the critical importance of using realistic evaluation frameworks.

Workflow for Real-World Benchmarking

The following diagram illustrates the standard workflow for benchmarking GRN inference methods using a framework like CausalBench, which relies on perturbation data to create a more biologically relevant evaluation.



Biological Validation: Insights from Evolutionary Developmental Biology

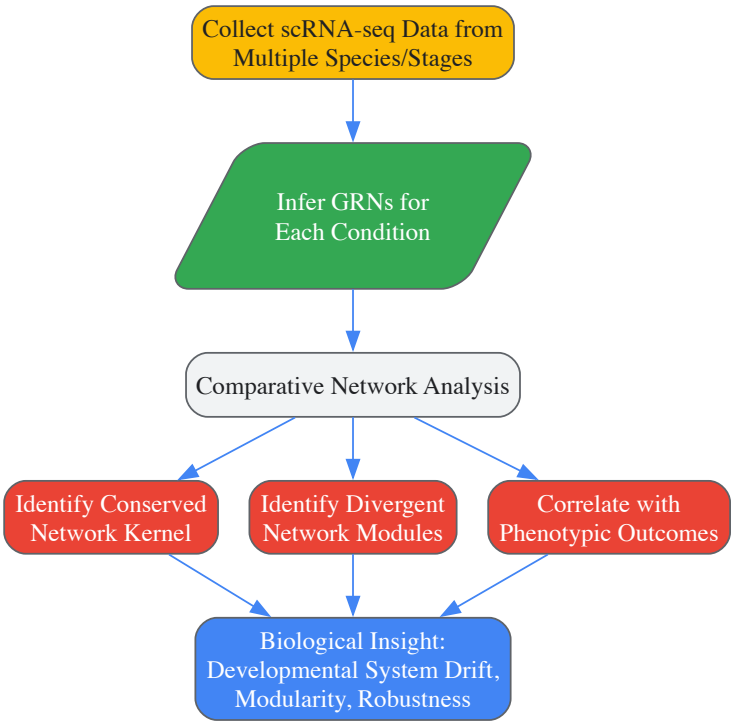
Beyond computational benchmarks, the ultimate validation of a GRN inference method lies in its ability to recapitulate and predict biological phenomena. Evolutionary developmental biology (Evo-Devo) provides a powerful context for this validation, as it allows for the comparison of GRNs across related species that exhibit both conserved and divergent traits.

A prime example is a study comparing gastrulation in two coral species of the genus *Acropora* (*A. digitifera* and *A. tenuis*), which diverged approximately 50 million years ago [3]. Despite the morphological conservation of gastrulation, comparative transcriptomics revealed that each species uses **divergent GRNs**, supporting the concept of **developmental system drift**. This provides a complex, real-world test case: a robust inference method should be able to identify both the **conserved regulatory "kernel"** (a subset of 370 genes co-regulated in both species) and the **species-specific peripheral rewiring** (differences in paralog usage and alternative splicing) that underlies the same morphological outcome [3].

This biological context is essential for moving beyond simple edge prediction accuracy. It forces methods to capture the **modularity**, **plasticity**, and **robustness** inherent in developmental GRNs. A method that only identifies a static, conserved core without capturing the lineage-specific differences would be of limited use for understanding evolutionary processes.

Workflow for Evolutionary GRN Analysis

The following diagram outlines a general workflow for using GRN inference in evolutionary developmental biology to study phenomena like developmental system drift.



A Practical Guide for Researchers and Practitioners

For researchers and drug development professionals aiming to infer GRNs, navigating the current landscape requires a strategic and critical approach. The following recommendations can serve as a guide:

- **Select Methods Based on Biological Question and Data Type:** There is no single "best" method. Choose an algorithm whose strengths align with your goal.
    - For an initial, genome-scale survey of potential regulators, a scalable tree-based method like **GRNBOOST2** is a robust choice [46] [47].
    - If distinguishing between activation and inhibition is critical, or if you are studying heterogeneous cell populations where context-specific networks are important, consider an emerging method like **scKAN** [47].
    - When working with perturbation data, prioritize methods that can natively incorporate interventional information.
- **Utilize Real-World Benchmarks in Method Selection:** Consult the results from benchmarks like **CausalBench** to understand how different methods perform on data that resembles yours. Prioritize methods that demonstrate strong performance in these realistic evaluations.
- **Employ a Multi-Method, Consensus Approach:** Given the lack of consensus, it is often prudent to run multiple inference methods on your dataset. The edges or network modules that are consistently predicted across different algorithms are often more reliable candidates for downstream experimental validation.
- **Validate Computationally and Biologically:** Before proceeding to wet-lab experiments, use held-out data (if available) and functional enrichment analysis to check the biological plausibility of your inferred network. The most powerful validation, however, comes from direct experimental perturbation of key predicted regulators to test their effect on target genes.

The Scientist's Toolkit: Essential Reagents for GRN Inference

Table 3: Key Research Reagents and Resources for GRN Inference

Resource / Reagent	Type	Function in GRN Research
CausalBench Suite [46]	Computational Benchmark	Provides a standardized framework and datasets (RPE1, K562) for evaluating GRN inference methods on real-world perturbation data.
BEELINE Benchmark [47]	Computational Benchmark	A widely used benchmark for evaluating GRN inference methods on synthetic and real single-cell data, providing standardized protocols.
scKAN Codebase [47]	Software Tool	An implementation of the KAN-based GRN inference method for inferring signed (activation/inhibition), context-specific networks.

Resource / Reagent	Type	Function in GRN Research
Large-Scale Perturbation Data (e.g., CRISPRi scRNA-seq) [46]	Dataset	Provides the causal anchor for training, testing, and validating inference methods, moving beyond purely correlative analyses.
Comparative Transcriptomic Data [3]	Dataset	Enables the evolutionary validation of inferred GRNs by testing for conserved kernels and divergent wiring across species.

The field of GRN inference is at a critical juncture. The lack of consensus among algorithms, once a source of confusion, is now being addressed through the development of sophisticated, biologically grounded benchmarking frameworks like CausalBench. The shift from synthetic to real-world benchmark data, particularly large-scale perturbation datasets, is revealing the true strengths and limitations of existing methods and is driving the development of more powerful, scalable, and nuanced algorithms like scKAN. For researchers in evolutionary development and drug discovery, this transition promises more reliable and biologically interpretable network models. By adopting a critical approach—leveraging robust benchmarks, employing multi-method consensus, and grounding inferences in biological context—we can move beyond the consensus problem and towards a deeper, more accurate understanding of the regulatory logic of life.

The comprehensive understanding of gene regulatory networks (GRNs) is a central goal in evolutionary developmental biology. While multi-omics technologies provide unprecedented insights into the multilayered regulation of gene expression, significant methodological and conceptual gaps persist in integrating data across genomic, epigenomic, transcriptomic, and three-dimensional architectural domains. This technical review examines current strategies for bridging cis-regulatory element function with 3D chromatin architecture data, focusing on computational integration frameworks, experimental technologies, and analytical pipelines. By synthesizing recent advances in multi-omic profiling, reference material development, and network modeling, we provide a roadmap for researchers seeking to elucidate the complex relationship between genome structure and gene regulatory function in developmental and evolutionary contexts.

Gene regulatory networks (GRNs) represent the complex circuitry of molecular interactions that control developmental processes, cellular differentiation, and evolutionary adaptations. The emergence of multi-omics technologies has revolutionized our ability to dissect these networks by providing complementary views of genomic variation, epigenetic states, chromatin architecture, and transcriptional outputs. However, the path from raw multi-omic measurements to mechanistic understanding of GRN operation remains fraught with integration challenges. These challenges are particularly pronounced in the interface between one-dimensional cis-regulatory annotations and the dynamic three-dimensional nuclear environment in which these elements physically interact.

Critical gaps persist in several domains: (1) the temporal coordination of epigenetic and architectural reorganization during development, (2) the functional interpretation of non-coding regulatory variation in spatial context, and (3) the computational integration of disparate data types into predictive models of regulatory function. Recent studies have revealed that the remodeling of DNA methylation and chromatin conformation are temporally separated during brain development, with chromatin reconfiguration preceding DNA methylation dynamics in differentiating astrocytes [48]. This asynchrony underscores the necessity of time-resolved multi-omic approaches for reconstructing developmental GRNs.

The convergence of these technological and analytical domains is particularly relevant for understanding the genetic basis of complex traits and diseases. Genome-wide association studies have identified numerous disease-associated variants in non-coding genomic regions, suggesting potential disruption of cis-regulatory elements [49]. Integrating these linear sequence variants with 3D chromatin interaction maps enables more accurate assignment of target genes and mechanistic interpretation of disease pathogenesis.

Technological Gaps and Solutions in Multi-Omic Profiling

Current Multi-Omics Technologies and Their Limitations

The multi-omics landscape encompasses diverse technologies that capture complementary aspects of genome structure and function, each with distinct strengths and limitations for GRN reconstruction. **Table 1** summarizes the primary technologies for profiling 3D chromatin architecture and their key characteristics.

Table 1: Technologies for Profiling 3D Chromatin Architecture and cis-Regulatory Elements

Technology	Principle	Resolution	Key Applications	Limitations
Hi-C [50] [49]	Proximity ligation-based genome-wide interaction mapping	1kb-1Mb	A/B compartments, TAD identification, chromatin loops	Sequence bias, low signal-to-noise ratio for specific interactions
ChIA-PET [50] [51]	Chromatin immunoprecipitation combined with proximity ligation	1-10kb	Protein-centric interactions (e.g., RNA Pol II, CTCF)	Antibody-dependent, limited to specific protein-mediated interactions
ChIA-Drop [50]	Microfluidics-based barcoding of multi-way interactions	Single-molecule	Complex multi-way interactions, multiplex transcriptional regulation	Technical complexity, lower throughput

Technology	Principle	Resolution	Key Applications	Limitations
Micro-C [50]	MNase-based chromatin digestion for nucleosome-resolution mapping	<1kb	Nucleosome-level interactions, fine-scale chromatin domains	Data complexity, computational challenges in analysis
SPRITE [50]	Split-pool barcoding without proximity ligation	Genome-wide	Higher-order nuclear organization, interchromosomal interactions	Cost, technical expertise requirements
snm3C-seq [48]	Single-nucleus methylome and chromatin conformation	Single-cell	Cell-type-specific 3D genome and DNA methylation simultaneously	Computational integration challenges, sparse data per cell

Each technology captures distinct aspects of chromatin organization, from the broad compartmentalization revealed by Hi-C to the specific protein-mediated interactions identified by ChIA-PET. However, significant limitations persist, including technology-specific biases, resolution constraints, and challenges in capturing transient interactions that are crucial for understanding dynamic GRNs. Micro-C offers improved resolution over Hi-C by utilizing micrococcal nuclease (MNase) instead of restriction enzymes, enabling nucleosome-level mapping of chromatin interactions [50]. Meanwhile, ligation-free methods like SPRITE and ChIA-Drop can reveal more complex multiway interactions that escape detection by paired ligation technologies [50].

Reference Materials and Standardization Approaches

A fundamental challenge in multi-omics integration is the lack of ground truth for validating measurements and integration methods. The **Quartet Project** addresses this by providing multi-omics reference materials derived from B-lymphoblastoid cell lines from a family quartet (parents and monozygotic twin daughters) [52]. This design provides built-in truth defined by genetic relationships and the central dogma of information flow from DNA to RNA to protein.

The project employs a **ratio-based profiling approach** that scales absolute feature values of study samples relative to a concurrently measured common reference sample, significantly improving data reproducibility and comparability across batches, laboratories, and platforms [52]. This paradigm shift from absolute to relative quantification addresses a root cause of irreproducibility in multi-omics measurements. Reference materials have been established for DNA, RNA, protein, and metabolites, enabling comprehensive quality assessment across omics layers [52].

Experimental Frameworks for Multi-Omic Data Generation

Integrated Workflows for Simultaneous Multi-Omic Profiling

Advanced methodologies now enable the simultaneous capture of multiple omic layers from the same biological sample, eliminating technical confounding factors and enabling direct correlation of different molecular features. The **single-nucleus methyl-3C sequencing (snm3C-seq)** method exemplifies this approach by jointly profiling chromatin conformation and DNA methylation in single nuclei [48]. This technology has revealed temporally distinct trajectories of epigenetic and architectural reorganization during human brain development, with chromatin reconfiguration preceding DNA methylation changes during astrocyte differentiation [48].

Diagram: Workflow for Integrated Single-Nucleus Multi-Omic Profiling

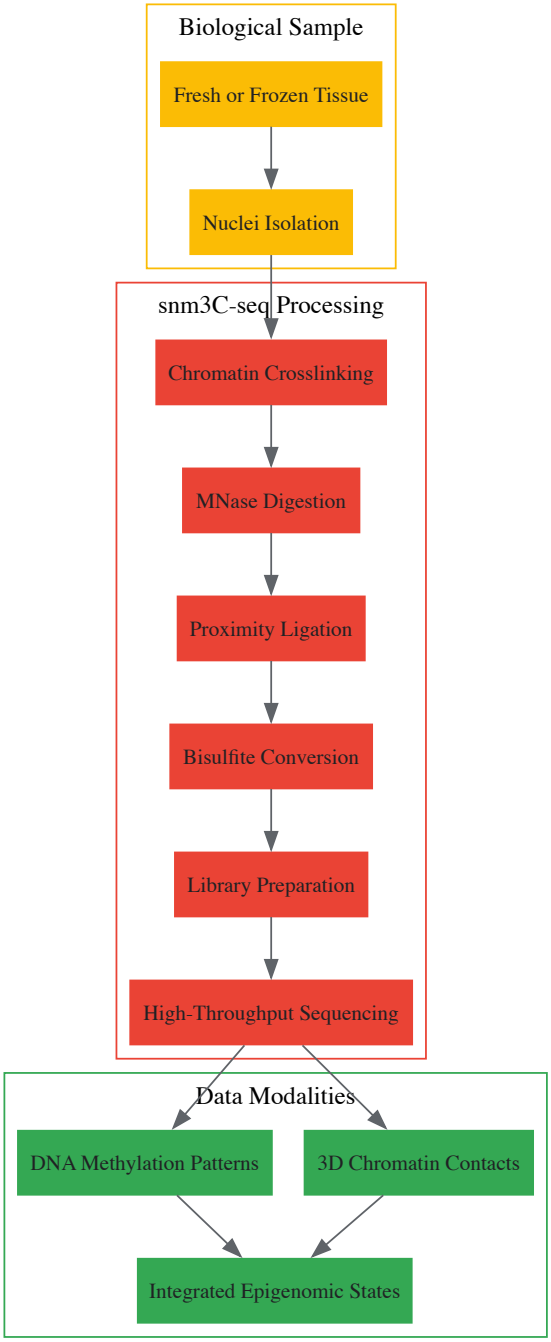


Table 2: Key Research Reagent Solutions for Multi-Omic GRN Studies

Reagent/Resource	Function	Application Examples
Quartet Reference Materials [52]	Multi-omics quality control and batch effect correction	Proficiency testing across DNA, RNA, protein, and metabolomic platforms
CTCF Antibodies [50] [51]	Immunoprecipitation of architectural protein interactions	ChIA-PET for CTCF-mediated chromatin looping, domain boundary identification
MNase Enzyme [50]	Nucleosome-resolution chromatin digestion	Micro-C for high-resolution chromatin interaction mapping
snm3C-seq Reagents [48]	Simultaneous profiling of methylation and conformation	Developmental trajectory analysis of epigenetic and architectural dynamics
Crosslinkers (Formaldehyde) [50]	Preservation of protein-DNA and chromatin interactions	Hi-C, ChIA-PET, and related proximity ligation methods
Barcoded Oligonucleotides [50]	Molecular tagging for multiplex assays	SPRITE, ChIA-Drop for multiway interaction detection

Computational Strategies for Multi-Omic Data Integration

Mathematical Frameworks for Data Integration

The computational integration of multi-omics data presents significant challenges due to differing dimensionalities, data distributions, and noise structures across omic layers. Integration methods can be broadly classified into **unsupervised approaches** that identify patterns without predefined outcomes, and **supervised methods** that predict specific biological endpoints [53]. These can be further categorized based on their mathematical foundations:

- **Network-based (NB) approaches** model interactions between biological variables using graph structures that incorporate known or predicted relationships [54]. These methods utilize graph measures (degree, connectivity, centrality) and algorithms for sub-network identification to extract biologically meaningful patterns.
- **Bayesian (BY) methods** employ statistical models that update prior assumptions about data probability distributions based on measured data using Bayes' rule [54]. Bayesian networks are particularly promising for modeling the probabilistic dependencies between different omic layers.
- **Network-free non-Bayesian (NF-NBY)** approaches include dimensionality reduction techniques like Multi-Block Partial Least Squares (MB-PLS) and Multiple Co-Inertia Analysis (MCIA) that identify covarying features across omic layers without explicit network modeling [54].
- **Matrix factorization methods** such as Joint Non-negative Matrix Factorization (jNMF) and MOFA (Multi-Omics Factor Analysis) decompose multiple omic data matrices into lower-dimensional representations that capture shared and specific patterns of variation [53].

Categories of Unsupervised Integration Methods

Table 3: Unsupervised Multi-Omics Data Integration Methods for GRN Analysis

Method Category	Key Methods	Underlying Principle	Applications in GRN Research
Regression/Association-based	CNAMet, MEMo, iPAC [53]	Sequential analysis based on presumed causal relationships	Identifying driver mutations and their transcriptional consequences
Canonical Correlation Analysis (CCA)	Sparse MCCA, BCCA, MCIA [53] [54]	Maximizing correlation between linear combinations of variables from different omics	Finding co-regulated gene sets across epigenetic and transcriptional layers
Factor Analysis-based	MOFA, Joint Bayesian Factor, BayRel [53]	Decomposition of multi-omics data into latent factors	Identifying shared biological axes of variation across omic layers
Kernel-based Clustering	SNF, mixKernel, NEMO [53]	Combining similarity matrices from different omics	Patient stratification based on multi-omic profiles
Matrix Factorization-based Clustering	iCluster, iNMF, JIVE [53]	Simultaneous decomposition of multiple omics data matrices	Disease subtyping, identification of molecular subtypes
Bayesian Clustering	PARADIGM, MDI, BCC [53]	Probabilistic modeling of cluster assignments	Pathway-centric integration of multi-omics data

These methods can be further classified by their integration strategy: **data-ensemble approaches** that concatenate multiple omics into a single matrix, **model-ensemble approaches** that analyze each omic separately then combine results, and **multi-step sequential analysis** that processes omics in a specified order based on presumed biological causality [53].

Framework for TE-Mediated Gene Regulatory Networks

Transposable elements (TEs) represent a significant component of cis-regulatory variation in vertebrate genomes, contributing approximately 43.4%, 47.5%, and 9.6% of pig, cattle, and chicken genomes, respectively [55]. A computational framework for constructing **TE-mediated gene regulatory networks (TE-GRNs)** has been developed to systematically analyze the contribution of TEs to regulatory evolution in livestock species [55]. This approach integrates TE annotation with transcriptomic and epigenomic data to identify tissue-specific TE expression and chromatin accessibility patterns.

Diagram: Computational Framework for TE-Mediated Regulatory Network Analysis

Biological Insights from Multi-Omic Integration

Temporal Dynamics in Developing Human Brain

Integrated analysis of chromatin conformation and DNA methylation during human hippocampal and prefrontal cortex development has revealed **temporally distinct trajectories** of epigenetic and architectural reorganization [48]. Specifically, chromatin conformation remodeling in radial glia cells occurs predominantly during mid-gestation, while CG methylation maturation extends into adulthood. This temporal separation suggests distinct regulatory mechanisms governing 3D genome reorganization versus DNA methylation patterning during neural development.

Single-cell multi-omic profiling has further identified **cell-type-specific patterns** of chromatin organization, with neurons exhibiting enriched short-range chromatin interactions compared to the long-range interactions predominant in glial cells [48]. These architectural differences likely reflect cell-type-specific gene regulatory strategies, with neuronal genomes optimized for precise, localized control of gene expression.

3D Genome Organization in Disease and Evolution

The integration of multi-omics data has been particularly fruitful in elucidating the role of 3D genome organization in disease pathogenesis. In cancer, disruptions of **topologically associating domains (TADs)** can lead to ectopic enhancer-promoter interactions and oncogene activation [51]. For example, somatic hemizygous microdeletions at 13q12.2 disrupt TAD boundaries and cause aberrant enhancer activity that contributes to FLT3 overexpression in acute lymphoblastic leukemia [51].

Evolutionary analyses leveraging multi-omics integration have revealed species-specific patterns of **transposable element amplification** and their contributions to regulatory innovation [55]. In pigs, cattle, and chickens, distinct TE families exhibit varying amplification histories and contemporary activities, with species-specific TE families substantially outnumbering other families. These differences suggest lineage-specific co-option of TEs for regulatory functions during domestication.

The integration of multi-omics data represents both a formidable challenge and tremendous opportunity for advancing our understanding of gene regulatory networks in evolutionary development. While significant progress has been made in computational methods, reference materials, and experimental technologies, several frontiers demand continued innovation. Future advances will likely focus on single-cell multi-omic technologies with enhanced throughput and molecular completeness, dynamic modeling of time-resolved regulatory networks, and machine learning approaches that can extract predictive models from increasingly complex and high-dimensional data. The integration of 3D chromatin architecture data with functional validation approaches will be particularly important for moving from correlative to causal understanding of gene regulation. As these methods mature, they will enable unprecedented insights into the evolutionary dynamics of regulatory genomes and their role in shaping phenotypic diversity.

Distinguishing Direct Regulation from Indirect Correlation in Network Reconstruction

Inference of gene regulatory networks (GRNs) is a fundamental challenge in systems biology, aiming to unravel the complex causal relationships between genes and their regulators [56]. A central obstacle in this process is the reliable distinction between **direct regulatory interactions** and **indirect correlations** induced by intermediate factors or shared regulatory inputs. This distinction is critical for generating accurate, biologically meaningful network models that can illuminate developmental processes, disease mechanisms, and evolutionary patterns [56] [13]. Within evolutionary development (evo-devo) research, where GRNs explain the emergence of morphological novelty and phenotypic plasticity, failing to resolve this ambiguity can lead to incorrect inferences about the core regulatory architecture underlying trait formation and diversification [13]. This guide provides an in-depth technical framework for addressing this challenge, integrating computational and experimental strategies tailored for researchers and drug development professionals.

Core Concepts and Challenges

A **direct regulatory interaction** typically involves a transcription factor (TF) binding to a cis-regulatory element (e.g., promoter or enhancer) to directly influence the transcription rate of a target gene [56]. In contrast, an **indirect correlation** arises when two genes show expression coordination without physical interaction, often because a third gene or a hidden variable regulates both. In a developing embryo, for instance, two genes might appear correlated because they both respond to the same morphogen gradient, not because one regulates the other.

The primary challenge stems from the nature of bulk and single-cell omics data, where measured correlations are a mixture of these direct and indirect effects. Transitive relationships pose a particular problem: if Gene A regulates Gene B, and Gene B regulates Gene C, a strong correlation between A and C may be detected, even in the absence of a direct physical interaction [57]. Furthermore, technical noise and the high dimensionality of data, where the number of genes (p) far exceeds the number of samples (n), complicate the accurate estimation of true dependencies.

Computational Methodologies and Their Foundations

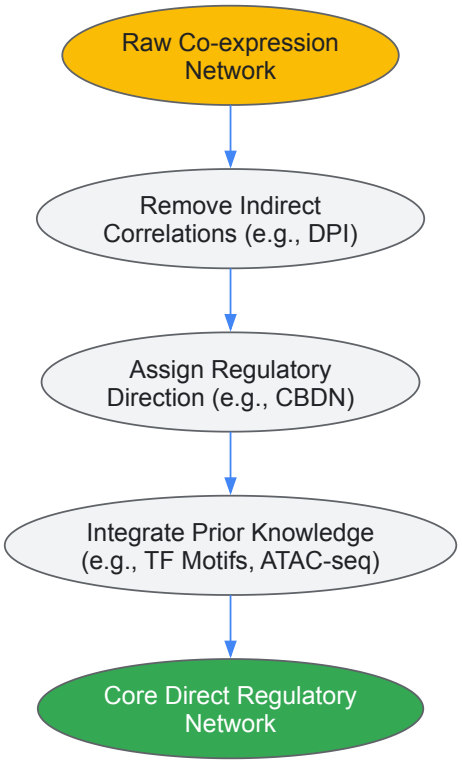
Computational methods for GRN inference employ diverse statistical and algorithmic principles to disentangle direct from indirect effects. **Table 1** summarizes the foundational approaches, their mechanisms for handling indirect effects, and key assumptions.

Table 1: Foundational Computational Approaches for Distinguishing Direct from Indirect Regulation

Method Category	Core Mechanism	Handling of Indirect Effects	Key Assumptions
Correlation & Information Theory [56]	Measures pairwise association (e.g., Pearson correlation, Mutual Information).	Limited; may detect transitive relationships. Data Processing Inequality (DPI) in methods like ARACNE removes the weakest edge in a triplet if below a threshold [57].	Co-expression implies co-regulation; DPI holds for true interactions.
Regression Models [56]	Models a target gene's expression as a function of potential regulator expression.	By conditioning on other genes, it attempts to isolate the direct effect of each predictor. Penalized regression (e.g., LASSO) enforces sparsity, retaining strongest predictors [56].	Linearity or known non-linear relationships; true network is sparse.
Probabilistic Graphical Models [56]	Constructs a network representing conditional dependence relationships between variables.	An edge exists if two variables are dependent conditioned on all other variables, thereby removing indirect paths [57].	Data follows an assumed distribution (e.g., Gaussian).
Context-Based Methods (e.g., CBDN) [57]	Computes an asymmetric influence function by assessing how a source gene affects the target's correlations with all other genes.	Uses a directed DPI to remove transitive interactions and infer directionality from gene expression data alone [57].	A regulator's influence is reflected in the global correlation context.
Dynamical Systems [56]	Models gene expression as a system evolving over time using differential equations.	The structure of the equations (e.g., which variables appear) can inherently represent direct causal influences.	Requires time-series data; model structure must be specified.

The following diagram illustrates the logical workflow a researcher might employ to progressively refine a co-expression network into a core direct regulatory network.

Figure 1: Workflow for Refining a Direct Regulatory Network



Detailed Experimental Protocols for Validation

Computational predictions require rigorous experimental validation. Below are detailed protocols for key validation experiments.

Perturbation-Based Validation (Knockdown/Knockout)

This protocol tests if the predicted regulator is necessary for the target gene's expression.

- **Design sgRNAs or siRNAs:** Design at least two independent sgRNAs (for CRISPR/Cas9) or siRNAs (for RNAi) targeting the transcription factor of interest. Include a non-targeting scrambled sequence as a negative control.
- **Deliver Perturbation:** Transfect siRNAs or transduce lentivirus carrying sgRNAs and Cas9 into the relevant cell line. Use a fluorescent reporter or antibiotic selection (e.g., puromycin) to enrich for successfully transfected/transduced cells.
- **Confirm Perturbation Efficiency:** 48-72 hours post-transfection/transduction, harvest cells. Extract RNA and protein.
  - Quantify TF mRNA knockdown/knockout efficiency using **RT-qPCR**.
  - Confirm reduction of TF protein levels using **Western Blotting**.
- **Measure Downstream Effects:** Using the same RNA samples, measure the expression of the putative direct target genes via **RT-qPCR** or **RNA-seq**.
- **Interpretation:** A direct regulatory relationship is strongly supported if the perturbation of the TF leads to a significant and consistent change in the target gene's expression. Indirect targets may show weaker or delayed effects.

Direct Physical Interaction Validation (Chromatin Immunoprecipitation - ChIP)

This protocol confirms the physical binding of a TF to a specific genomic locus.

- **Cross-link and Lyse Cells:** Treat cells with 1% formaldehyde for 10 minutes at room temperature to cross-link DNA and bound proteins. Quench with glycine. Harvest cells and lyse them to extract nuclei.
- **Chromatin Shearing:** Sonicate chromatin to fragment DNA into 200-500 bp fragments. This can be verified by agarose gel electrophoresis.
- **Immunoprecipitation:** Incubate the sheared chromatin with a specific antibody against the TF of interest. Use an isotype-control antibody for a negative control. Protein A/G beads are then added to pull down the antibody-TF-DNA complexes.
- **Wash, Reverse Cross-link, and Purify DNA:** Wash beads stringently to remove non-specific binding. Reverse the cross-links by heating at 65°C. Treat with RNAse and proteinase K, and purify the immunoprecipitated DNA.
- **Analysis:**
  - **ChIP-qPCR:** Design primers flanking the predicted TF binding site (e.g., from motif analysis). Enrichment is calculated relative to the input DNA and control IgG sample.

- **ChIP-seq:** Prepare a sequencing library from the purified DNA. Align sequenced reads to the reference genome and call peaks to identify all genomic binding sites.

A Scientist's Toolkit: Essential Research Reagents

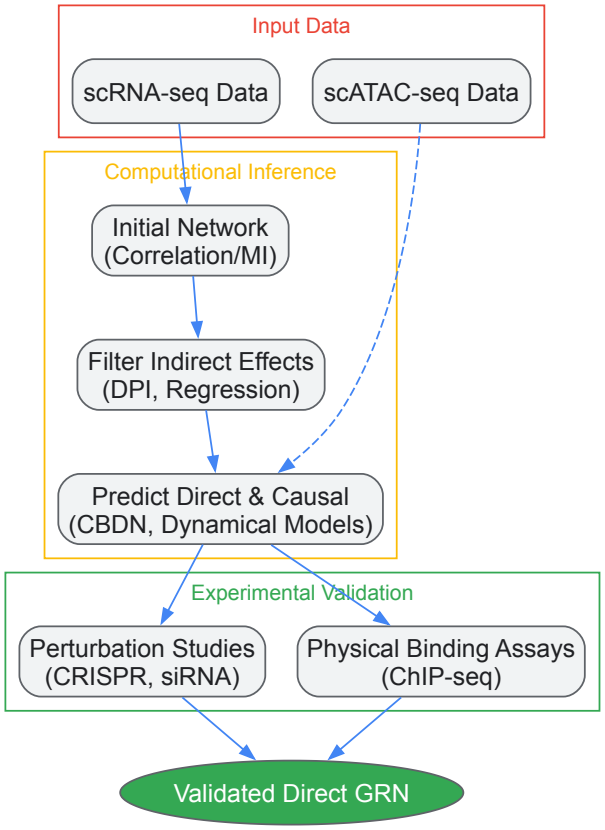
Table 2: Key Research Reagents for GRN Inference and Validation

Reagent / Solution	Function	Key Considerations
Single-cell Multi-ome Kit (e.g., 10x Multiome) [56]	Simultaneously profiles RNA expression (scRNA-seq) and chromatin accessibility (scATAC-seq) in the same single cell.	Provides matched regulatory and transcriptional data, crucial for linking TFs to target genes.
Validated TF Antibodies	Essential for ChIP experiments to specifically pull down the TF of interest.	Antibody specificity is critical; validate with knockout cell lines if possible.
CRISPR/Cas9 Knockout Kit (e.g., lentiviral sgRNA)	Enables targeted gene knockout for perturbation studies.	Controls for off-target effects are necessary; use multiple sgRNAs.
siRNA/shRNA Libraries	Enables transient or stable gene knockdown for perturbation studies.	Potential for off-target effects; requires careful control design.
Dual-Luciferase Reporter Assay System	Tests the ability of a DNA sequence to enhance transcription in the presence of a TF.	Co-transfect the TF and the reporter construct; normalize for transfection efficiency.

Integrated Workflow and Visualization

A robust strategy for reconstructing direct regulatory networks involves the integration of computational inference with multi-omic data validation. The following diagram outlines this comprehensive, multi-step workflow.

Figure 2: Integrated Workflow for Direct GRN Reconstruction



Distinguishing direct regulation from indirect correlation is not merely a technical refinement but a prerequisite for constructing accurate and predictive models of gene regulation. As research increasingly focuses on the role of GRNs in evolution and development [13] [16]—where subtle changes in network architecture can lead to macroevolutionary phenotypic changes—the precision afforded by the integrated methods described here becomes paramount. The future of this field lies in the continued development of sophisticated algorithms that leverage multi-omic single-cell data [56] [15] and in the close, iterative dialogue between computational prediction and experimental validation.

Validating GRN Models and Comparative Cross-Species Analysis

The validation of dynamical models of gene regulatory networks (GRNs) is a critical step in systems biology, ensuring that theoretical predictions accurately reflect biological reality. A powerful validation technique involves comparing the co-expression patterns generated by a model against experimental co-expression data. This guide details a methodology that leverages the solution of the Fokker-Planck equation to derive a theoretical epigenetic landscape, whose stationary probability distribution is used to calculate a simulated co-expression matrix. This matrix is then directly compared to an experimental co-expression matrix, providing a robust, quantitative means to discriminate between competing dynamical models of GRNs. This approach is particularly relevant in evolutionary developmental biology, where understanding the dynamics of GRNs can illuminate the mechanisms behind morphological evolution and phenotypic diversity.

In evolutionary developmental (evo-devo) biology, Gene Regulatory Networks (GRNs) represent the core computational systems that control developmental processes and, consequently, evolutionary change. A GRN is a dynamic system comprising genes, their RNA and protein products, and their regulatory interactions. The temporal evolution of protein concentrations within a GRN can be described by a system of ordinary or stochastic differential equations, whose solutions define an **epigenetic landscape** [39]. This landscape, a modern formalization of Waddington's metaphor, is characterized by basins of attraction representing stable cell states (e.g., different phenotypic fates in a developing flower) and the paths between them.

Countless dynamical models can be proposed for a given GRN, but their biological relevance must be rigorously tested. Model validation bridges the gap between theoretical mathematics and empirical observation. Matching **simulated co-expression** to **experimental co-expression** data provides a powerful, discriminating validation test [39]. A co-expression matrix quantifies the correlation in expression levels between pairs of genes across a population of cells or samples. A valid dynamical model should be able to recapitulate the core co-expression patterns observed in nature, confirming that the model's underlying logic and parameters accurately capture the system's biology.

Core Methodology: From Dynamical System to Co-expression Validation

This section outlines the primary workflow for validating a continuous dynamical model of a GRN by comparing simulated and experimental co-expression.

The Continuous Dynamical Model

The process begins with a continuous-time model describing the evolution of protein concentrations. For a GRN with N genes, the system can be described by a set of differential equations. For instance, for each gene i, the concentration of its associated protein, P_i, can be modeled as emerging from the production of its corresponding mRNA, m_i, which is itself regulated by other proteins in the network [39].

A typical formulation derived from biochemical reaction schemes is:

- $dm_i / dt = \Sigma_j (\alpha_{ij} * k_{ij} * P_j^{n_{ij}}) / (1 + k_{ij} * P_j^{n_{ij}}) - \gamma_i * m_i$
- $dP_i / dt = \beta_i * m_i - \delta_i * P_i$

Where:

- α_{ij} is the maximum transcription rate.
- k_{ij} is the binding constant.
- n_{ij} is a Hill coefficient representing cooperativity.
- γ_i and δ_i are the degradation rates of mRNA and protein, respectively.
- β_i is the translation rate.

The regulatory effect (activation or repression) is embedded in the parameters and the summation over the regulators j.

The Fokker-Planck Equation and the Epigenetic Landscape

To account for stochastic fluctuations inherent in biological systems, the deterministic dynamics are translated into a probabilistic framework using the **Fokker-Planck equation (FPE)**. The FPE describes the time evolution of the probability density function of protein concentrations in the network [39].

Let $P(x, t)$ be the probability density of finding the system in a state with protein concentrations $x = (x_1, ..., x_N)$ at time t. The FPE is: $\partial P(x, t) / \partial t = - \Sigma_i \partial/\partial x_i [\mu_i(x) P(x, t)] + \Sigma_i \Sigma_j \partial^2 / [\partial x_i \partial x_j] [D_{ij}(x) P(x, t)]$

Here, $\mu_i(x)$ represents the deterministic drift (given by the right-hand side of the dynamical system equations), and $D_{ij}(x)$ is the diffusion tensor, which captures the noise intensity.

The **epigenetic landscape** is formally defined by the stationary solution of the FPE, $P_s(x)$, which is the probability distribution obtained when $\partial P/\partial t = 0$. This solution defines the free energy potential of the system via $F(x) = -\ln(P_s(x))$. The valleys (minima) of $F(x)$ correspond to the attractors

(stable cell states) of the landscape [39].

Calculating Simulated Co-expression

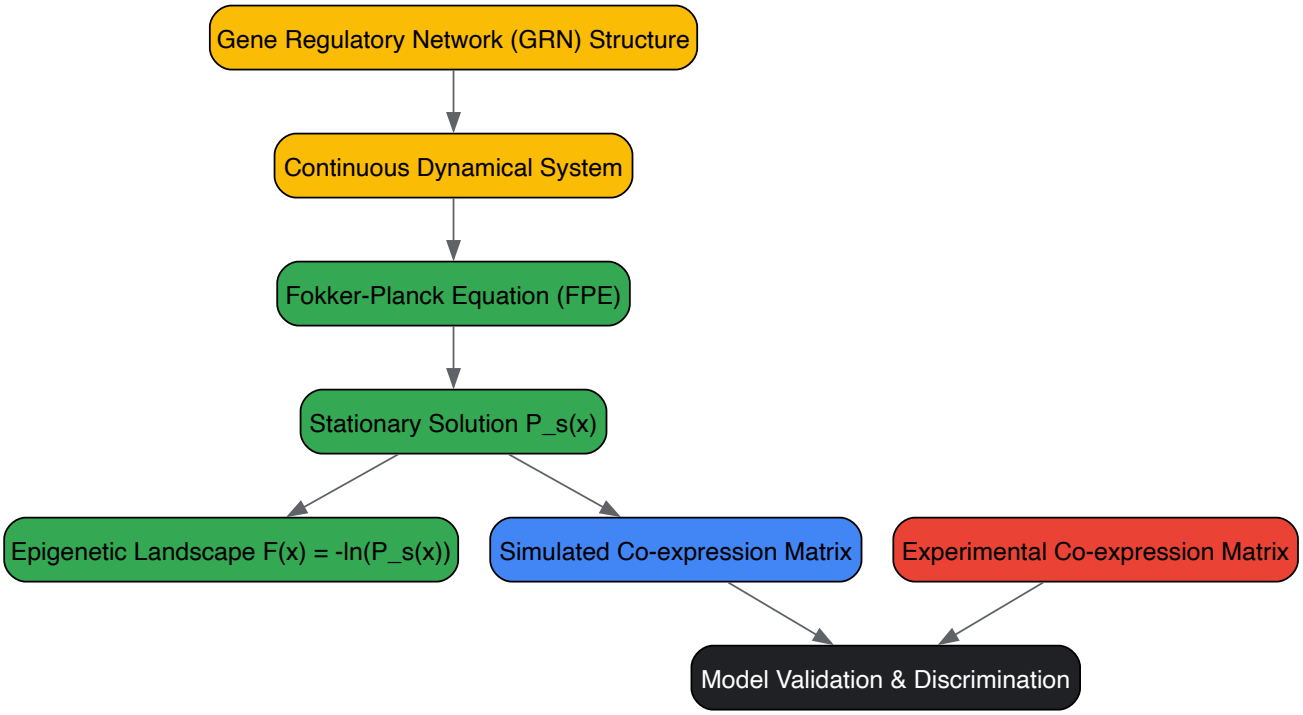
From the stationary distribution $P_s(x)$, one can calculate the theoretical co-expression matrix. The co-expression between gene i and gene j is defined as the statistical correlation (e.g., Pearson correlation) between their protein concentrations, x_i and x_j, under the stationary distribution [39].

The covariance is calculated as: $Cov(x_i, x_j) = \int (x_i - \mu_i)(x_j - \mu_j) P_s(x)\, dx$ where μ_i and μ_j are the mean concentrations. The correlation coefficient ρ_{ij} is then: $\rho_{ij} = Cov(x_i, x_j) / (\sigma_i \sigma_j)$ where σ_i and σ_j are the standard deviations of x_i and x_j, respectively. The matrix of all ρ_{ij} values is the **simulated co-expression matrix**.

Validation via Comparison with Experimental Co-expression

The final, critical step is to compare the simulated co-expression matrix with an **experimental co-expression matrix**. The experimental matrix is typically derived from techniques like microarray or RNA-sequencing across multiple samples or single cells [39] [56]. A good agreement between the simulated and experimental matrices, assessed through metrics like mean squared error or correlation between the matrix elements, provides strong evidence that the dynamical model accurately captures the real system's behavior. This methodology directly relates the theoretical model to experimental data, enabling the discrimination between competing model structures and parameters [39].

The following diagram illustrates this core workflow.



Detailed Experimental Protocols for Key Assays

Generating high-quality experimental data is paramount for robust model validation. Below are detailed protocols for key assays that generate co-expression and chromatin interaction data.

Microarray Analysis for Co-expression Data

Microarray technology allows for the simultaneous measurement of expression levels of thousands of genes.

Protocol:

- **RNA Extraction:** Isolate total RNA from biological samples (e.g., different tissues, developmental stages, or treatments) using a guanidinium thiocyanate-phenol-chloroform extraction method (e.g., TRIzol). Assess RNA quality and integrity using an Agilent Bioanalyzer.
- **cDNA Synthesis and Labeling:** Reverse-transcribe purified mRNA into complementary DNA (cDNA). Incorporate fluorescent dyes (e.g., Cy3 and Cy5) into the cDNA during amplification.
- **Hybridization:** Mix the labeled cDNA samples and hybridize them to a microarray chip containing immobilized DNA probes for the genes of interest. Incubate for 14-16 hours at 65°C in a specialized hybridization oven.

- **Washing and Scanning:** Wash the microarray slide with a series of stringency buffers (e.g., SSC and SDS solutions) to remove non-specifically bound cDNA. Scan the slide using a laser scanner to detect the fluorescence intensity at each probe.
- **Data Normalization and Co-expression Calculation:** Process the raw intensity files using software like R/Bioconductor. Perform background correction, within-array normalization (e.g., Loess), and between-array normalization (e.g., Quantile). Calculate the co-expression matrix by computing pairwise correlation coefficients (e.g., Pearson or Spearman) between the normalized expression profiles of all genes across the samples.

Chromatin Conformation Capture (3C) Assays

3C-based methods are crucial for understanding the 3D genome architecture, which underlies regulatory interactions that drive co-expression.

Protocol (Systematic Evaluation of Hi-C Variants) [58]:

- **Cross-linking:** Cross-link cells with 1-2% formaldehyde (FA) to fix protein-DNA and DNA-DNA interactions. For enhanced cross-linking and reduced random ligation, a two-step protocol using 1% FA followed by 3 mM DSG (Disuccinimidyl glutarate) is recommended.
- **Chromatin Fragmentation:** Lyse cells and digest chromatin with a restriction enzyme. The choice of enzyme determines resolution:
  - **HindIII:** Lower resolution, larger fragments (5-20 kb). Better for detecting compartment strength.
  - **DpnII/MboI:** Higher resolution, smaller fragments (0.5-5 kb). Standard for loop detection.
  - **MNase (for Micro-C):** Highest resolution, digests to mononucleosomes (~150 bp). Requires size selection.
- **Proximity Ligation:** Under dilute conditions, the digested DNA ends are ligated. This step preferentially joins DNA fragments that are spatially proximal in the nucleus.
- **Reversal of Cross-linking and DNA Purification:** Reverse cross-links by incubating with Proteinase K at 65°C. Purify the DNA via phenol-chloroform extraction and ethanol precipitation.
- **Library Preparation and Sequencing:** Prepare a sequencing library from the ligated DNA. This typically involves shearing the DNA, size selection, and adding sequencing adapters. Sequence on an Illumina platform (e.g., 150-200 million read pairs per library for mammalian genomes).
- **Data Processing:** Map sequenced reads to the reference genome using dedicated pipelines (e.g., Distiller). Process mapped reads with tools like `pairtools` and `cooler` to generate normalized, multi-resolution contact matrices.

Table 1: Impact of 3C Protocol Choices on Data Output [58]

Experimental Parameter	Effect on Compartment Detection	Effect on Loop Detection	Effect on Random Ligation
Fragmentation: HindIII	Stronger signal	Poorer resolution	Lower
Fragmentation: DpnII	Standard signal	Good resolution	Standard
Fragmentation: MNase	Weaker signal	Best resolution	Higher
Cross-link: FA only	Standard signal	Standard signal	Higher
Cross-link: FA+DSG	Stronger signal	Enhanced signal	Lower

A Case Study: Flower Morphogenesis inArabidopsis thaliana

A specific implementation of this validation framework was applied to the GRN governing flower morphogenesis in *Arabidopsis thaliana* [39].

The GRN Model

The network consists of 12 key genes (e.g., AP1, AG, LFY). A Boolean model was first established, defining the activation rules for each gene based on the state of its regulators [39]. This discrete model was then generalized into a continuous dynamical system using a framework of ordinary differential equations that describe the production rates of mRNAs and proteins for each node, as outlined in Section 2.1.

Table 2: Key Genes in the Arabidopsis thaliana *Flower Morphogenesis GRN [39]*

Node Number	Gene Name	Node Number	Gene Name
1	EMF1	7	UFO
2	TFL1	8	BFU
3	LFY	9	AG
4	AP1	10	AP3

Node Number	Gene Name	Node Number	Gene Name
5	CAL	11	PI
6	LUG	12	SUP

Implementation of the FPE and Co-expression Calculation

The Fokker-Planck equation associated with the continuous model was formulated. Given the high dimensionality of the system (12 genes), an analytical solution was infeasible. The authors employed a **gamma mixture model** to approximate the stationary solution *P_s(x)*, transforming the problem of solving the FPE into a tractable optimization problem [39]. From the resulting *P_s(x)*, the theoretical co-expression matrix was calculated.

Results and Validation

The simulated co-expression matrix derived from the FPE solution showed "good agreement" with the experimental co-expression matrix obtained from microarray data [39]. This successful comparison validated the continuous dynamical model and the inferred epigenetic landscape, demonstrating that the model could accurately recapture the correlational structure of gene expression observed in real flowers.

The Scientist's Toolkit: Essential Reagents and Materials

The following table lists key reagents and materials required for the experiments and analyses described in this guide.

Table 3: Research Reagent Solutions for GRN Model Validation

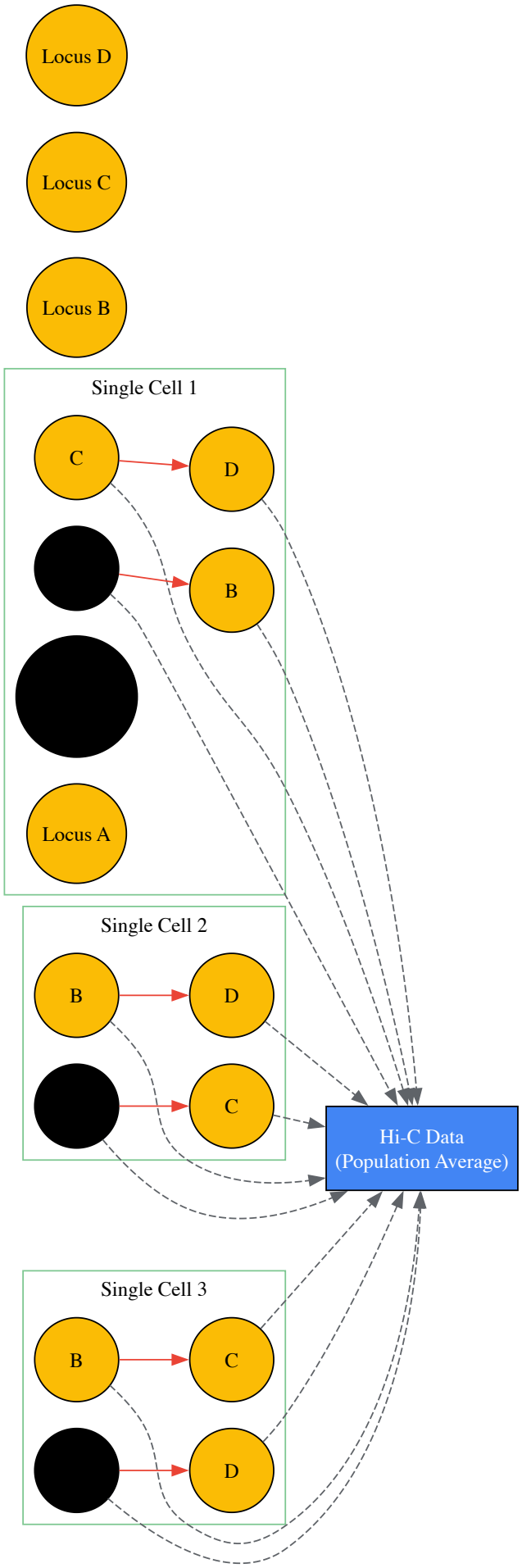
Item Name	Function/Application	Specific Examples / Notes
Formaldehyde (FA)	Cross-linking agent for 3C protocols. Fixes protein-DNA and DNA-DNA interactions.	Used at 1-2% concentration. A foundational cross-linker in most 3C methods [58].
Disuccinimidyl Glutarate (DSG)	Protein-protein cross-linker. Used in combination with FA to improve cross-linking efficiency.	3 mM DSG following 1% FA fixation reduces random ligation and enhances compartment signal in Hi-C [58].
Restriction Enzymes (DpnII, HindIII)	Chromatin fragmentation in Hi-C. Cuts DNA at specific sequences to generate fragments for ligation.	DpnII (4-cutter) provides higher resolution. HindIII (6-cutter) provides larger fragments, better for compartments [58].
Micrococcal Nuclease (MNase)	Chromatin fragmentation in Micro-C. Digests linker DNA, yielding mononucleosomal fragments.	Enables nucleosome-resolution interaction maps. Requires a size selection step post-digestion [58].
TRIzol Reagent	Monophasic solution of phenol and guanidine isothiocyanate for the simultaneous isolation of RNA, DNA, and proteins from biological samples.	Standard for total RNA extraction for microarray or RNA-seq co-expression analysis.
Gamma Mixture Model	A probabilistic model used to approximate the high-dimensional stationary solution of the Fokker-Planck equation.	Transforms the intractable problem of solving the FPE into an optimization problem, enabling landscape calculation [39].
BioTapestry / Jupyter Notebooks	Software for modeling and visualizing GRNs and for interactive computational analysis, respectively.	BioTapestry is used for GRN layout and visualization [38]. Jupyter Notebooks provide an environment for dynamical modeling and data analysis [38].

Visualizing Key Relationships and Workflows

The following diagrams illustrate the probabilistic nature of chromatin conformation data and the foundational methodological approaches for GRN inference.

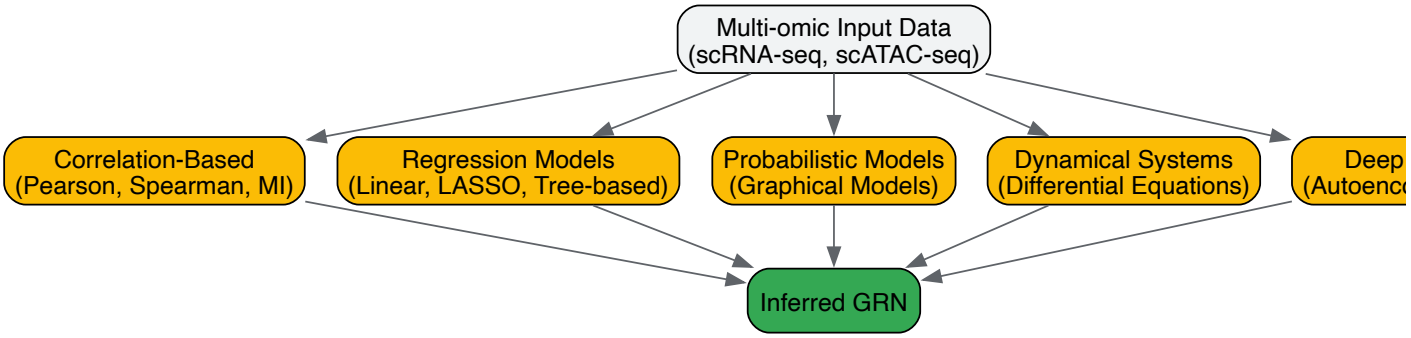
The Probabilistic Nature of Chromatin Conformation

Data from 3C-based methods like Hi-C represents a population average of many possible chromatin conformations within a cell population, not a single, static structure [59]. This concept is crucial for interpreting the data used to inform and validate models.



Methodological Foundations of GRN Inference

GRN inference methods employ diverse mathematical and statistical principles to reconstruct regulatory connections from data. Understanding these foundations is key to selecting the right tool for model building and validation [56].



Quantifying Gene Module Stability and Topological Change Across Cell States

The dynamics of gene regulatory networks (GRNs) across various cellular states are fundamental to understanding developmental biology, disease progression, and evolutionary processes. This technical guide synthesizes cutting-edge methodologies for quantifying stability and topological alterations in gene modules, providing researchers with robust frameworks for analyzing GRN architecture. We detail computational and mathematical approaches that move beyond traditional differential expression analysis to capture the intricate rewiring of regulatory interactions that underlie cellular transitions. By integrating these quantitative measures with evolutionary developmental biology principles, we enable deeper investigation into how topological constraints shape phenotypic possibilities and evolutionary trajectories.

Gene regulatory networks (GRNs) represent the complex circuits of interactions where transcription factors regulate target genes, ultimately controlling development, phenotypic plasticity, and evolutionary processes. In evolutionary developmental biology, understanding the topological organization of GRNs is crucial as it creates specific constraints that both channel and limit morphological evolution [60]. The hierarchical nature of GRN topology serves as a primary mechanism for establishing these evolutionary constraints while simultaneously facilitating phenotypic innovation within defined parameters [60].

Recent advances have revealed that physical genome organization, particularly through topologically associating domains (TADs), plays a crucial role in determining developmental possibilities and limitations [60]. The three-dimensional architecture of chromatin creates specific interaction domains that facilitate or prevent certain genetic regulatory interactions, embodying the physical constraints on development first proposed by Pere Alberch [60]. This intersection of developmental biology and differential topology provides a powerful framework for understanding vertebrate ontogenetic constraints and how the topological organization of GRNs influences developmental trajectories.

Quantitative Frameworks for Topological Analysis

Topological Feature Extraction from GRNs

The topological properties of GRNs provide crucial information about their organization and function. Research has consistently identified three primary topological features as most relevant for understanding GRN architecture: Knn (the average nearest neighbor degree), PageRank, and degree [61]. These features are conserved along evolution and remain relevant in pluripotent cells, suggesting their fundamental importance in network organization [61].

Knn (average nearest neighbor degree) measures the average degree of a node's neighbors, revealing whether highly-connected nodes tend to connect with other highly-connected nodes [61]. In GRNs, regulators typically exhibit low Knn values, meaning their targets have low connectivity, while targets often show high Knn values [61]. **PageRank** identifies nodes with high influence through a recursive algorithm that considers both the quantity and quality of connections [61]. **Degree** simply represents the number of connections a node has, with transcription factors often serving as hubs with high degree centrality [61].

These topological features are not randomly distributed but correlate with biological function. Life-essential subsystems are governed mainly by transcription factors with intermediate Knn and high PageRank or degree, whereas specialized subsystems are primarily regulated by transcription factors with low Knn [61]. This distribution suggests that high probability of transcription factors being accessed by random signals and high probability of signal propagation to target genes ensures the robustness of life-essential subsystems [61].

Table 1: Key Topological Features in Gene Regulatory Networks

Feature	Mathematical Definition	Biological Interpretation	Associated Subsystems
Knn (Average Nearest Neighbor	$K{nn}(i) = \frac{1}{Ni} \sum{j \in N(i)} kj$ where Ni is number of neighbors, kj is degree of	Measures connectivity pattern of a node's neighborhood; Low Knn indicates	Essential: Intermediate Knn; Specialized: Low

Feature	Mathematical Definition	Biological Interpretation	Associated Subsystems
Degree)	neighbor j	connections to low-degree nodes	Knn
PageRank	$PR(i) = \frac{1-d}{N} + d \sum_{j \in M(i)} \frac{PR(j)}{L(j)}$ where d is damping factor, M(i) are neighbors linking to i, L(j) is outbound links	Recursive measure of node influence based on quantity and quality of connections	Essential subsystems (high PageRank)
Degree	$ki = \sum{j} A_{ij}$ where A is adjacency matrix	Number of direct connections; identifies network hubs	Essential subsystems (high degree)

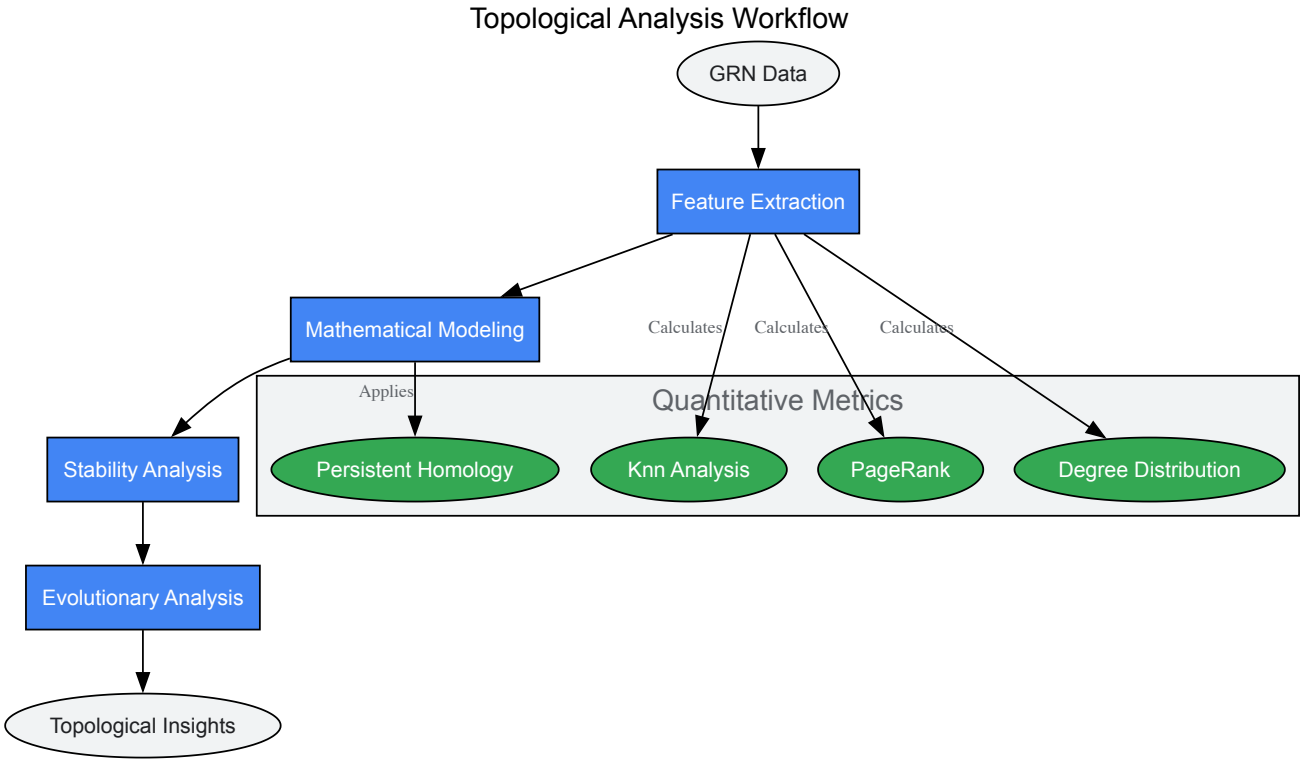
Mathematical Framework for Topological Analysis

A rigorous mathematical framework enables comprehensive analysis of GRN topology. Let X be the set of all possible gene expression states, where each state is represented as a point in R^n, with n being the number of genes in the network [60]. We equip X with the standard topology $τ$ induced by the Euclidean metric: $$τ = {U ⊆ X | U is open in the standard topology of R^n}$$

The gene regulatory network can be represented as a continuous mapping $f: X → X$, defined component-wise by: $$f(x1, x2, ..., xn) = (f1(x), f2(x), ..., fn(x))$$ where $x = (x1, x2, ..., xn) ∈ X$, and each $fi: X → R$ models the regulation of gene i based on the entire system state x [60].

Differential topology provides powerful tools for analyzing expression landscapes. We introduce a smooth scalar function $E: X → R$ representing the potential energy landscape of the gene expression system [60]. The critical points of E correspond to stable or unstable gene expression states, with system dynamics described by the gradient flow: $$\frac{dx}{dt} = -∇E(x)$$ where $∇E(x)$ is the gradient of E at point x [60].

Using Morse theory, we analyze the topology of X by studying the critical points of E and their indices. For a critical point $p ∈ X$, the Morse index $μ(p)$ is defined as the number of negative eigenvalues of the Hessian matrix $HE(p)$ at p: $$μ(p) = number of negative eigenvalues of HE(p)$$ This index equals the dimension of the unstable manifold $W^u(p)$ at p [60].



Persistent Homology for Multi-Scale Analysis

Persistent homology provides a powerful method for analyzing the hierarchical and multi-scale structure of regulatory networks [60]. Consider a filtration of simplicial complexes: $$∅ = K0 ⊆ K1 ⊆ ··· ⊆ KN = K$$ where Ki is a simplicial complex corresponding to the network at a certain threshold or scale [60].

The p-th persistence diagram $Dgmp(f)$ is defined as a multiset of points (bi, di) in R^2, where bi and d_i represent the birth and death times of p-dimensional homological features during the filtration [60]. This approach allows researchers to identify topological features that persist across multiple scales, distinguishing robust network properties from transient ones.

Methodologies for Quantifying Gene Module Stability

Gene Homeostasis Z-Index

Gene expression stability within a homeostatic cell can be quantified through the gene homeostasis Z-index, a measure that highlights genes under active regulation in response to internal and external stimuli [62]. Traditional measures of gene expression variability, such as variance or coefficient of variation, often fail to distinguish between genes with widespread variability across cells and genes whose variability is attributable to a small proportion of cells with sharply upregulated expression [62].

The Z-index addresses this limitation through a "k-proportion" statistic, which represents the percentage of cells with expression levels below an integer value k determined by the mean gene expression count [62]. Under equivalent mean expressions, regulatory genes display significantly higher k-proportions compared to their homeostatic counterparts because extreme expression levels in a limited subset of cells skew the mean expression upward [62].

The relationship between k-proportion and mean expression can be visualized on a "wave plot," where regulatory genes surface as anomalies appearing as distinct outliers above the general trend [62]. The k-proportion inflation test compares the observed k-proportion with the expected value from a set of negative binomial distributions with a shared dispersion parameter, yielding a Z-score for each gene (the gene homeostasis Z-index) that serves as a relative measure of gene expression instability [62].

Table 2: Stability and Variability Metrics Comparison

Metric	Calculation Method	Application Context	Advantages	Limitations
Gene Homeostasis Z-index	k-proportion inflation test against negative binomial distribution	Identifying genes under active regulation in cell subsets	Detects regulatory heterogeneity; robust to noise	Requires single-cell resolution data
scran	Decomposes variance into biological and technical components	Single-cell RNA-seq variability analysis	Effective for capturing cell-to-cell variability	Less sensitive to subset-specific regulation
Seurat VST	Variance stabilizing transformation	Single-cell RNA-seq feature selection	Stabilizes variance across expression levels	May miss genes with subset-specific expression
Persistent Entropy	$E(Dgm) = -\sum_i p_i \log p_i$ *where* $p_i = (d_i - b_i)/T$	Quantifying significance of topological features in networks	Multi-scale analysis; robust to noise	Computationally intensive for large networks

Benchmarking Stability Metrics

Simulation studies demonstrate that the Z-index exhibits competitive or superior performance across a range of conditions, particularly in scenarios with higher noise levels or larger proportions of cells with upregulated genes [62]. When the percentage of cells with upregulated genes increases, the Z-index maintains stable performance while other methods degrade or shift, suggesting better resilience against increasing biases [62].

In practical applications, the Z-index successfully identifies genes with significant regulatory patterns that traditional methods overlook. For example, in CD34+ cells, the Z-index revealed organ-specific patterns such as heightened synaptic transmission activities in islets and regulatory patterns for neuropeptides like insulin and somatostatin, which exhibit extreme values within a limited number of cells [62].

Comparative Analysis of Gene Modules Across Cell States

Gene2role: Role-Based Gene Embedding

Gene2role presents a novel gene embedding approach that leverages multi-hop topological information from genes within signed GRNs [63]. This method addresses limitations of current comparative analytical approaches that focus on simple topological information such as the degree of genes, which are limited in their ability to fully capture the similarities and differences among complex GRNs [63].

The Gene2role framework enables identification of genes with significant topological changes across cell types or states, offering a fresh perspective beyond traditional differential gene expression analyses [63]. Additionally, it quantifies the stability of gene modules between two cellular states by measuring changes in gene embeddings within these modules, providing a powerful approach for probing the dynamic regulatory landscape [63].

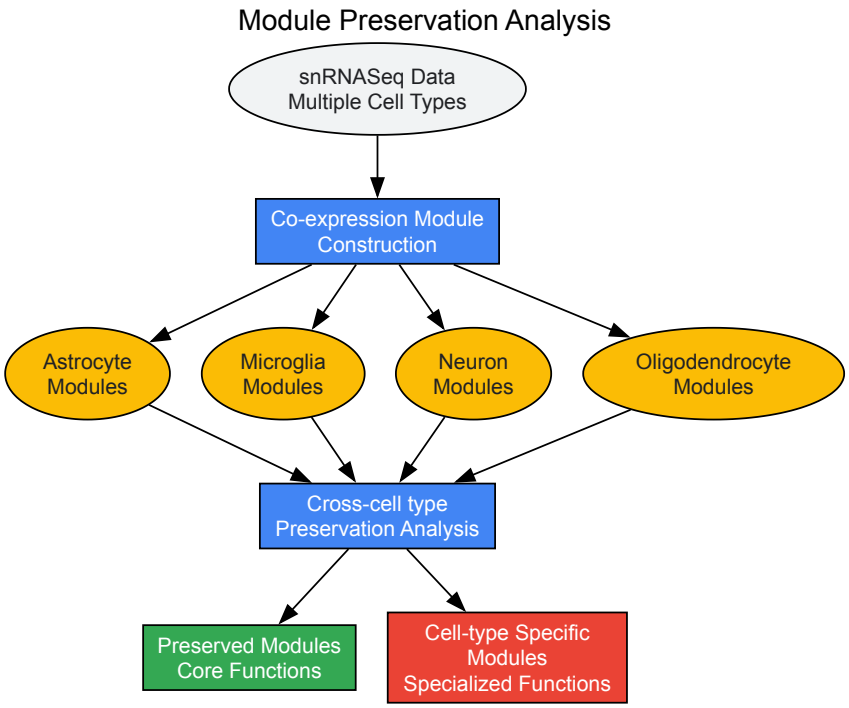
Single-Nucleus Coexpression Module Analysis

The module-trait network (MTN) approach enables finding and modeling cell-specific gene regulatory networks by constructing single-nucleus coexpression modules for each major cell type [64]. This method involves: (1) constructing coexpression modules; (2) identifying groups of

coexpressed genes that represent molecular systems; and (3) modeling directional relationships between modules and traits using Bayesian networks [64].

Application of this approach to snRNASeq data from human dorsolateral prefrontal cortex tissues revealed 193 modules with at least 30 genes each across major cell types, comprising 26 modules for astrocytes, 26 for endothelial cells, 29 for excitatory neurons, 24 for inhibitory neurons, 30 for microglial cells, 30 for oligodendrocytes, and 28 for oligodendrocyte precursor cells [64]. These coexpression networks followed a scale-free topology where a few hub genes present a large number of connections, as expected in real-world biological data [64].

Comparative analysis showed that single-nucleus modules exhibited distinct communities with altered connectivity compared to bulk RNASeq, suggesting cell-specific gene co-regulation [64]. A total of 56 modules were not preserved in bulk RNASeq, with numbers ranging from 6 to 11 specific modules depending on the cell type population [64]. For example, nine microglia modules and 11 excitatory neuron modules were not preserved in bulk RNASeq, highlighting the importance of single-cell resolution for capturing cell-type-specific regulation [64].



Evolutionary Perspectives on Module Organization

Analysis of the relationship between evolutionary history and topological properties reveals that slowly evolving ("cold"), old genes tend to interact with each other, as do rapidly evolving ("hot"), young genes [65]. This natural segregation creates community structures with relatively homogeneous evolutionary histories [65].

Gene duplication appears to have placed old, cold genes and communities at the center of networks, and young, hot genes and communities at the periphery [65]. This organization can be quantified through single-node centrality measures and two specialized measures: the set efficiency and interset efficiency, which provide new perspectives for understanding evolutionary genetics [65].

Table 3: Gene Module Classification by Evolutionary Properties

Module Type	Evolutionary Rate	Evolutionary Age	Network Position	Functional Characteristics
Cold Modules	Slow evolution	Old genes	Network center	Essential cellular functions; high connectivity
Hot Modules	Rapid evolution	Young genes	Network periphery	Specialized functions; environmental response
Mixed Modules	Variable	Mixed ages	Intermediate positions	Integrated functions bridging core and periphery

Experimental Protocols and Methodologies

Protocol 1: Gene2role Analysis for Comparative GRN Studies

Objective: To identify genes with significant topological changes across cell types or states using role-based gene embedding [63].

Materials:

- Signed gene regulatory networks from at least two cellular states
- Gene expression data corresponding to each state
- Computational environment with Gene2role implementation

Procedure:

- **Network Preparation**: Compile signed GRNs for each cellular state of interest, ensuring consistent node (gene) identification across networks.
- **Multi-hop Feature Extraction**: For each gene in each network, extract topological features that capture information beyond immediate neighbors, including:
  - Signed walk profiles up to 3-4 hops
  - Role-based similarity metrics
- **Embedding Generation**: Generate gene embeddings for each network using the Gene2role algorithm, which maps genes to a low-dimensional space based on their topological roles.
- **Cross-state Comparison**: Align embeddings across cellular states and calculate distance metrics between each gene's representations.
- **Change Quantification**: Identify genes with significant embedding changes, indicating altered topological roles between states.
- **Module Stability Assessment**: For predefined gene modules, calculate aggregate embedding changes to quantify module stability/instability.

Validation: Apply to GRNs inferred from four distinct data sources to demonstrate effectiveness in capturing intricate topological nuances [63].

Protocol 2: Single-Nucleus Module-Trait Network Analysis

Objective: To construct cell-type-specific coexpression modules and model their relationships with phenotypic traits [64].

Materials:

- Single-nucleus RNA sequencing data from relevant tissues
- Phenotypic trait data for same samples
- High-performance computing resources for large-scale network analysis

Procedure:

- **Data Preprocessing**:
  - Quality control and normalization of snRNASeq data
  - Cell type annotation using established markers
  - Creation of participant-level normalized pseudo-bulk matrices for each cell type

- **Coexpression Module Construction**:

  - Apply Speakeasy community detection algorithm to identify modules of co-regulated genes
  - Filter modules to retain those with at least 30 genes
  - Validate scale-free topology property

- **Module Characterization**:

  - Functional enrichment analysis (Gene Ontology, pathways)
  - Cell-type specificity assessment using Human Protein Atlas
  - Identification of hub genes within each module

- **Cross-Module Comparison**:

  - Calculate module preservation statistics between cell types
  - Compute normalized mutual information between modules
  - Identify cell-type-specific versus conserved modules

- **Trait Association**:

  - Associate modules with relevant phenotypic traits
  - Construct Bayesian networks to model directional relationships
  - Identify key driver modules for traits of interest

Application Note: This protocol was successfully applied to snRNASeq data from 424 participants in the ROSMAP study, identifying astrocytic module 19 (ast_M19) as associated with cognitive decline through a subpopulation of stress-response cells [64].

Table 4: Research Reagent Solutions for Gene Module Analysis

Reagent/Resource	Function	Application Context	Example Sources
Single-nucleus RNA-seq Platforms	High-resolution transcriptomic profiling	Cell-type-specific module identification	10X Genomics, Parse Biosciences
Speakeasy Algorithm	Community detection in large networks	Coexpression module construction	Available as software package [64]
Gene2role Implementation	Role-based gene embedding	Comparative topological analysis across cell states	Available as computational method [63]
Hi-C Chromatin Conformation	3D genome architecture mapping	Topologically Associating Domain (TAD) identification	Arima Genomics, Dovetail Genomics
Persistent Homology Software	Multi-scale topological analysis	Network stability quantification	JavaPlex, GUDHI, R-TDA
Bayesian Network Tools	Directional relationship modeling	Module-trait network construction	BNLearn, WinMine, Bayesialab

The quantitative frameworks and methodologies presented in this guide enable rigorous analysis of gene module stability and topological changes across cellular states. By integrating concepts from differential topology, network science, and evolutionary biology, researchers can now move beyond traditional gene-centric analyses to understand how the rewiring of regulatory networks underlies developmental and disease processes.

The emerging picture suggests that topological constraints play crucial roles in both enabling and limiting evolutionary possibilities. The hierarchical organization of GRNs creates different levels of developmental constraint, with core regulatory circuits showing remarkable conservation across vertebrates while allowing for specialized adaptations at the periphery [60]. As single-cell technologies continue to advance and computational methods become more sophisticated, we anticipate increasingly precise quantification of these topological principles across diverse biological contexts.

Future developments will likely focus on dynamic network modeling that captures temporal transitions between cellular states, integration of multi-omic data sources for more comprehensive network reconstruction, and application of these principles to therapeutic development through targeted network interventions. The tools and protocols outlined here provide a foundation for these advances, empowering researchers to decode the topological language of gene regulatory networks.

Identifying Differentially Topological Genes (DTGs) Beyond Expression Analysis

In the field of evolutionary developmental biology, understanding the molecular drivers of cellular identity and phenotypic variation has traditionally relied on differential gene expression (DGE) analysis. However, genes function within complex interconnected systems where their **regulatory roles** extend beyond expression levels to include their position and influence within gene regulatory networks (GRNs). This technical guide introduces the concept of Differentially Topological Genes (DTGs)—genes whose **structural roles** within GRNs change significantly across cellular states, such as between species, during development, or in disease progression. We present Gene2role, a novel **role-based embedding method** that leverages multi-hop topological information to identify DTGs, providing researchers with a sophisticated toolkit for probing the dynamic regulatory landscape underlying evolutionary and developmental processes.

Gene regulatory networks represent the complex wiring diagrams of cellular regulation, comprising genes as nodes and their regulatory interactions as edges. While traditional DGE analysis identifies genes with significant expression changes, it fails to capture genes that may maintain constant expression levels but undergo critical **functional shifts** in their network connectivity. These topological changes can alter a gene's regulatory influence without corresponding expression changes, representing a hidden layer of regulatory evolution.

The limitation of conventional approaches stems from their focus on simple topological metrics or proximity-based embeddings, which cannot capture the nuanced structural roles genes play across multi-hop neighborhoods in signed GRNs (where edges denote activation or inhibition) [66] [35]. Role-based network embedding methods address this limitation by projecting genes from separate networks into a unified space based on structural similarity, enabling direct comparison of genes across different cellular states or evolutionary contexts [66] [35].

Theoretical Foundation: From Differential Expression to Differential Topology

Conceptual Framework of Differentially Topological Genes

DTGs are defined as genes that exhibit statistically significant changes in their topological properties within GRNs across different biological conditions. Unlike differentially expressed genes, which show quantitative changes in transcript abundance, DTGs may undergo **qualitative changes** in their network relationships, potentially representing:

- **Hub transitions**: Genes that gain or lose central positioning in regulatory hierarchies
- **Role shifts**: Changes in a gene's structural equivalence within network motifs
- **Pathway rewiring**: Alterations in a gene's connectivity patterns within functional modules
- **Bottleneck changes**: Modifications in a gene's importance as a connector between network modules

These topological alterations can drive significant phenotypic changes in evolution and development without necessarily altering gene expression levels, representing an important mechanism for evolutionary innovation and developmental plasticity.

Mathematical Representation of Gene Topology

In signed GRNs, topological representation begins with encoding both the connectivity and regulatory nature of interactions. For a signed GRN represented as $G = (V, E^{+}, E^{-})$, where V represents genes, and E^{+} and E^{-} represent activating and repressing interactions, respectively, we define the **signed-degree vector** for topological representation [66] [35]:

$$\textbf{d} = [d^{+}, d^{-}]$$

where d^{+} and d^{-} represent positive and negative degrees, respectively. This two-dimensional representation captures both the connectivity and regulatory character of each gene, providing a foundation for more sophisticated topological comparisons.

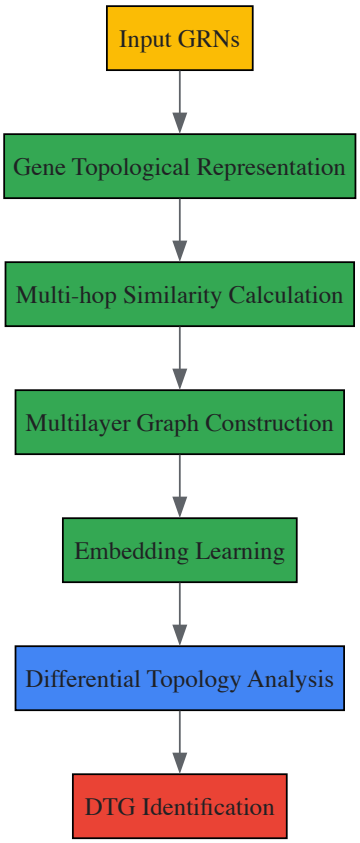
Table 1: Key Mathematical Formulations for Gene Topology Analysis

Concept	Mathematical Representation	Biological Interpretation
Signed-degree vector	$\textbf{d} = [d^{+}, d^{-}]$	Captures both connectivity and regulatory nature of a gene
Exponential Biased Euclidean Distance (EBED)	$\text{EBED}(\textbf{d}_u, \textbf{d}_v) = \text{exp}\left(\sqrt{\left(\log\frac{d_u^{+}+1}{d_v^{+}+1}\right)^2 + \left(\log\frac{d_u^{-}+1}{d_v^{-}+1}\right)^2}\right)$	Measures topological similarity adjusted for scale-free network properties
k-hop neighborhood	$R_k(u)$ = sorted degree sequence of genes k hops from u	Extends topological analysis beyond immediate connections
Multi-layer similarity	$f_k(u,v) = f_{k-1}(u,v) + D_k(u,v)$	Recursively computes topological similarity across network depths

Methodological Approach: The Gene2role Framework

Core Algorithm and Workflow

Gene2role implements a sophisticated pipeline for identifying DTGs through role-based graph embedding. The method adapts and extends frameworks from struc2vec and SignedS2V to handle the specific challenges of signed GRNs [66] [35]. The overall workflow comprises three major components: network construction, embedding generation, and downstream analysis for DTG identification.



Graph 1: Gene2role workflow for DTG identification. The process transforms raw GRNs into topological embeddings enabling comparative analysis across cellular states.

Technical Implementation

Gene Topological Similarity Calculation

The Exponential Biased Euclidean Distance (EBED) function addresses the scale-free nature of GRNs, where degree distributions often follow power-law distributions [66] [35]. The EBED applies logarithmic transformation to mitigate distribution effects, computes Euclidean distance, then applies an exponential function to preserve original distance proportionality:

[D{o}(u,v) = \text{EBED}(\textbf{d}u, \textbf{d}v) = \text{exp}\left(\sqrt{\left(\log\frac{du^{+}+1}{dv^{+}+1}\right)^2 + \left(\log\frac{du^{-}+1}{d_v^{-}+1}\right)^2}\right)]

For capturing higher-order topological relationships, Gene2role defines (Rk(u)) as the sorted sequence of degrees for genes that are k hops away from gene u, employing dynamic time warping (DTW) to calculate k-hop distance ((Dk, k>0)) between genes u and v [66] [35].

Multilayer Graph Construction and Embedding

Gene2role constructs a multilayer weighted graph that encodes topological similarities between genes across different neighborhood radii. Each layer contains all genes, with weights in the k-th layer between genes u and v computed as [66] [35]:

[wk(u,v) = e^{-fk(u,v)}]

where (f_k(u,v)) represents the k-hop topological similarity. This multilayer graph enables the use of random walk-based approaches to generate gene embeddings that capture structural similarities at multiple scales, projecting genes from different networks into a unified embedding space for comparative analysis.

Experimental Protocol for DTG Identification

Data Preparation and GRN Construction

- **Network Sources**: GRNs can be constructed from diverse data sources depending on research goals and data availability:
    - **Manually curated networks**: Literature-derived networks with validated regulatory information (e.g., HSC, mCAD, VSC, GSD from BEELINE) [66] [35]

- **Single-cell RNA-seq data**: Cell type-specific GRNs inferred using tools like EEISP or Spearman correlation from scRNA-seq data [66] [35]
- **Single-cell multi-omics data**: Integrated GRNs from platforms like CellOracle combining scATAC-seq and scRNA-seq data [66] [35]

- **Network Processing**:
  - Filter connections based on statistical significance (e.g., p-value < 0.01)
  - Retain top edges based on absolute coefficient values (e.g., top 2000 edges)
  - Establish edge signs based on regulatory coefficient values

Embedding Generation and DTG Analysis

- **Parameter Configuration**:
  - Set embedding dimensions based on network complexity (typically 128-256 dimensions)
  - Define random walk parameters (length: 80-100, number: 10-20 per node)
  - Specify neighborhood scope (k-hop, typically 2-4 layers)

- **Differential Topology Calculation**:
  - Project genes from different cellular states into unified embedding space
  - Calculate Euclidean distances between matched genes across states
  - Apply statistical testing (e.g., permutation testing) to identify significant topological changes
  - Adjust for multiple testing using Benjamini-Hochberg FDR control

Table 2: Research Reagent Solutions for DTG Analysis

Resource Category	Specific Tools/Databases	Function in DTG Analysis
GRN Construction	EEISP [66], CellOracle [66] [35], LINGER [25]	Infer regulatory networks from expression and accessibility data
Benchmark Networks	BEELINE [66] [35]	Provide standardized GRNs for method validation
Expression Data Sources	GEO [67], SCAR [68]	Supply raw data for network inference
Embedding Algorithms	struc2vec [66] [35], SignedS2V [66] [35]	Generate role-based node embeddings
Validation Resources	ChIP-seq data [25], eQTL datasets [25]	Provide experimental validation of regulatory predictions

Validation and Interpretation Framework

Biological Validation of DTGs

Validating the biological significance of identified DTGs requires multiple approaches:

- **Experimental Validation**:
  - Chromatin immunoprecipitation sequencing (ChIP-seq) to confirm predicted changes in TF binding [25]
  - Expression quantitative trait loci (eQTL) consistency checks for cis-regulatory predictions [25]
  - Functional perturbation experiments (CRISPR, RNAi) to validate predicted functional importance

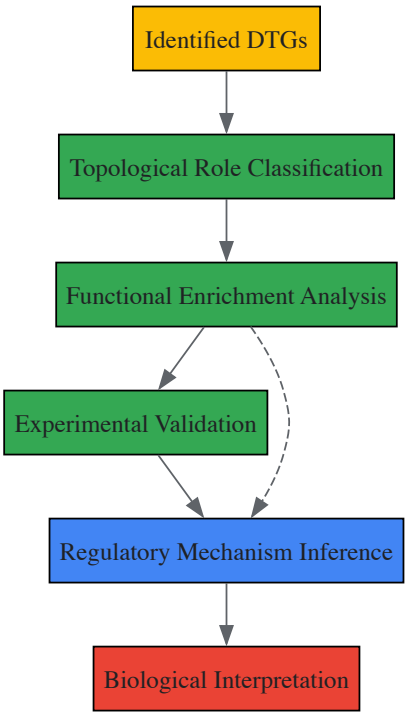
- **Computational Validation**:
  - Enrichment analysis of DTGs in known functional pathways and processes
  - Comparison with established gold standards (e.g., RegNet, TRRUST)
  - Cross-validation with independent datasets and network inference methods

Interpretation of DTG Functional Significance

Interpreting DTGs requires moving beyond identification to understanding their biological implications:

- **Role Transition Analysis**: Classify DTGs based on the nature of their topological changes:
  - **Hub emergence/gain**: Genes acquiring central network positions
  - **Role specialization**: Genes developing more specific structural positions
  - **Module switching**: Genes changing their primary module affiliation
  - **Regulatory sign flipping**: Genes changing predominant regulatory influence

- **Functional Integration**: Relate topological changes to known biological processes through enrichment analysis of DTG sets in:
    - Gene Ontology biological processes
    - Kyoto Encyclopedia of Genes and Genomes pathways
    - Disease association databases
    - Evolutionary conservation metrics



Graph 2: DTG interpretation framework. The process extends from identification to biological mechanism inference through classification and validation.

Applications in Evolutionary Developmental Biology

Comparative Analysis Across Species

The DTG framework enables systematic investigation of regulatory network evolution by identifying genes that have undergone significant topological changes between species. Applications include:

- **Developmental Program Evolution**: Identify regulatory genes that have changed network roles in the evolution of novel developmental pathways
- **Constraint and Plasticity Analysis**: Detect topological conservation in core developmental processes versus diversification in adaptive traits
- **Cell Type Evolution**: Trace the evolutionary trajectory of cell type specification by analyzing topological changes in cell type-specific GRNs

Analysis of Developmental Processes

During ontogeny, cells undergo precisely orchestrated changes in gene regulation. DTG analysis can reveal:

- **Lineage Commitment Decisions**: Identify topological changes in master regulators at developmental branch points
- **Morphogenetic Transitions**: Detect genes undergoing role transitions during tissue patterning and organogenesis
- **Stem Cell Differentiation**: Track topological progression of regulatory genes during differentiation trajectories

Integration with Disease Mechanisms

DTG analysis provides novel insights into pathological processes:

- **Disease Initiation and Progression**: Identify early topological changes preceding phenotypic manifestation
- **Therapeutic Target Discovery**: Detect structurally critical genes that may not show expression changes
- **Network Medicine Applications**: Map disease-associated genetic variants onto DTGs to interpret mechanistic consequences

Comparative Analysis with Traditional Methods

Advantages Over Differential Expression Analysis

While DGE analysis focuses exclusively on expression level changes, DTG analysis provides complementary information that often reveals different aspects of gene regulation:

Table 3: Comparison of DGE and DTG Analytical Approaches

Analytical Dimension	Differential Expression Analysis	Differential Topology Analysis
Primary Focus	Changes in gene expression levels	Changes in network position and connectivity
Detection Capability	Genes with significant expression differences	Genes with structural role changes regardless of expression
Regulatory Insight	Identifies regulated genes	Identifies regulatory influencers and hub transitions
Methodological Tools	DESeq2 [69], edgeR [69], limma [69]	Gene2role [66] [35], LINGER [25]
Biological Interpretation	Direct functional consequences via abundance	Indirect effects through altered regulatory influence
Technical Requirements	Expression data (RNA-seq, microarrays)	Inferred or experimentally derived GRNs

Integration with Multi-Omics Approaches

The most comprehensive understanding of regulatory dynamics emerges from integrating DTG and DGE analyses:

- **Complementary Gene Sets**: DTGs and differentially expressed genes often represent distinct gene sets with partial overlap, together providing a more complete picture of regulatory changes
- **Hierarchical Regulation**: DTG analysis can identify master regulators whose topological changes drive expression changes in downstream genes
- **Multi-scale Modeling**: Combine topological and expression information for predictive models of cellular behavior and fate decisions

Future Directions and Technical Challenges

Methodological Advancements

Several frontiers promise to extend DTG analysis capabilities:

- **Temporal Network Analysis**: Developing methods to track topological changes across continuous developmental or evolutionary time series
- **Single-Cell Topology Inference**: Creating approaches to infer cell-to-cell variation in gene regulatory topology from single-cell multi-omics data
- **Integrated Machine Learning**: Incorporating topological features into deep learning models for enhanced prediction of phenotypic outcomes
- **Cross-Species Alignment**: Improving methods for comparative topological analysis across divergent species

Technical Considerations and Limitations

Current challenges in DTG analysis include:

- **Network Quality Dependence**: The accuracy of DTG identification is constrained by the quality of input GRNs
- **Computational Complexity**: Role-based embedding methods require substantial computational resources for large networks
- **Validation Bottlenecks**: Experimental validation of predicted topological changes remains technically challenging
- **Context Specificity**: GRNs and thus topological roles are condition-specific, requiring careful experimental design

The identification of Differentially Topological Genes represents a paradigm shift in how we conceptualize and analyze gene regulatory changes in evolution and development. By moving beyond expression-level analysis to examine changes in network architecture, researchers can discover previously hidden regulatory changes that drive phenotypic variation. The Gene2role framework provides a robust methodological foundation for DTG identification, enabling researchers to probe the dynamic regulatory landscape with unprecedented resolution. As network biology continues to mature, integrating topological perspectives with traditional molecular profiling promises to unlock deeper insights into the evolutionary developmental biology of complex traits and diseases.

The growing understanding of **gene regulatory networks (GRNs)** provides a mechanistic foundation for interpreting how specific genetic alterations disrupt normal development and cause disease. In evolutionary developmental biology, GRNs are defined as the functional circuitry of regulatory genes and their interactions that direct developmental processes. The new "Plausible Mechanism Pathway," announced by the U.S. Food and Drug Administration (FDA) in November 2025, represents a transformative regulatory framework that directly applies this mechanistic understanding [70] [71]. Proposed by FDA Commissioner Martin Makary and Chief Medical and Scientific Officer Vinay Prasad, this pathway

addresses a critical challenge in modern therapeutics: how to approve **bespoke therapies** for ultra-rare diseases when traditional randomized clinical trials are not feasible [70] [72] [73].

This pathway marks a significant shift from symptom-based to **mechanism-based regulation**, prioritizing biologic plausibility over population-level evidence [71]. It is particularly relevant for therapies targeting rare genetic diseases where the molecular causality is precisely understood, often involving disruptions in well-characterized GRNs. The framework acknowledges that for conditions with numerous causative mutations, each potentially requiring a customized therapeutic approach, traditional development pathways are economically and practically unsustainable [72] [71].

Core Elements of the Plausible Mechanism Pathway

The Plausible Mechanism Pathway establishes five rigorous criteria that therapies must meet to qualify for this regulatory approach. These elements create a structured framework for evaluating bespoke treatments while maintaining scientific rigor.

Table 1: Core Eligibility Elements for the Plausible Mechanism Pathway

Element	Description	GRN Research Correlation
Defined Molecular Causality	Identification of specific molecular or cellular abnormality, not broad diagnostic criteria [74] [72]	GRN models identify key regulatory nodes whose disruption causes phenotypic consequences [13] [12]
Targeted Therapeutic Mechanism	Product acts directly on underlying or proximate biological alterations [74] [73]	Therapeutics designed to specifically correct dysregulated network nodes
Established Natural History	Well-characterized disease progression in untreated populations [74] [72]	Provides baseline against which network-level interventions are measured
Evidence of Target Engagement	Confirmation that the target was successfully drugged or edited [74] [73]	Demonstration of successful intervention at the specific network node
Observable Clinical Improvement	Consistent clinical improvement excluding regression to the mean [74] [72]	Restoration of normal network function manifesting as phenotypic improvement

The pathway operationalizes these elements through a structured approval process. After demonstrating success in **several consecutive patients** with different bespoke therapies, FDA may move toward granting marketing authorization [70] [74]. Manufacturers can then leverage **platform data** from these personalized products to support approvals for similar therapies targeting additional genetic conditions [74] [71]. This approach is conceptually similar to how GRN research identifies conserved regulatory motifs that function across different developmental contexts [13] [12].

The case of "Baby K.J." exemplifies this pathway in practice. This newborn with a rare liver condition received a customized base-editing therapy developed within seven months, with the FDA processing a single-patient expanded access application in approximately one week [70] [74] [73]. The rapid intervention was possible because the **precise genetic defect** was understood, the therapy directly targeted this abnormality, and the natural history of the condition provided a clear baseline against which to measure outcomes [74].

Comparison with Existing Regulatory Pathways

The Plausible Mechanism Pathway differs fundamentally from existing expedited programs, though it may interface with them. Unlike the Breakthrough Therapy designation, which accelerates development and review of drugs for serious conditions, the Plausible Mechanism Pathway does not include a formal designation process [74] [72]. It also differs from the Accelerated Approval pathway, which requires post-approval confirmatory trials, though products approved under the Plausible Mechanism Pathway may utilize either accelerated or traditional approval pathways [72] [71].

Table 2: Comparison of Expedited Regulatory Pathways

Pathway	Evidence Standard	Post-Marketing Requirements	Applicability
Plausible Mechanism	Mechanistic plausibility + clinical improvement in sequential patients [74] [72]	RWE collection for efficacy preservation, off-target effects, long-term safety [74] [72]	Ultra-rare diseases with known biologic cause [70] [71]
Accelerated Approval	Effect on surrogate endpoint reasonably likely to predict clinical benefit [72]	Confirmatory trials to verify clinical benefit [72]	Serious conditions with unmet need [72]
Breakthrough Therapy	Preliminary clinical evidence substantial improvement over available therapies [72]	Standard post-market monitoring	Serious conditions [72]

Pathway	Evidence Standard	Post-Marketing Requirements	Applicability
Rare Disease Evidence Principles (RDEP)	Single adequate and well-controlled trial with confirmatory evidence [72]	Standard post-market monitoring	Rare diseases (<1,000 U.S. patients) with known genetic defect [72]

The Plausible Mechanism Pathway incorporates substantial **post-market evidence generation** requirements. Sponsors must collect real-world evidence (RWE) to demonstrate continued preservation of efficacy, monitor for off-target effects (particularly relevant for gene-editing therapies), study the effect of early treatment on childhood development, and detect unexpected safety signals [74] [72] [73]. This approach aligns with the understanding that for bespoke therapies, evidence accumulation will necessarily extend beyond pre-market evaluation.

Integration with Gene Regulatory Network Research

The Plausible Mechanism Pathway shares fundamental principles with contemporary GRN research, particularly in its emphasis on understanding system-level interventions. In evolutionary biology, GRN models explain how phenotypic plasticity emerges from regulatory architecture [13] [12]. Similarly, the pathway requires understanding how therapeutic interventions alter disease trajectories by targeting specific network nodes.

Recent GRN research provides methodological frameworks relevant to the pathway's implementation. Studies of plasticity in model organisms like *Pristionchus pacificus* have identified environmentally responsive "switch genes" within GRNs that determine phenotypic outcomes [12]. This mirrors the pathway's requirement for targeting proximate biological alterations. The network-based approach to understanding dispersal evolution [13] offers computational frameworks for predicting how interventions might propagate through biological systems.

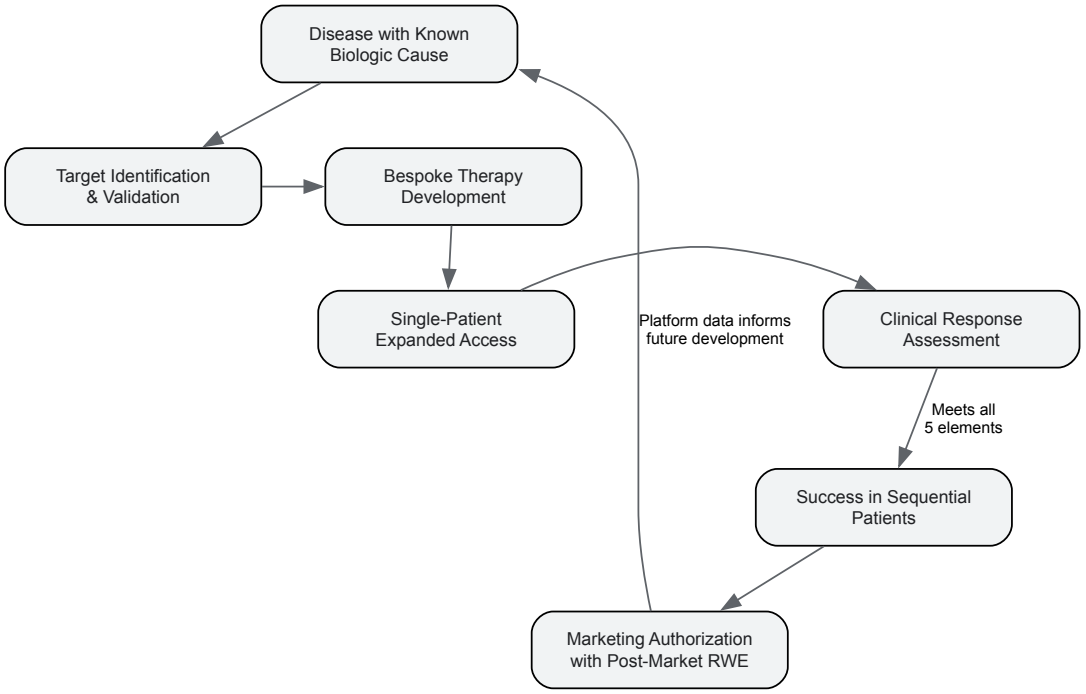


Diagram 1: The Plausible Mechanism Pathway workflow illustrates the sequential process from disease characterization to marketing authorization, highlighting the iterative nature of evidence generation.

The pathway's requirement for "confirmation that the target was successfully drugged or edited" [74] [72] parallels experimental approaches in GRN research that validate network perturbations. For example, in studying density-dependent dispersal evolution, researchers use gene-regulatory network models to confirm that theoretical inputs produce expected phenotypic outputs [13]. Similarly, developmental transcriptomics in *Pristionchus* establishes causal relationships between environmental cues, gene expression changes, and mouth-form plasticity [12].

Experimental Protocols and Methodological Framework

Natural History Characterization Protocol

Objective: Establish comprehensive baseline of disease progression in untreated populations.

Methodology:

- **Retrospective data collection**: Aggregate historical clinical data from patient registries, medical records, and published literature [74] [72]

- **Prospective observation**: For ultra-rare diseases without established natural history, implement structured monitoring of untreated patients using standardized assessment protocols [72]
- **Endpoint definition**: Identify and validate clinically meaningful endpoints that reflect disease-specific functional impairments [72]
- **Statistical modeling**: Develop disease progression models that account for variability in expression and progression rates [72]

GRN Correlation: This protocol mirrors approaches in evolutionary developmental biology where wild-type developmental trajectories are characterized before analyzing mutants [13] [12].

Target Engagement Validation Protocol

Objective: Confirm that the therapeutic intervention successfully modulates its intended molecular target.

Methodology:

- **Molecular confirmation**: When clinically feasible and appropriate, obtain tissue biopsies for direct assessment of target engagement [74] [72]
- **Biomarker development**: Identify and validate surrogate biomarkers that correlate with target modulation [72]
- **Functional assays**: Implement ex vivo or in vitro functional assays using patient-derived cells or tissues [74]
- **Imaging approaches**: When applicable, utilize molecular imaging techniques to visualize target engagement non-invasively [72]

GRN Correlation: Similar to validation experiments in GRN research where chromatin immunoprecipitation or reporter assays confirm transcription factor binding and regulatory effects [12] [14].

Clinical Outcome Assessment Protocol

Objective: Document clinically meaningful improvements attributable to the intervention.

Methodology:

- **Patient-as-own-control design**: Compare pre- and post-treatment course for individual patients, accounting for the established natural history [74] [72]
- **Standardized assessment**: Implement objective, quantifiable measures of disease-relevant functions [72]
- **Blinded evaluation**: When feasible, incorporate blinded endpoint adjudication to minimize bias [72]
- **Statistical analysis**: Employ methods that account for regression to the mean and natural disease variability [74] [72]

GRN Correlation: Parallels approaches in evolutionary biology where isogenic backgrounds are used to isolate the phenotypic effects of specific genetic perturbations [13] [12].

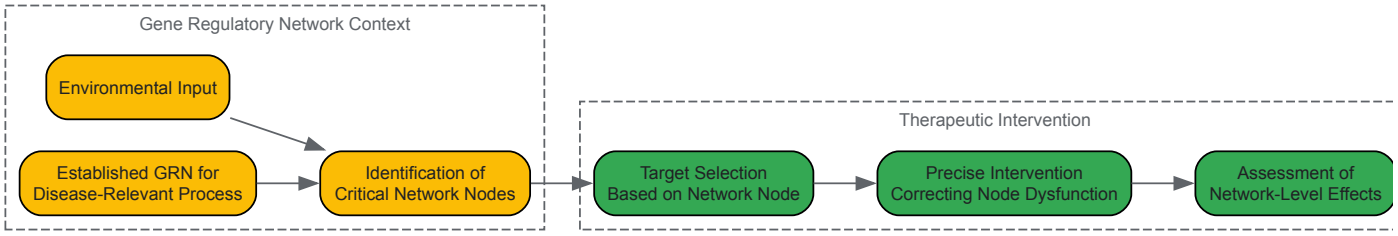


Diagram 2: The integration of GRN research with therapeutic development shows how understanding network architecture informs target selection and validation strategies.

Research Reagent Solutions for Pathway Implementation

Table 3: Essential Research Tools for Plausible Mechanism Pathway Applications

Tool Category	Specific Technologies	Research Applications
GRN Analysis Tools	BioTapestry [38]; TEKRABber [14]	Network modeling and visualization; Cross-species TE-gene correlation analysis
Transcriptomic Methods	Developmental transcriptomics [12]; RNA-seq across genetic backgrounds [12]	Identification of environmentally sensitive genes; Differential expression analysis
Experimental Models	Nematode (*P. pacificus*) mouth-form plasticity [12]; Individual-based metapopulation models [13]	Plasticity gene identification; Modeling evolution of complex traits

Tool Category	Specific Technologies	Research Applications
Computational Approaches	Jupyter Notebooks for modeling [38]; Bipartite network analysis [14]	Basic modeling approaches; KRAB-ZNF and TE interaction mapping

The Plausible Mechanism Pathway represents a fundamental shift in therapeutic regulation that aligns with principles from gene regulatory network research. By emphasizing **mechanistic understanding** over traditional statistical thresholds, it creates a viable pathway for treating ultra-rare diseases where conventional trials are impossible [70] [72]. This approach acknowledges that when the causal chain from molecular defect to clinical phenotype is sufficiently established, and when interventions precisely target known defects, rigorous evidence of effectiveness can be obtained through carefully documented individual responses rather than population-level statistics [74] [71].

The integration of GRN research into therapeutic development will likely accelerate as this pathway is implemented. Understanding how genetic perturbations disrupt regulatory networks, and how interventions restore normal network function, provides the scientific foundation for the pathway's mechanistic requirements [13] [12]. Conversely, data generated through this pathway will expand our understanding of human GRNs by documenting how targeted interventions alter network behavior in actual patients rather than model systems [14].

As the pathway evolves, key areas for development include standardizing evidence requirements for different therapeutic modalities, establishing thresholds for "success" in sequential patients, and creating efficient platforms for collecting and analyzing post-market real-world evidence [74] [72]. The FDA has indicated that formal guidance on the pathway is forthcoming, which should provide additional clarity on implementation details [74] [72]. For the research community, the pathway creates new opportunities to translate basic discoveries in gene regulatory networks into clinical benefits for patients with rare diseases, ultimately fulfilling the promise of precision medicine.

Conclusion

The study of Gene Regulatory Networks provides a unifying framework that bridges evolutionary biology and clinical medicine. Key takeaways include the understanding that core developmental processes are governed by deeply conserved GRN kernels, while evolutionary innovation and disease often arise from rewiring of their peripheral components. The methodological revolution powered by single-cell multi-omics and AI is rapidly translating GRN blueprints into actionable insights for drug discovery, as evidenced by the 'plausible mechanism' pathway for bespoke therapies. Future efforts must focus on creating generalizable, cross-cell-type models of the cis-regulatory code and standardizing the validation pipeline for GRN models. For biomedical researchers, this evolving knowledge base is critical for identifying robust therapeutic targets within complex regulatory architectures, ultimately paving the way for a new class of network-based interventions in developmental disorders and cancer.

References

- 1. Modeling GRNs with a Probabilistic Categorical Framework [https://arxiv.org/html/2508.13208v1]
- 2. A logic -incorporated gene regulatory network deciphers principles in... [https://elifesciences.org/articles/88742]
- 3. Developmental system drift and modular gene regulatory ... [https://www.life-science-alliance.org/content/8/11/e202503293]
- 4. CRISPR-GPT for agentic automation of gene-editing ... [https://www.nature.com/articles/s41551-025-01463-z]
- 5. Developmental system drift and modular gene regulatory ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12360431/]
- 6. Functional divergence of conserved developmental ... [https://www.nature.com/articles/s41598-025-14207-5]
- 7. Divergent evolutionary strategies pre-empt tissue collision ... [https://www.nature.com/articles/s41586-025-09447-4]
- 8. The divergent intron-containing actin in sponge ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12146513/]
- 9. Modeling the epigenetic attractors landscape: toward a post ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC4407578/]
- 10. Not just a colourful metaphor: modelling the landscape of ... [https://www.nature.com/articles/npjsba20161]
- 11. The geometry of cell fate [https://www.sciencedirect.com/science/article/pii/S2405471221004841]
- 12. Developmental transcriptomics in Pristionchus reveals the ... [https://pubmed.ncbi.nlm.nih.gov/40490376/]
- 13. A gene-regulatory network model for density-dependent ... [https://peercommunityjournal.org/articles/10.24072/pcjournal.626/]
- 14. Regulatory networks of KRAB zinc finger genes and ... [https://elifesciences.org/articles/103608]
- 15. Gene regulatory networks | Nature Reviews Genetics [https://www.nature.com/subjects/gene-regulatory-networks/nrg]
- 16. The interplay of tissue mechanics and gene regulatory ... [https://www.sciencedirect.com/science/article/pii/S1084952125000643]
- 17. The GRN concept as a guide for evolutionary developmental ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC9515236/]
- 18. Modularity and hierarchy in biological systems: Using gene ... [https://pubmed.ncbi.nlm.nih.gov/33602494/]
- 19. Modularity, criticality, and evolvability of a developmental ... [https://elifesciences.org/articles/42832]
- 20. Developmental system drift and modular gene regulatory ... [https://pubmed.ncbi.nlm.nih.gov/40813231/]
- 21. Paralogous Genes Involved in Embryonic Development [https://pmc.ncbi.nlm.nih.gov/articles/PMC9690401/]
- 22. Problems with Paralogs: The Promise and Challenges of ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11406157/]
- 23. On the effects of the modularity of gene regulatory networks ... [https://www.sciencedirect.com/science/article/abs/pii/S0303264721002264]
- 24. Paralog-divergent Features May Help Reduce Off-target ... [https://www.sciencedirect.com/science/article/pii/S1672022917300840]

- 25. Inferring gene regulatory networks from single-cell ... [https://www.nature.com/articles/s41587-024-02182-7]
- 26. Emerging technologies of single-cell multi-omics - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC12130758/]
- 27. Inference of cell type-specific gene regulatory networks on ... [https://www.nature.com/articles/s41467-023-38637-9]
- 28. Inference of cell type-specific gene regulatory networks on cell ... [https://pubmed.ncbi.nlm.nih.gov/37244909/]
- 29. Recovering gene regulatory networks in single-cell ... [https://pubmed.ncbi.nlm.nih.gov/41067887/]
- 30. Inferring gene regulatory networks from single-cell multi- ... [https://www.sciencedirect.com/science/article/abs/pii/S0303264725001418]
- 31. Single-cell genomics and regulatory networks for 388 human ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11365579/]
- 32. Single cell multiomics approach to analyze replication ... [https://www.nature.com/articles/s42003-025-08694-5]
- 33. Evolution of Gene Regulatory Networks that Control ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC3076009/]
- 34. Evolution of the gene regulatory network of body axis by ... [https://elifesciences.org/articles/89615]
- 35. Gene2role: a role-based gene embedding method for ... [https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-025-06128-x]
- 36. Discovering Causal Gene Regulatory Networks with LLMs [https://arxiv.org/html/2410.15828v1]
- 37. A comprehensive evaluation of large language models in ... [https://pubmed.ncbi.nlm.nih.gov/39364206/]
- 38. 2025 GERN Schedule [https://www.mbl.edu/education/advanced-research-training-courses/course-offerings/gene-regulatory-networks-development/schedule]
- 39. A fast validation test of gene regulatory network models via ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12089004/]
- 40. Towards Identifiability of Interventional Stochastic ... [https://arxiv.org/html/2505.15987v2]
- 41. A Tutorial on Analysis and Simulation of Boolean Gene ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC2808677/]
- 42. 9.7 Dynamic modeling of biological systems - Bioinformatics [https://fiveable.me/bioinformatics/unit-9/dynamic-modeling-biological-systems/study-guide/JPPa2d8urlsOuXBK]
- 43. Solving Fokker-Planck-Kolmogorov Equation by ... [https://arxiv.org/html/2510.09422v1]
- 44. Holimap: an accurate and efficient method for solving ... [https://www.nature.com/articles/s41467-024-50716-z]
- 45. Stochastic hybrid models of gene regulatory networks [https://www.sciencedirect.com/science/article/pii/S0025556418302098]
- 46. A large-scale benchmark for network inference from single ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11897147/]
- 47. Kolmogorov-Arnold Network for Gene Regulatory ... [https://arxiv.org/html/2506.13740v1]
- 48. Temporally distinct 3D multi-omic dynamics in the ... [https://www.nature.com/articles/s41586-024-08030-7]
- 49. Gene regulation in the 3D genome - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC6061806/]
- 50. Advances in the multimodal analysis of the 3D chromatin ... [https://www.nature.com/articles/s12276-024-01246-7]
- 51. 3D chromatin architecture and transcription regulation in cancer [https://jhoonline.biomedcentral.com/articles/10.1186/s13045-022-01271-x]
- 52. Multi-omics data integration using ratio-based quantitative ... [https://www.nature.com/articles/s41587-023-01934-1]
- 53. Unsupervised Multi-Omics Data Integration Methods [https://pmc.ncbi.nlm.nih.gov/articles/PMC8981526/]
- 54. Methods for the integration of multi-omics data: mathematical ... [https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-015-0857-9]
- 55. Multi-omics analysis reveals critical cis-regulatory roles of ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11914811/]
- 56. harnessing the power of single-cell multi-omic data [https://www.nature.com/articles/s41540-023-00312-6]
- 57. Reconstructing directed gene regulatory network by only ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC5001240/]
- 58. Systematic evaluation of chromosome conformation ... [https://www.nature.com/articles/s41592-021-01248-7]
- 59. The statistical-mechanics of chromosome conformation ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC3899129/]
- 60. Determinants of Topological and Their... Gene Regulatory Networks [https://www.preprints.org/manuscript/202411.0113/v1]
- 61. Three topological features of regulatory control... networks [https://www.nature.com/articles/s41598-021-03625-w?error=cookies_not_supported&code=a6bdf0a5-fe1c-40a6-a24e-18c87b06da23]
- 62. A novel gene expression stability metric to unveil homeostasis ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12512952/]
- 63. Gene2role: a role-based gene embedding method ... [https://pubmed.ncbi.nlm.nih.gov/40413377/]
- 64. Gene module-trait network analysis uncovers cell type specific ... [https://actaneurocomms.biomedcentral.com/articles/10.1186/s40478-025-02143-4]
- 65. and Evolutionary Properties of Topological and Community... Genes [https://pubmed.ncbi.nlm.nih.gov/27359334/]
- 66. Gene2role: a role-based gene embedding method for ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12103023/]
- 67. Identification of key interactive genes and their potential ... [https://www.nature.com/articles/s41598-025-14044-6]
- 68. Protocol for identifying differentially expressed genes using ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC10940175/]
- 69. Differential gene expression analysis pipelines and ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC10951429/]
- 70. FDA unveils new regulatory roadmap for bespoke drug ... [https://www.biopharmadive.com/news/fda-plausible-mechanism-pathway-n-of-1-crispr/805235/]
- 71. FDA Announces New Plausible Mechanism Pathway [https://www.pharmexec.com/view/fda-announces-plausible-mechanism-pathway]
- 72. FDA's New “Plausible Mechanism Pathway” and Other ... [https://www.arnoldporter.com/en/perspectives/advisories/2025/11/fda-plausible-mechanism-pathway-and-other-initiatives-drugs-for-ultra-rare-conditions]

- 73. FDA Leaders Propose New Regulatory Pathway for ... [https://www.aabb.org/news-resources/news/article/2025/11/19/fda-leaders-propose-new-regulatory-pathway-for-bespoke-therapies]
- 74. FDA announces “plausible mechanism” approval pathway ... [https://www.hoganlovells.com/en/publications/fda-announces-plausible-mechanism-approval-pathway-for-certain-personalized-therapies-with-few-detai]